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Review

Ultra-high-power laser welding of thick-section steel: Current research progress and future perspectives

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Outline

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Highlights

- An introduction to two ultra-high-power laser welding techniques and their respective advantages and disadvantages are presented.
- Formation of macroscopic weld defects in ultra-high-power laser welding process of thick-section steel and their current prevention methods are sorted and elaborated.
- Insufficient filler wire distribution, inhomogeneous microstructural characteristics, and poor mechanical properties of ultra-high-power laser welded thick-section steel joints are discussed.

- The future research perspectives of ultra-high-power laser welding process of thick-section steel are prospected.

Abstract

Thick-section steel is widely used in large structural applications, taking advantage of its characteristics, including high strength, good toughness, and low cost. Ultra-high-power laser welding (UHPLW), with laser power of 10kW or higher, offers a new approach for manufacturing large thick-section steel structures due to its core ability to provide deep penetration. However, UHPLW of thick-section steel faces several welding defects problems such as spatter, root hump, porosity, and solidification cracks. Additionally, inhomogeneous microstructure and poor mechanical properties remain challenging issues. These problems impede the widespread use of UHPLW technology in thick-section steel applications and require continuous efforts to address. This review aims to provide an introduction to UHPLW technology, describe the macroscopic defects problems and research progress in prevention methods, and highlight the microstructure inhomogeneity along with the detrimental effects on mechanical properties. Finally, four future research perspectives and a conclusion are presented to provide new ideas for the current dilemma and guide future development towards achieving high-quality welded joints of thick-section steel.

Keywords

Thick-section steel; Ultra-high-power laser welding; Welding defects; Microstructure; Mechanical properties

4. Future research perspectives

5. Conclusion

Declaration of Competing Interest

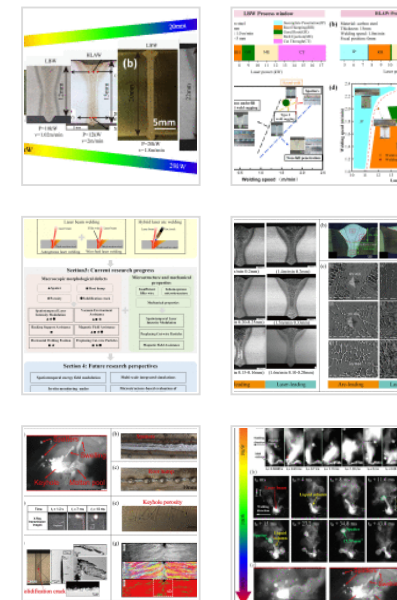
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Nomenclature

Table 2. An overview of some of the maj...

UHPLW

Ultra-high-power laser welding

EBW

Electron beam welding

LBW

Laser beam welding

HLAW

Hybrid laser arc welding

HAZ

Heat affected zone

SLIM

Spatiotemporal laser intensity modulation

HWP

Horizontal welding position

PCP

Preplacing cut-wire particles

MFA

Magnetic field assistance

VEA

Vacuum environment assistance

BSA

Backing support assistance

ALW

Autogenous laser welding

WFLW

Wire feed laser welding

LBS

Laser beam shaping

HSC

High speed camera

FKW

Front keyhole wall

RKW

Rear keyhole wall

CMT

Cold metal transfer

SEM

Scanning electron microscope

EBSD

Electron backscatter diffraction

CFD

Computational fluid dynamics

VOF

Volume of fluid

FZ

Fusion zone

BM

Base metal

AF

Acicular ferrite

NMI

Non-metallic inclusions

PWHT

Post weld heat treatment

PF

Phase field

CP

Crystal plasticity

FE

Finite element

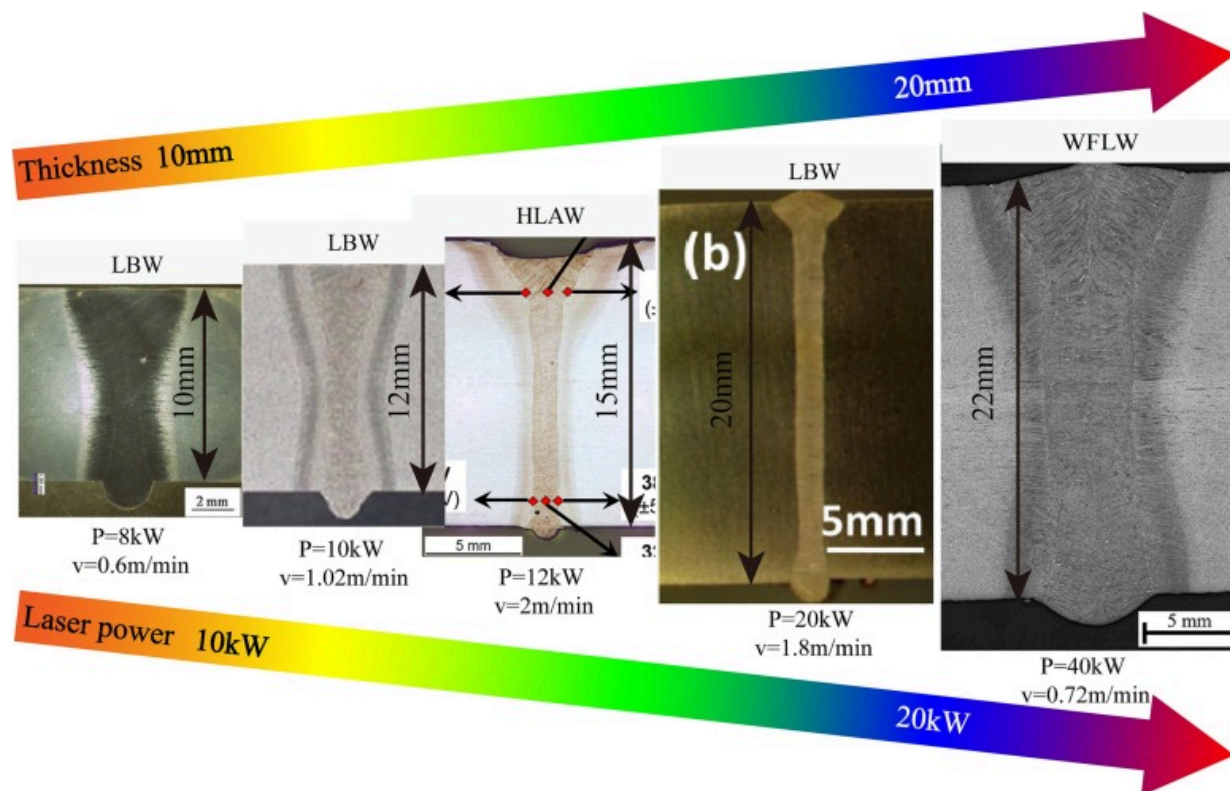
LBM

Lattice Boltzmann method

1. Introduction

Since its invention, steel has been the predominant engineering material for structural construction applications, with various grades and thicknesses available [1]. Among the steel family, thick-section steel, typically used as the main body of large-scale structures, has been widely employed with increasing demand in industries such as oil and gas pipeline, power plants, construction engineering, shipbuilding, and ocean engineering [2],

[3], [4], [5], [6]. In the manufacturing of large-scale structures, the welding technique selected has a significant impact on safety, design flexibility, environmental considerations, and economic factors. High-power laser welding is a favored option for large thick-section components due to its high energy density [7], fast welding speed [8], deep penetration [9], low distortion [10], and ability to produce high-quality joint [11]. The most noteworthy advantage is that high-power laser welding can significantly improve efficiency by reducing the number of weld layers required, sometimes even to a single pass. To the best of the authors' knowledge, in the laser power range up to 20kW, the penetration is approximately 1 mm per 1 kW, while it drops to approximately 0.5mm per 1 kW from 20kW to 100kW [12], [13]. With the thickness range up to 20mm, Fig. 1 demonstrates that thick-section steel can be welded on a single pass with high-quality and high welding speed using a high-power laser. Hence, thick-section steel and high-power laser welding are naturally linked together. However, recent reviews on laser welding of steel have usually concentrated on special steels [14], advanced processes [15], or industrial scenarios [16]. Relatively little attention has been paid to presenting a view of the steel thickness and laser power level. Therefore, current research progress and future perspectives for high-power laser welding of thick-section steel should be reviewed.



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Fig. 1. Cross-sections of full penetration joints in one single pass [17], [18], [19], [20], [21].

1.1. Scope of this review

This review aims to fill the gap by providing an overview of current research progress in UHPLW of thick-section steel. Our focus is on macroscopic defects and suppression methods, as well as microstructure and mechanical properties of thick-section steel joints fabricated by the UHPLW process. We have intentionally refrained from discussing laser welding in all laser power ranges and have limited our review to high power ($\geq 10\text{kW}$) laser welding process. The majority of the reviewed studies herein are related to thick-section steel and thick-section iron-based alloys with a thickness exceeding 10mm. The

majority (>80%) of the references are concerned with UHPLW of thick-section steel. Except in a few references, laser welding of thin-section steels and other metal alloys has been cited. It is important to note that narrow-gap laser welding [22] is not extensively covered in this review. A discussion of all laser welding techniques would further lengthen this review and dilute its focus. Thus, we have made a conscious decision to limit our scope to UHPLW of thick-section steels.

This review also highlights future research perspectives for achieving good properties with a view to engineering applications. By providing a concise and informative review of the latest advancements in UHPLW of thick-section steel, we hope to contribute to the ongoing development of this field.

1.2. Thick-section steel

Thick-section steel, an indispensable class of steel products, is comprised of a variety of grades. These grades are used in various applications where the requirements are as follows. (1) Cryogenic or elevated temperature service, (2) Weather resistance or corrosion resistance under harsh environments, (3) Good metallurgical and mechanical properties, such as high strength, good toughness, and excellent weldability, (4) Low price. The properties of thick-section steel make it suitable for use in large structural applications. A taxonomic classification of thick-section steels along with their varieties and applications is shown in [Table 1](#).

Table 1. Types of thick-section steels and their applications.

Type	Commonly used grade name	Application
Structural steel	S355JR	Construction industry
	S355G7	Offshore industry
	AH36 DH36 EH36	Shipbuilding industry

Type	Commonly used grade name	Application
High strength low alloy steel	S700MC	Load bearing structures
	S960QL	Heavy transport, lifting and mining industry
Chrome molybdenum steel	ASTM A387	Oil and gas industry, nuclear industry, and fossil fuel power stations
Weathering steel	S355J2W	Load bearing or heavy structures, bridge construction
Pressure vessel steel	ASTM SA285	Oil industry (including inshore and offshore), petrochemical industry, and gas production industry
	ASTM SA516	
	ASTM SA537	
Pipeline steel	X42, X52, X60, X80, X100, X120	Oil and gas transportation

The definition of thickness of thick-section steel can vary depending on the standards organizations, industrial application, and academic research. ISO divides steel plate into thin steel plates (less than 3 mm) and thick steel plates (not less than 3 mm) in thickness [23]. ASTM considers that the customary definition of thick-section steel strip or sheet vary from 6 mm through 25 mm [24]. In industrial production, the definition of thick-section steel can vary depending on the application and industry. For example, in shipbuilding, thick-section steel typically refers to plates with a thickness greater than 40 mm, while in construction, it may refer to sections with a thickness greater than 20 mm. Academic research has focused on understanding the mechanical properties and welding characteristics of thick-section steel. It is generally accepted that steel plates with a thickness exceeding 10 mm belong to the category of thick-section steel plates. Combined with the majority research objects about laser welding of thick-section steel

plates in the literatures, the scope of this review is focused on thick-section steel with a plate thickness of 10mm and above.

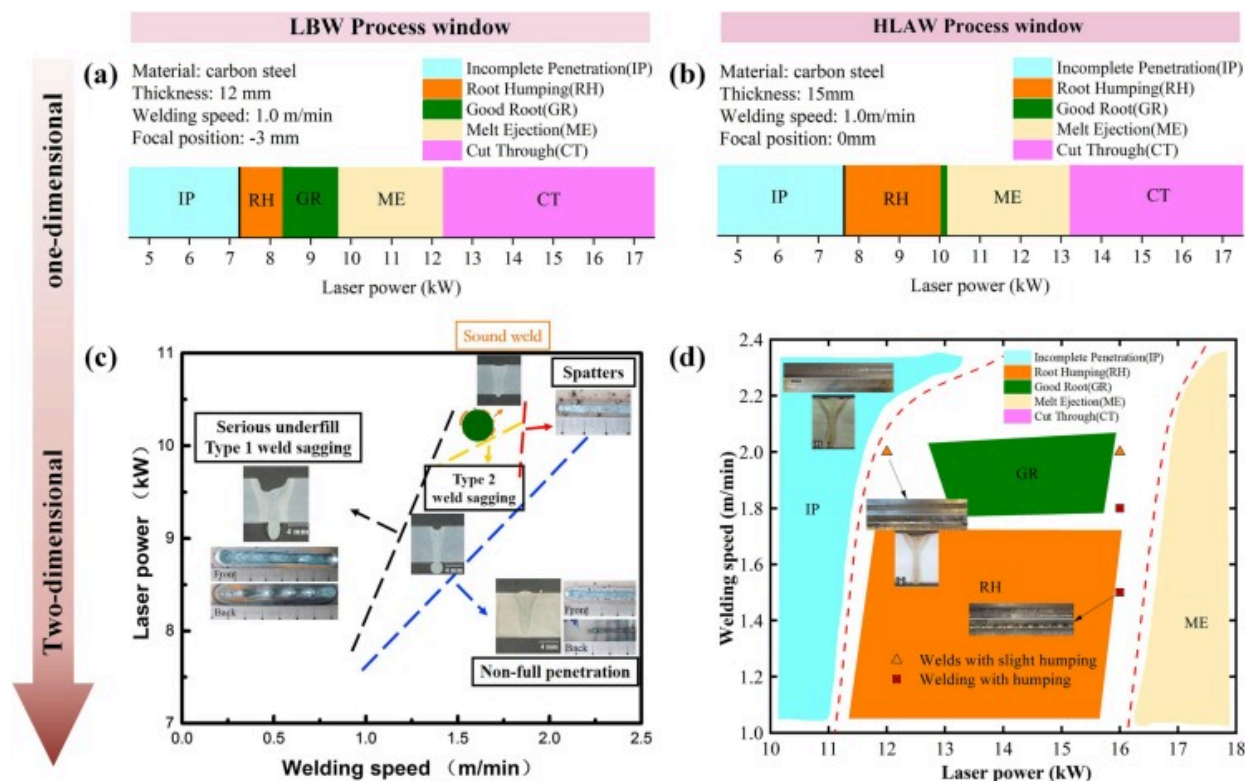
1.3. Ultra-high-power laser welding

Up to now, laser welding of relatively thick steel up to 10mm in one single pass, supported by kW-class laser with good stability [25], has been well established. Whether in academic research or engineering applications, the acceptable macroscopic morphology microstructure and mechanical properties meet the requirements of relevant standards. Guo et al. [26] achieved a single-pass weld of 8mm thick steel with a laser power of 6.5kW. Free of macroscopic defects were observed in the joint, which showed better tensile properties than the base material. Cao et al. [27] successfully applied hybrid laser-arc welding technique to completely penetrate a nearly 10mm thick butt joint in a single-pass process using a maximum laser power of 5.2kW. According to publicly available information [28], some advanced shipyards have actively applied laser welding technology to the production of ship panels with a thickness of up to 7mm.

In recent years, UHPLW has become one of the most popular welding technologies in the world due to the stability and maximum output power of the laser with high-quality and acceptable cost. With the high-power density of 10kW-level laser, high depth-to-width ratio welds can be obtained. UHPLW can greatly reduce the number of thick-section steel weld pass layers even to one-single-pass, which can improve the welding efficiency significantly [21]. It should be noted here that the metal alloys with a thickness of exceeding 10mm, which have a wide range of engineering applications, are mainly steels. Thick-section steels and UHPLW are then naturally linked together in the fabrication of large structures, particularly in the configuration of butt joints and T-joints [20], [21], [29], [30], [31]. Fang et al. [20] obtained defect-free butt-weld joints in 20mm thick austenitic stainless steel at welding speeds 1.8m/min with a defocusing distance of -10mm using a laser power of 20kW. Churiaque et al. [31] produced full-penetration HLAW T-joints in 15mm thick EH36 steel with square grooves by single-sided welding using a laser power of 12.5kW. UHPLW, either standalone or coupled with GMAW, could

prove to be a game-changing technology for large thick-section steel structure fabrication. Therefore, research on UHPLW technology is of great scientific significance and application value.

However, UHPLW of thick-section steel suffers from severe spatters [32], root hump [33], porosity [34], solidification crack [35], and poor mechanical properties [36], although there are some successful reports [7], [37] in macroscopic weld profile. These welding defects problems are the main obstacles that limit the application of UHPLW on thick-section steels owing to the narrow welding process window. As shown in Fig. 2(a, c), the LBW process window for 10mm and 12mm thick-section steel is a small zone (indicated by olive green color) with good root cases. The HRAW process window for 15mm thick-section steel in Fig. 2(b, d) is a potentially acceptable zone (olive green color) without satisfactory root cases [38], [39], [40]. This crucial information drive researchers to focus on these welding defects. The cause of welding defects and corresponding prevention methods have been extensively studied. The aiding methods have achieved good improvements such as spatiotemporal laser intensity modulation (SLIM) [41], horizontal welding position (HWP) [42], preplacing cut-wire particles (PCP) [43], magnetic field assistance (MFA) [44], vacuum environment assistance [45], backing support assistance (BSA) [46], etc. Nevertheless, the solidification cracking in the root and the inhomogeneous microstructure along the thickness direction has still not been well resolved. The challenge facing the international welding academia and industry is how to effectively and properly use ultra-high-power lasers to achieve high-quality welds of thick-section steel. Thus, new theories and methods that can address these challenges and optimize the UHPLW process of thick-section steel are urgent.



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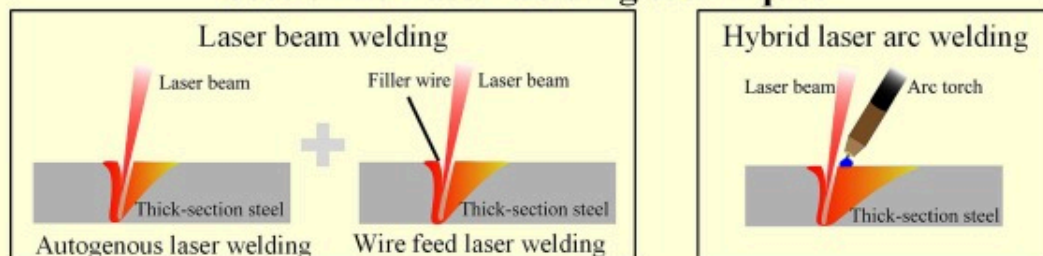
Fig. 2. One-dimensional process window of (a) LBW process is a small zone near the laser power of 9kW, and (b) HLAW process window is almost non-existent, two-dimensional process window of (c) LBW process is a small zone near the parameter of 10kW laser power and 1.0m/min welding speed, and (d) HLAW process was potentially acceptable zone in the range of 14 to 16kW laser power and 1.8 to 2.0m/min welding speed combined with limited arc heat input ($\sim 0.4\text{kJ/mm}$) [38], [39], [40].

1.4. Organization of this article

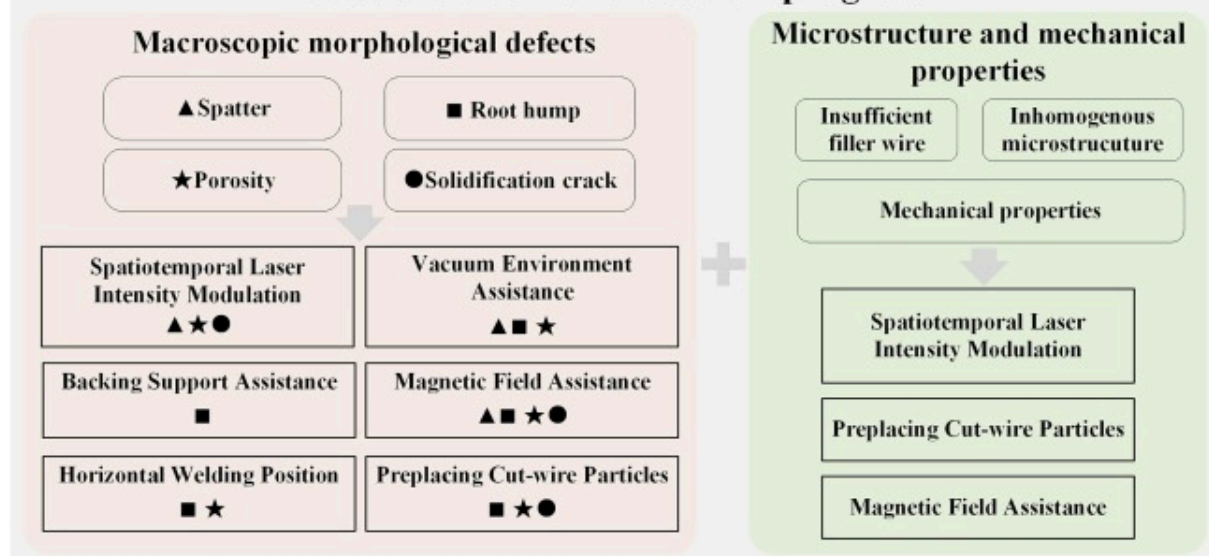
To provide the readers with new ideas for the current dilemma and guide future development of high-quality welded joints of thick-section steel. This review

concentrates on thick-section steel with a thickness of 10mm or more and divides the UHPLW technique into laser beam welding (LBW) and hybrid laser arc welding (HLAW). This paper aims to review the current welding defects problems, development progress on prevention methods, and future perspectives for UHPLW of thick-section steel. In this paper, an overview of UHPLW of thick-section steel was arranged in detail. [Fig. 3](#) shows the content structure, which can be divided into three main parts. It begins with an introduction of the two welding techniques in [Section 2](#). Then, a comprehensive discussion on current progress is divided into two parts in [Section 3](#): the first one presents the problem of macroscopic morphological defect formation and their prevention methods sorted and elaborated; the second one discusses microstructure inhomogeneity and the affected mechanical properties, as well as potential improvement methods. Further, four future research perspectives for achieving high-quality welding of thick-section steel are presented in [Section 4](#). The final section contains a conclusion about the whole described process of current research progress and future perspectives regarding UHPLW of thick-section steel.

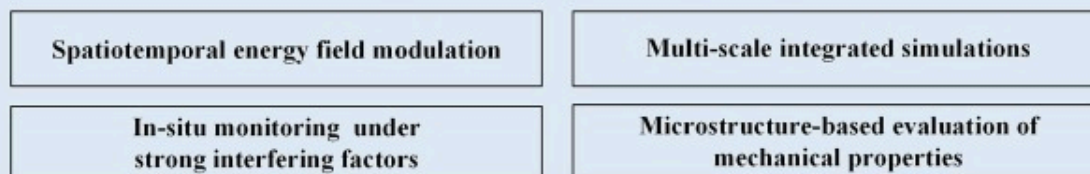
Section 2: Laser welding techniques



Section 3: Current research progress



Section 4: Future research perspectives



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Fig. 3. The structure of this review and the flow chart of UHPLW of thick-section steel.

2. Laser welding techniques

For UHPLW of thick-section steel, LBW and HLAW can be applied, offering deep penetration depths in a single pass. LBW can divide into autogenous laser welding (ALW) and wire feed laser welding (WFLW). These techniques are briefly discussed in terms of their advantages and disadvantages for a better understanding of the thick-section steel welding process.

2.1. Laser beam welding

To achieve deep penetration in LBW, the keyhole mode must be used to allow the high-power laser beam to drill into the inner part of the molten pool. With the ongoing improvement of the maximum power and stability of the laser, satisfactory penetration is now attainable through the use of appropriate processing parameters [7], [9], [20].

ALW uses laser energy to melt the base metal itself to realize joining thick-section steel. The typical ALW is unrealistic with several limitations for industrial application, the strict requirement for the gap tolerance restricted by the small laser spot and a possible loss of important alloying elements due to severe metal evaporation [9], [47]. ALW of thick-section steel with a single pass often results in mistracking of the weld seam, spattering at the weld surface, root humping, porosity, and cracking in the weld zone [42], [48]. Even some series of thick-section steel undergo solid-state transformation at high cooling rates after ALW, generating brittle microstructure [49] which has inappropriate hardness and poor toughness [7].

Part of the limitations can be overcome by WFLW, in which the filler wire is melted by the laser energy and then transferred into the molten pool [18], [50], [51], [52]. WFLW improved process stability and got a sound weld shape to reduce spatter, underfill, root

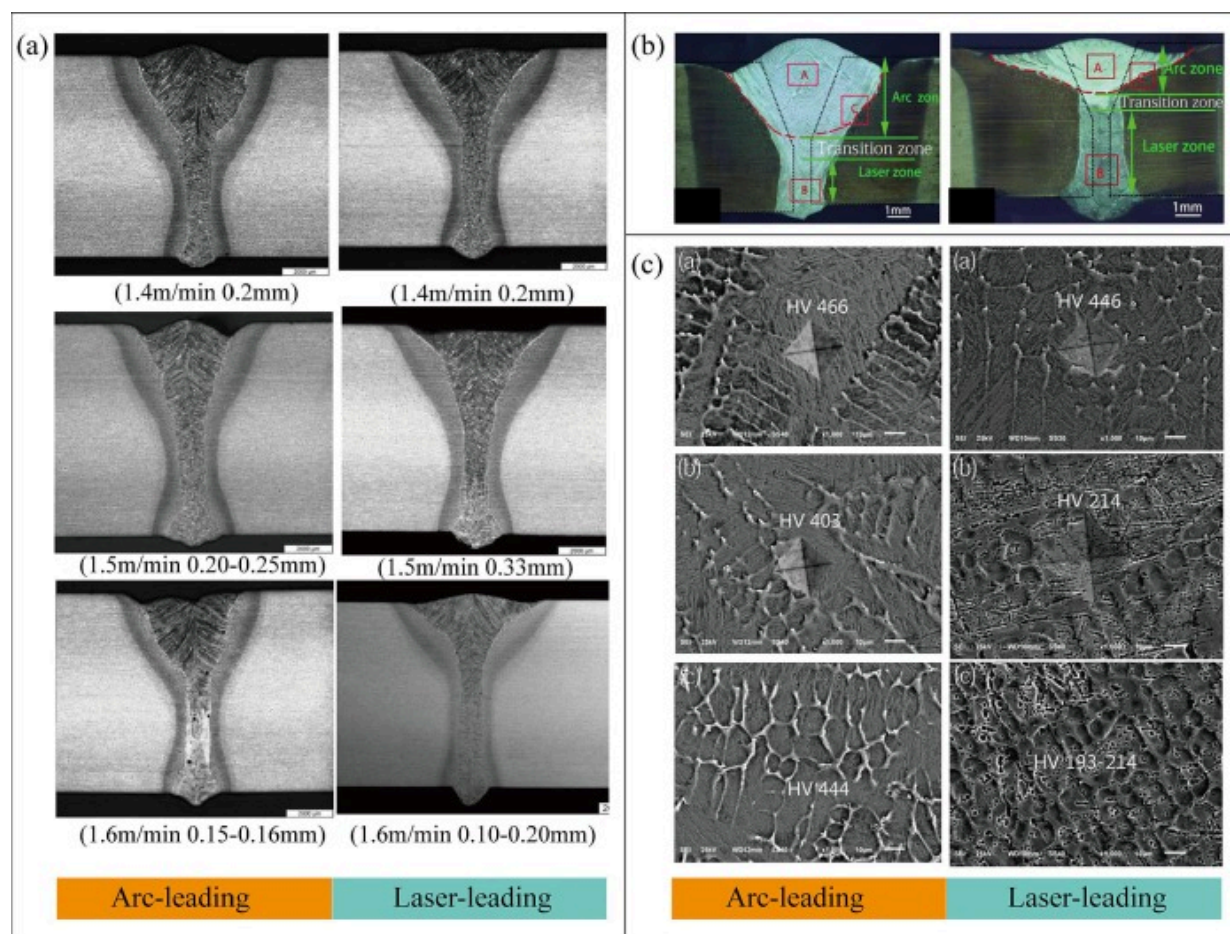
hump, and other defects [18], [51]. With the increase in thickness, welding of thick-section steel plates was more challenging, and the process was sensitive even with a slight parameter change. Insufficient metal mixing is another issue for thick-section steel in the WFLW. The filler wire may concentrate at the upper part of the weld, which is caused by the steep shape of the molten pool and the high cooling rate. The non-uniform element distribution may cause defects such as solidification cracking, then undermine the metallurgical and mechanical properties.

2.2. Hybrid laser arc welding

Hybrid laser arc welding (HLAW) is to combine a laser beam and an electric arc to achieve high-quality and efficient welding under a synergistic effect [53], [54]. HLAW inherits the advantages of laser and arc welding, and can compensate for the drawbacks of ALW [36], [55]. HLAW of thick-section steel has attracted considerable attention as a promising alternative for the replacement of conventional processes [56].

Nevertheless, proper selection of welding parameters is crucial to avoid welding defects and obtain a better weld shape with superior microstructure and mechanical properties [27], [57]. For instance, the relative position of laser and arc source is a critical parameter that produces different results, as illustrated in Fig. 4. The laser-leading HLAW process can produce higher quality welds than arc-leading HLAW with less underfill and porosity defects due to lower evaporative losses of alloying elements and lower cooling rate [27]. In contrast, the arc-leading HLAW process results in a better weld shape and a more uniform microstructure, which exhibits smaller grain coarsening in HAZ due to fast cooling rate [57]. Nevertheless, the process increases the risk of keyhole collapse due to droplet impingement on the keyhole opening [58] and is not conducive to pores escaping when the laser-arc distance is short [27]. It must however be noted that there is no conclusive experimental evidence for these alternative explanations. Due to numerous process parameters, it is crucial to conduct an initial analysis to determine the parameter with the greatest effect on the formation, such as the filler wire rate, the laser-arc distance, and the leading configuration. Additionally, it is necessary to consider the

coupling relationship between the process parameters before exploring the impact on the microstructure and mechanical properties of the thick-section steel welded joints.



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Fig. 4. Comparison between Arc-leading and Laser-leading mode on macrostructure (a), (b) and microstructure(c) of the welded joints in HLAW [27], [57].

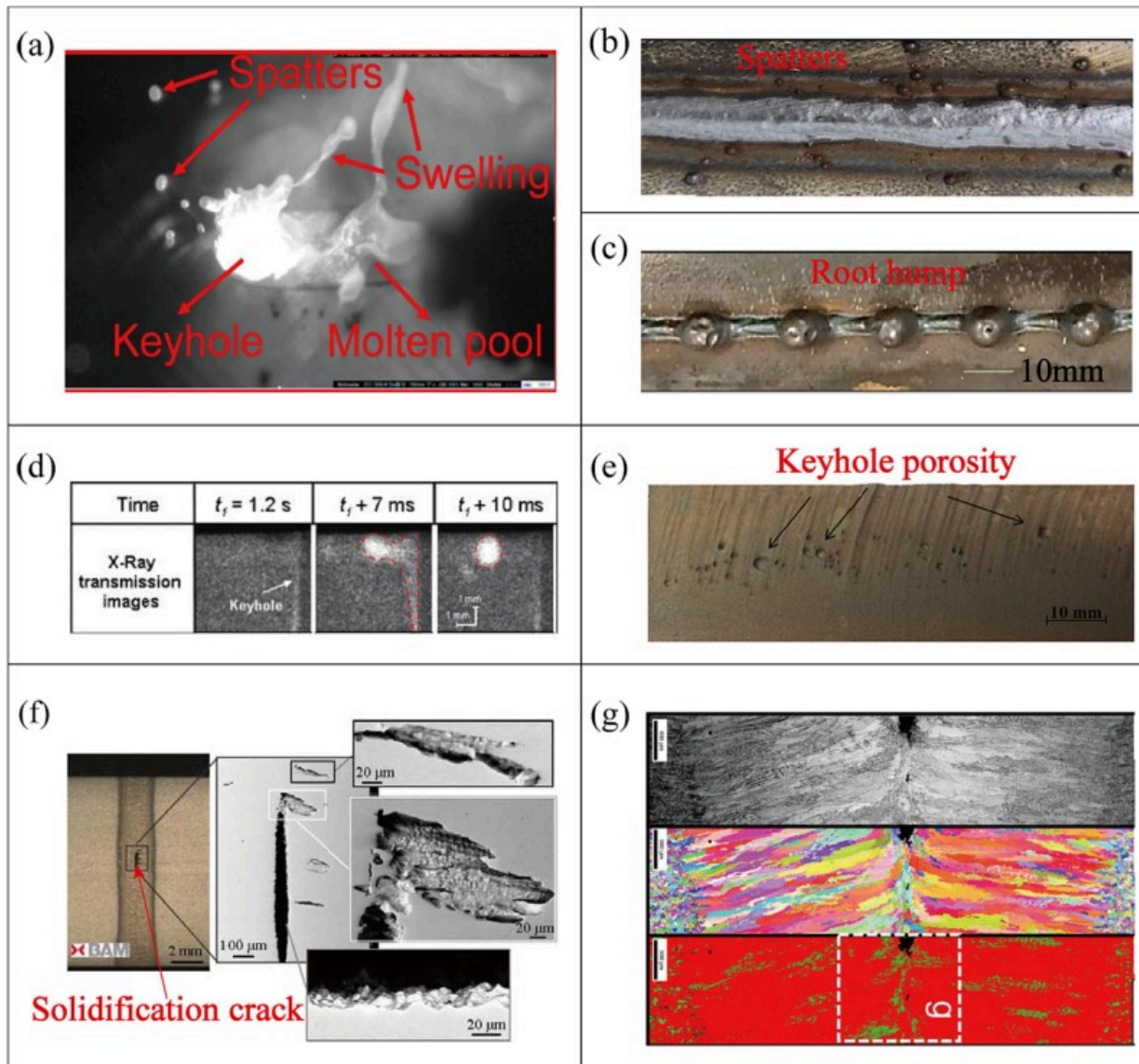
In fact, the origin mechanisms of the laser-arc synergistic effect, the laser induced metal transfer effect, and the shielding gas effect lack scientific explanation [59], [60], [61], [62],

[63]. For instance, the penetration depth does not always increase with the shortening of the laser-arc distance, but reaches a maximum at an appropriate distance. When the distance is too small, part of the laser energy is reflected and absorbed due to the droplet impingement, and the droplets interfere with the keyhole, causing the keyhole to collapse [64]. As the distance increases, the arc stability strengthens the energy absorption rate increased due to the induced laser beam, resulting in an increase in the penetration depth. However, this positive effect caused by the laser-arc synergistic effect diminishes as the distance becomes too large. Although the aforementioned observations have been reasonably explained, there is still a lack of information regarding melt pool flow information to directly prove the explanation. Additionally, the synergistic effect weakens as the thickness of thick-section steel increases in the welded root with one single pass. Therefore, the problems such as solidification cracking and brittle phase formation encountered at the root of the thick-section steel weld remain an open question [65], [66], [67]. HLAW of thick-section steel requires further research on this point, which contributes to the process of reliability and efficiency.

3. Current research progress

3.1. Macroscopic morphological defects

During the UHPLW process of thick-section steel, distinctive features such as highly unstable keyhole behavior [68], extremely narrow (depth/width>10) keyhole morphology [69], dense wrinkle surface structures [70], and bulge formation [71] can be observed. Welding defects generated such as spatter, root hump, porosity, and solidification cracks, can compromise the overall quality of the final weld joint. Fig. 5 illustrates the different possible defects that can occur during the welding process. These defects formation are unusual under conventional welding conditions, particularly root humping and solidification cracking. The generation mechanism of each morphological defect is investigated through experiments and simulations, followed by various suppression methods are proposed and validated.



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Fig. 5. An overview of different types of defects that may form during UHPLW of thick-section steel. (a) and (b) show spattering via HSC and spatters on the welded surface [12], [72]. (c) shows root hump [19]. (d) and (e) show pore formation via X-ray observation and

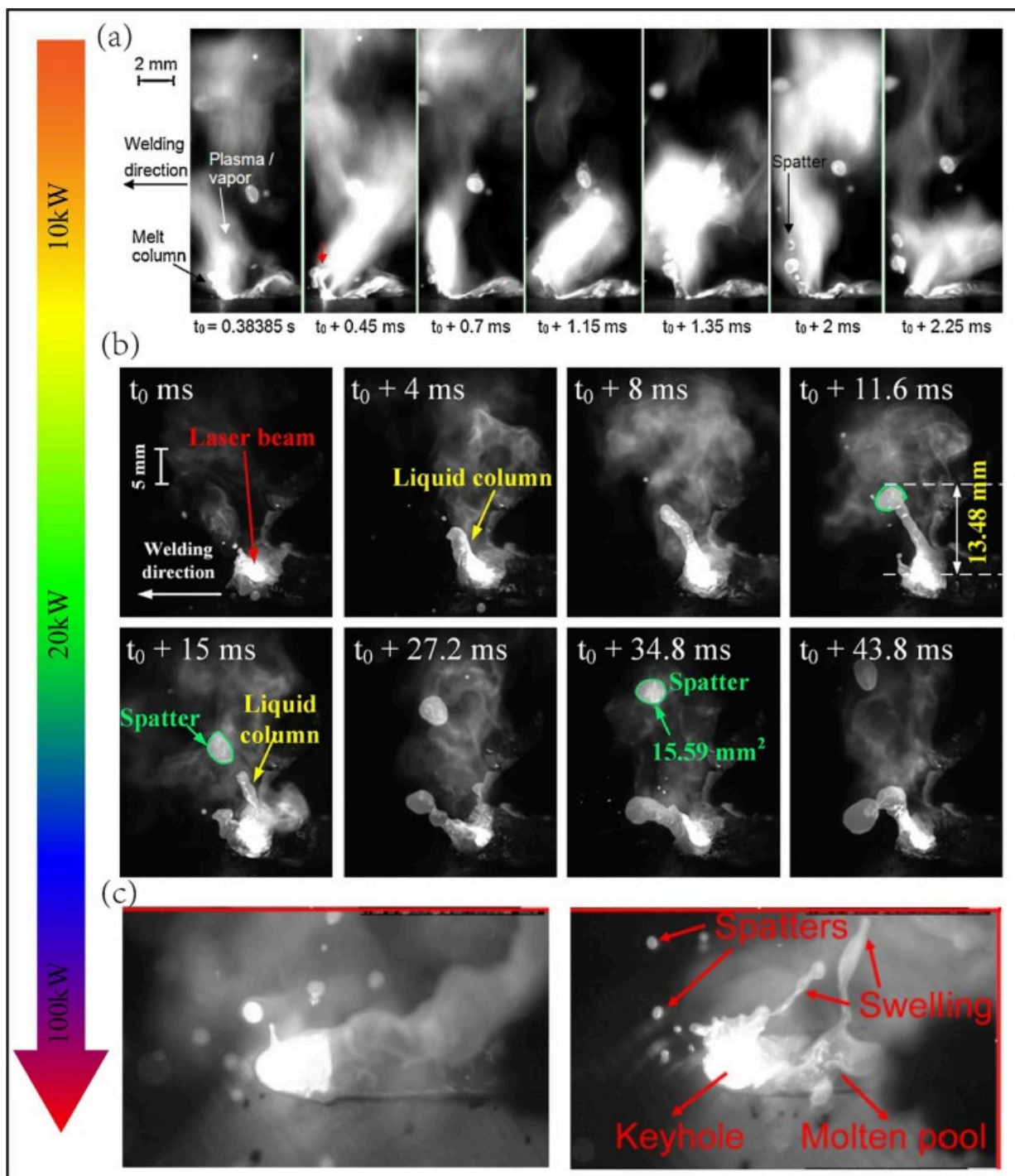
keyhole porosity [11]. (f) and (g) show the solidification cracking formation via SEM and EBSD analysis [12].

3.1.1. Spatter

Spatter, referring to the ejection of droplets from the melt pool, is one of the major problems in UHPLW of thick-section steel. Spatters have the potential to contaminate the focusing lens and negatively impact the welding process's stability by altering the laser beam's focusing characteristics. Severe spattering can lead to undesirable weld defects, such as underfill and undercuts [11], [13], [73], which can have a detrimental effect on the weld's mechanical properties.

It is well acknowledged that spatters are ejected due to the momentum of the droplets being sufficient to overcome the molten pool surface tension [11], [32], [74]. This spatter phenomenon depends on the process variables such as laser power, welding speed, focal point position, and beam inclination angle. High speed camera (HSC) is the main tool to observe spatter generation and study their influence mechanism. Fig. 6 illustrates the dynamic distribution of spatter generation observed during experiments at various laser power levels. Different from relatively-low-power laser welding, the spatter is characterized by the liquid column in UHPLW process. At the laser power below 10kW, droplets are eject from the observable liquid column at the front keyhole wall (FKW), leading to the spatter formation [75]. When the power reaches 20kW, a significantly higher liquid column induces large particle spatters at the FKW [76]. During the UHPLW process at 100kW, there is no clear distribution tendency of the liquid column, and spattering becomes severe [12]. The primary factor affecting spatter the location of spatter ejection around the keyhole is welding speed [11], [74], [77], [78], [79]. At low welding speed (below 3 m/min) more spatters tends to occur at the FKW due to the laser-induced evaporation that squeezes the front molten metal layer. In contrast, at high welding speeds (above 3 m/min), the ratio and the size of spatters occurring behind the rear keyhole wall (RKW) increase. Other factors affecting the position of the liquid column and spatter distribution include the inclination angle of the laser beam [11], [80]

or the focal point position is [81], [82]. In order to gain a deeper understanding of the spatter generation mechanism, a steel/glass “sandwich” specimen [75], [83] and an X-ray imaging system [73] were utilized to observe the keyhole and melt pool. The primary driving forces for spattering are attributed to the strong shear stream of the plume blowing out from the keyhole and the recoil pressure of metal vapor inside the keyhole [75], [78], while the former factor may play a dominant role [11]. Further, simulation can give us a clearer picture to reveal the physics of spatter formation quantitatively [83]. Some new features such as the intense localized boiling of the mesoscale keyhole wall [79] and the elongated liquid column [76] have emerged for UHPLW of thick-section steel. Unfortunately, the model that can capture the different physics process through the above features between UHPLW and conventional laser welding has not been reported yet. The evolution law of the liquid column in different areas around the keyhole can be further studied.



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Fig. 6. Spatter was identified by HSC observation during bead-on-plate UHPLW process with parameters (a) thickness 12mm, laser power 10kW, welding speed 3m/min [75], (b) thickness 20mm, laser power 21 kW, welding speed 1.2m/min [76] and (c) thickness 100mm, laser power 21 kW, welding speed 3–5m/min [12].

Since spatter is attributed to the metal vapor, suppression of the metal vapor sources is essential. The metal vapor is derived from the laser beam, and is influenced by the ambient pressure. As previously mentioned, forward laser beam inclination [13] and the negative focal spot position [81] can have a suppressing effect. SLIM may offer extra strategy for spatter prevention such as laser beam oscillation [84], dual laser beam [85], and superimposed-mode beam [86], that have been demonstrated in galvanized steel and aluminum alloy welding. Laser beam shaping (LBS), as an advanced SLIM technique, has been investigated in spatter suppression. For instance, a novel adjustable-ring mode was used to achieve less spattering laser welding of stainless steel due to a relatively opening stable keyhole [87]. Under VEA, Borner et al. [88] and Jiang et al. [89] obtain excellent weld bead without spatters.

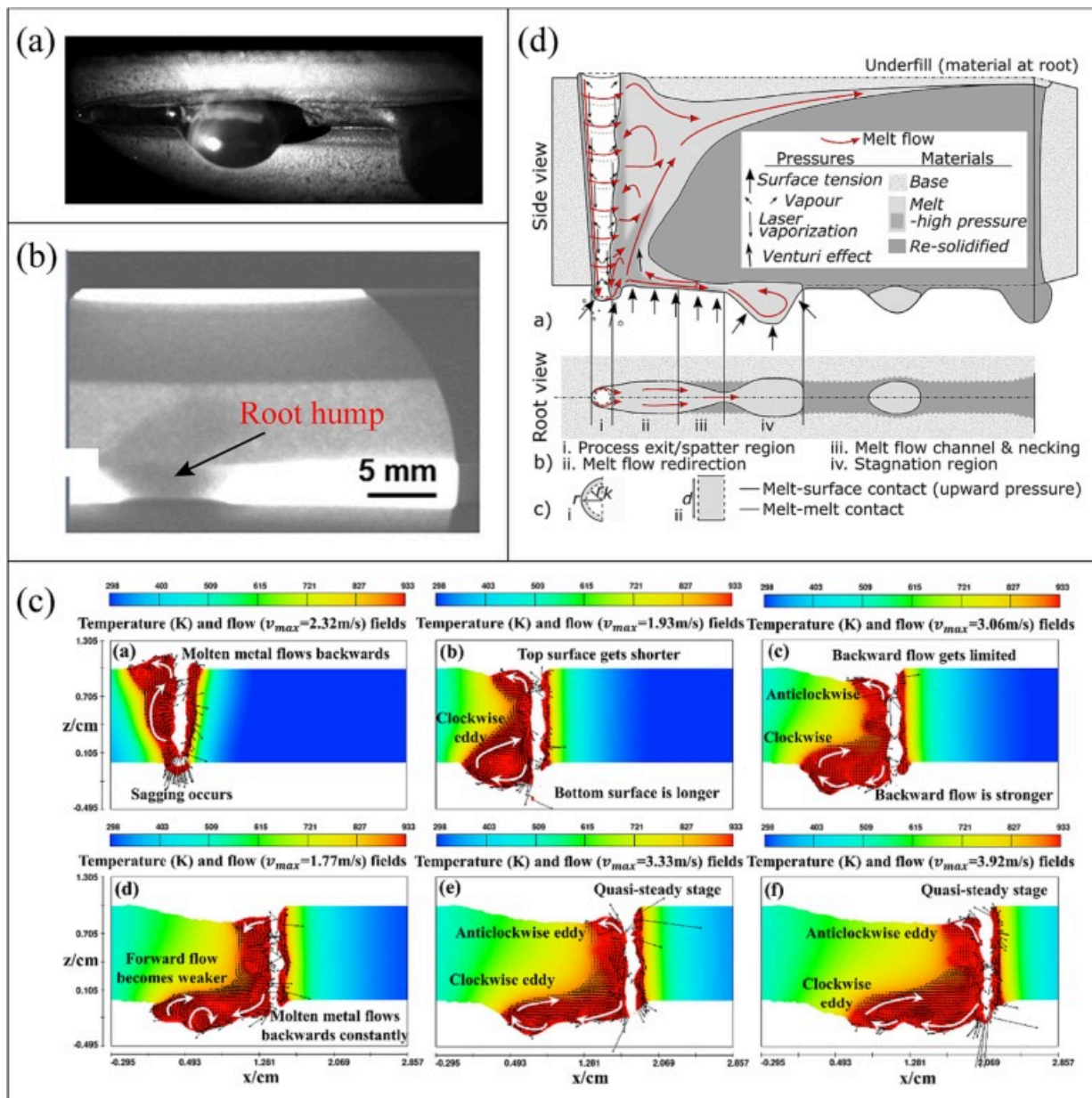
The spatter formation mechanism of HLAW is more complicated than that of LBW thanks to the new added arc effect, the shielding gas effect [90] and the laser-arc synergistic effect [91]. Spattering is additionally affected by the metal droplet transfer effect [92], [93] due to the different arc modes [58], [94], the inappropriate arc power, i.e. the voltages and current range, and the composition of the shielding gas [72], [90], [95], etc. The analysis of the force acting on the droplet was the key point that the competitive dominance between gravitational force, electromagnetic force, plasma drag force, metal vapor reaction force and surface tension [72], [93], [96], [97]. Compared three arc modes, it was found that cold metal transfer (CMT) mode produced less spatter than standard and pulse modes due to surface tension governed drop transfer through controlling wire feeding [94]. When the globular mode is transferred a buried arc mode through

optimizing the arc voltage, spattering significantly reduced [90], [95]. Under different shielding gas mixtures ratio, 20% CO₂+80% Ar was used to produce a stable spray transfer mode and little spatter were achieved due to stable comprehensive force [72]. Arc-leading mode resulted in larger keyholes than the laser-leading mode, thus reducing the fluctuation of metal vapor and the height of the liquid column [76], and inhibiting the formation of spatters [98]. Selection of arc-leading mode that also produce sufficient liquid metal and the larger molten pool is considered to be beneficial to reduce spatters during UHPLW process. However, ultra-high-power laser induced strong action force has a significant effect on the droplet transfer, relevant force analyses on spatter formation are still lacking.

3.1.2. Root hump

An attractive solution for joining thick-section steel, above 10mm or up to 20mm thickness is single pass full penetration laser welding [99], [100], [101]. However, weld root hump is usually formed in full penetration laser welding, even plates thickness just 10mm [42]. In this paper, root hump is used to represent a certain type of descriptive defect such as continuous or intermittent root sagging, dropping out or chain of pearls [102]. Root humping is also linked to porosity [103], underfill [104] and undercuts [105] which reduce the fatigue life of the welded component [106] and hinder the application of UHPLW of thick-section steel.

To reveal the causes of the root hump, HSC [33], [38], [104], [107], X-ray technologies [108], [109] and simulation methods [110], [111], [112], [113] were used. The growth of the hydrostatic pressure with increasing thickness of thick-section steel making it difficult to balance surface tension. And this phenomenon will drive the melt to flow to the rear end [110], and it will continue to gather before solidification, thereby forming a root hump. Therefore, the backflow driving force of the molten pool and the solidification rate of the backflow melt are necessary conditions for the root humps (Fig. 7). The periodicity characteristics of root hump is ascribed to the periodical small volume of the metal pool the periodical melted Delta-shape region [114].



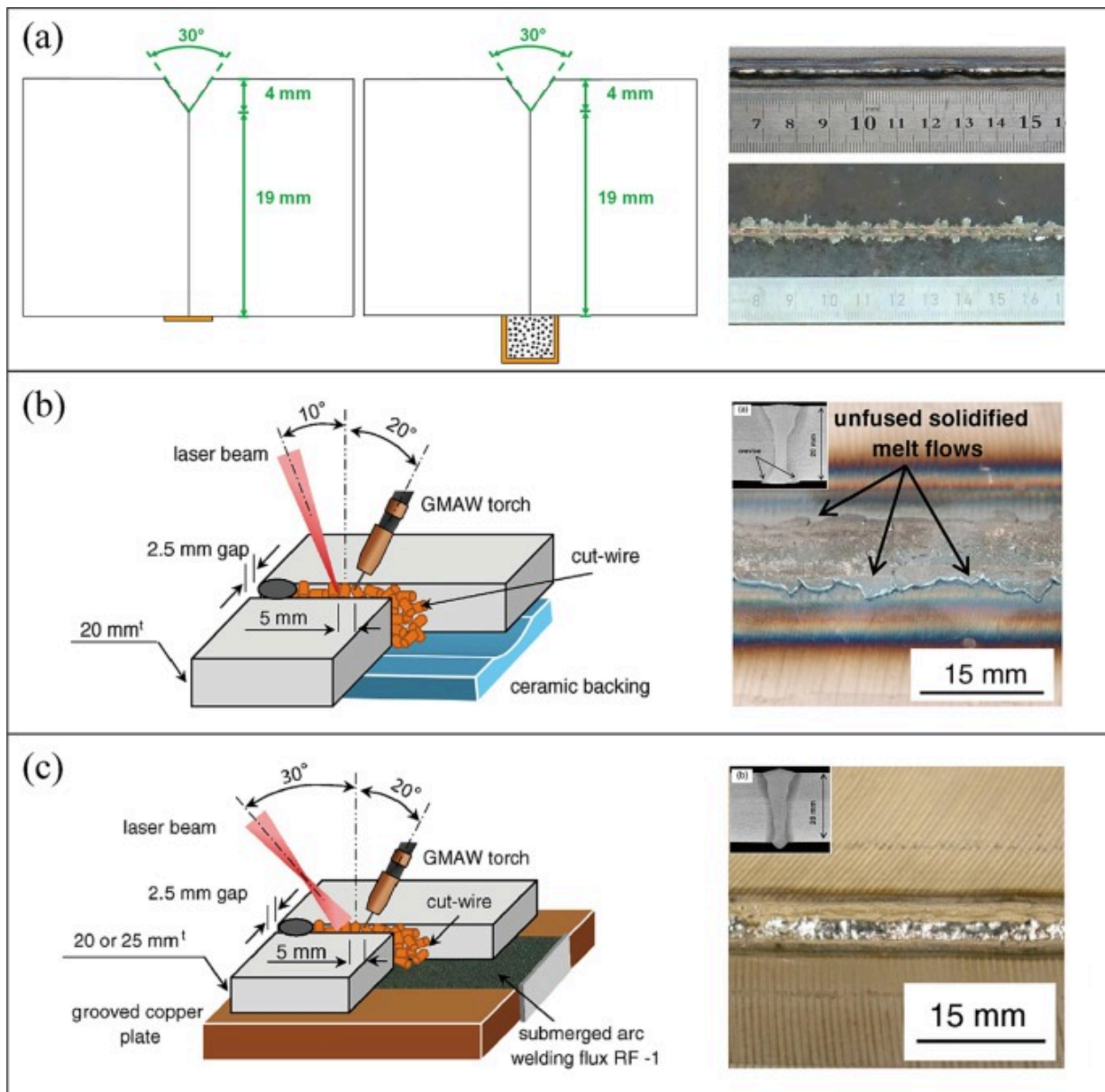
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Fig. 7. The generation mechanism of root hump: (a) HSC observation during LBW [104] and (b) X-ray observation during HLAW [108] of the bottom melt pool dynamics; (c) modeling result during WFLW of the melt flow fields [113] and (d) schematic illustration [33].

Suppressing root hump requires a delicate balance of optimized welding parameters with limitations. BSA, MFA and VEA are considered effective methods for achieving good weld beads during full penetration welding. Seffer et al. [46] used a metal strip or powder for BSA in HLAW of pipeline steel and successfully suppressed root hump formation. Wahba et al. [43] used ceramic strip and ceramic submerged arc welding flux for BSA and achieved full penetration in HLAW of 20mm thick-section steel. Fig. 8 shows the acceptable bottom surface appearance after HLAW with different backing support materials. Bachmann et al. [48], [115] found that single-pass laser beam welding of thick-section steels with MFA can suppress root humping to achieve high quality joints under certain conditions of oscillation frequency and magnetic flux density, and, and provided a theoretical explanation for this innovative method. Fig. 9 shows the scheme of the MFA system and the good weld beads results of UHPLW of 20mm thick-section steels. The introduction of additional oscillating and permanent MFA method can also bring several beneficial effects, such as porosity suppression and grain refinement. However, when welding ferromagnetic steels, the material exhibits a strong hysteresis loop phenomenon, resulting in additional power being converted into heat loss, making cooling of the MFA system challenging. Additionally, the material loses its ferromagnetic properties when heated above the Curie temperature, resulting in the coexistence of areas with and without hysteresis behavior, making the search for process windows difficult. LBW under VEA is another promising method. Studies by Katayama et al. [45], Reisgen et al. [116], and Jiang et al. [40], [89] have demonstrated that this approach can significantly increase process stability, resulting a high-quality weld seam and deep penetration without root hump. Fig. 10 depicts the experimental setup for VEA and presents the results of LBW under low vacuum environment. The VEA method can effectively inhibit the plume, lower the boiling point of the material, and thereby increase penetration efficiency [117]. HWP

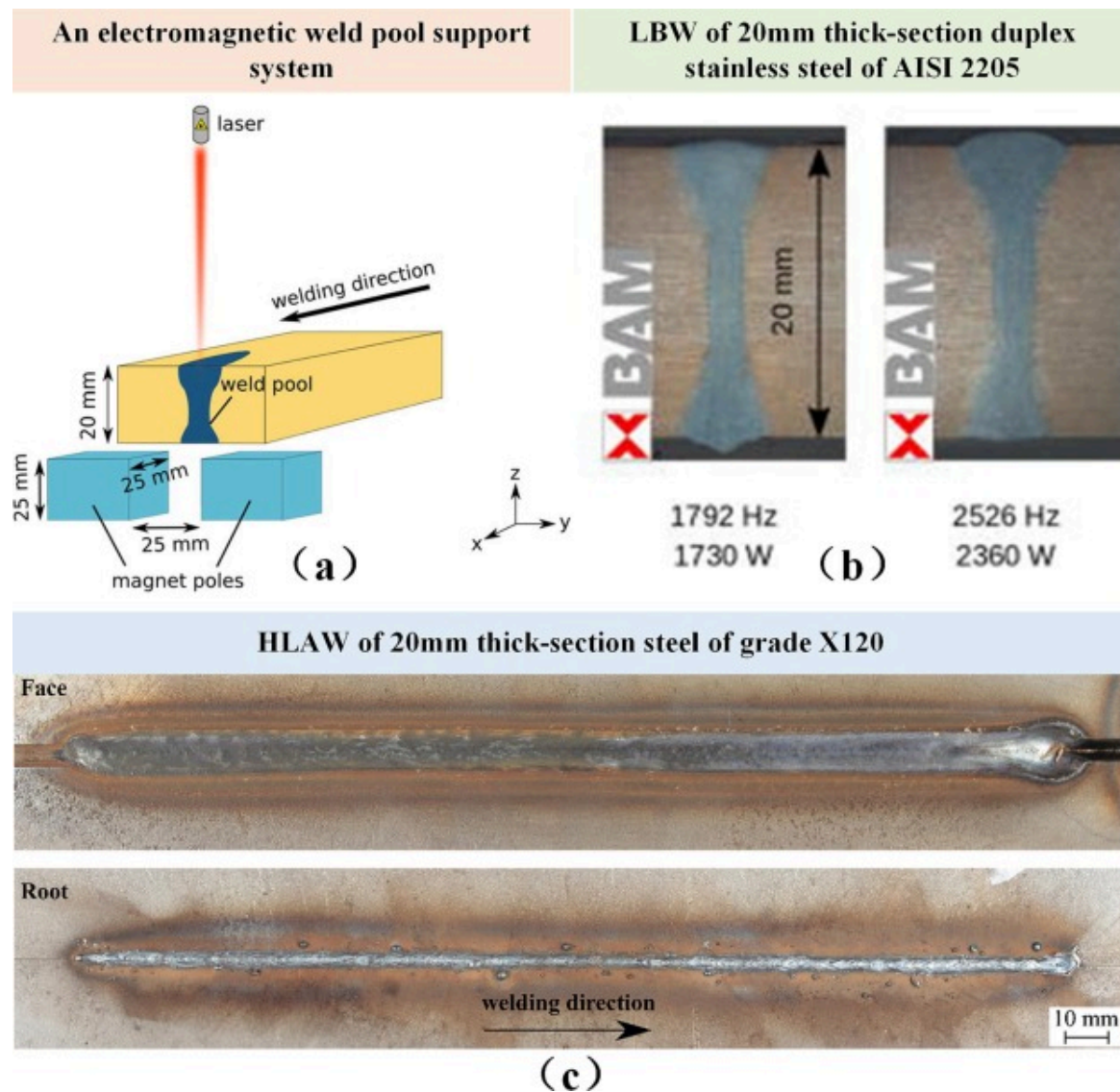
can also achieve good results while reducing the gravitational pressure towards the root [17], [42], [47], [118], [119]. In the case of HLAW process, the arc-leading mode is advantageous in preventing root hump formation due to the longer droplet transfer cycle and the smaller droplet diameter, which helps to slow down the increase of hydrostatic pressure [120]. Besides the prevention methods mentioned above, the most common alternative are using double sided single pass welding [121], [122] or novel multi-pass welding [123], [124] for full penetration and avoidance of root humps. If humping can no longer be avoided, post welding techniques such as grinding [125] or laser re-melting [110], [126], are also available to achieve the service requirements.



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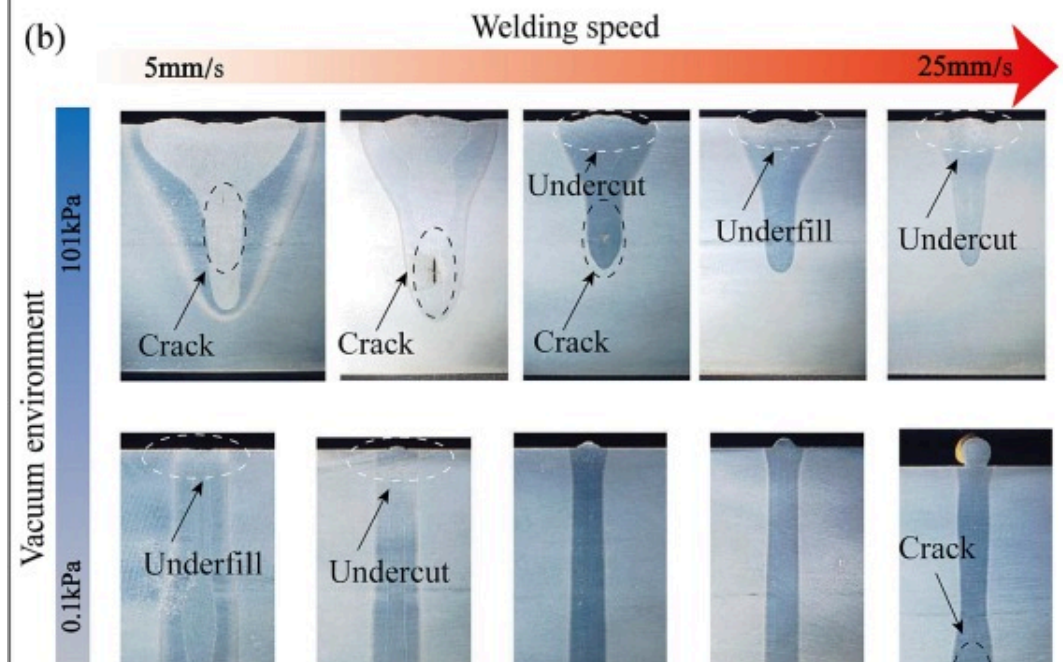
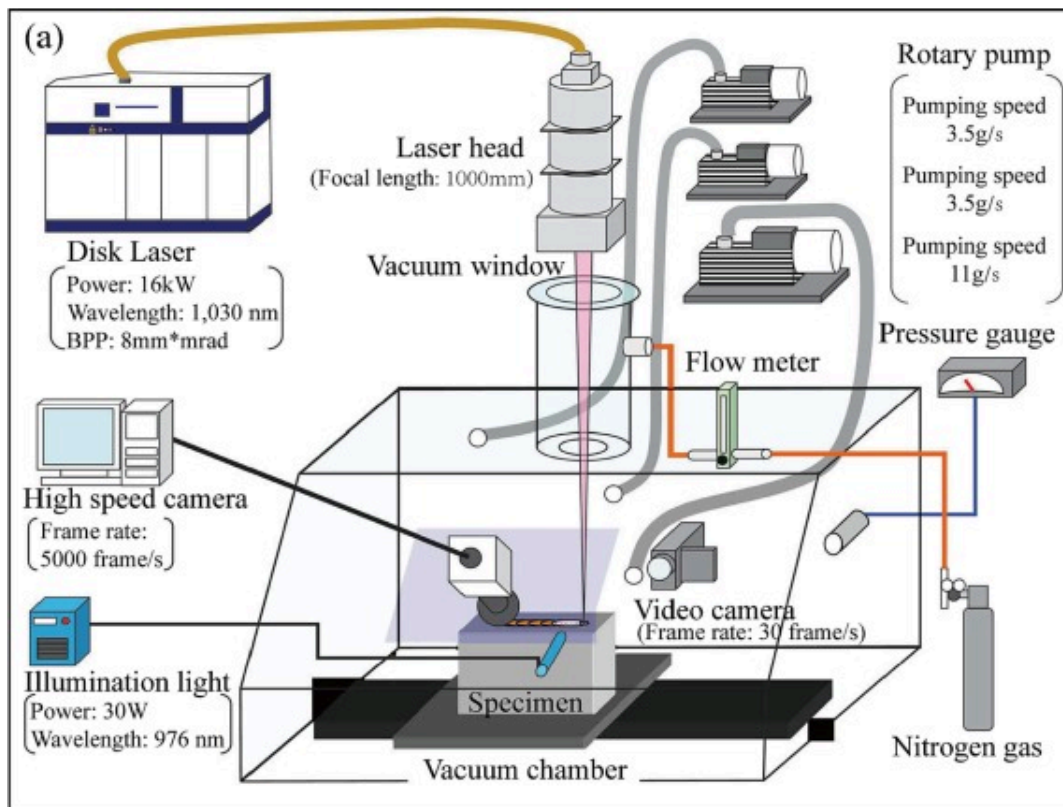
Fig. 8. Bottom surface appearance by using (a) a metal strip or powder [46], (b) ceramic backing and (c) submerged arc welding flux [43].

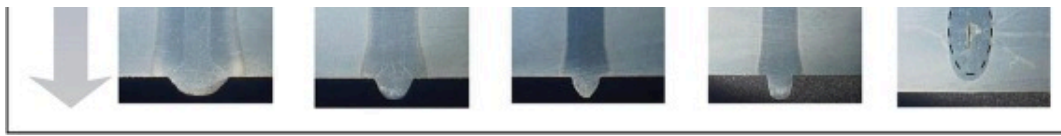


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Fig. 9. (a) Sketch of the electromagnetic weld pool support system, (b) cross sections of the welded joint using 13.9kW laser power at a welding speed of 0.5m/min and (c) macroscopic morphology of the welded joint using 16.5kW laser power at a welding speed of 1 m/min with MFA method [\[48\]](#).





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Fig. 10. (a) Schematic experimental setup under VEA, Weld cross sections at 16kW laser power and different welding speed at (b) atmospheric environment and 0.1 kPa low vacuum environment [45].

A general summary of the above suppress methods and the underlying physics can be provided. The stability criteria of the weld pool is given as follows:

$$p_h + p_{\text{down}} \leq p_\gamma + p_{\text{up}}$$

Where p_h is the hydrostatic pressure which can be estimated as ρgh , where ρ is the density of the melt, g is the gravity acceleration, and h is the thickness of thick-section steel.

p_{down} is the any additional potential downward pressure,

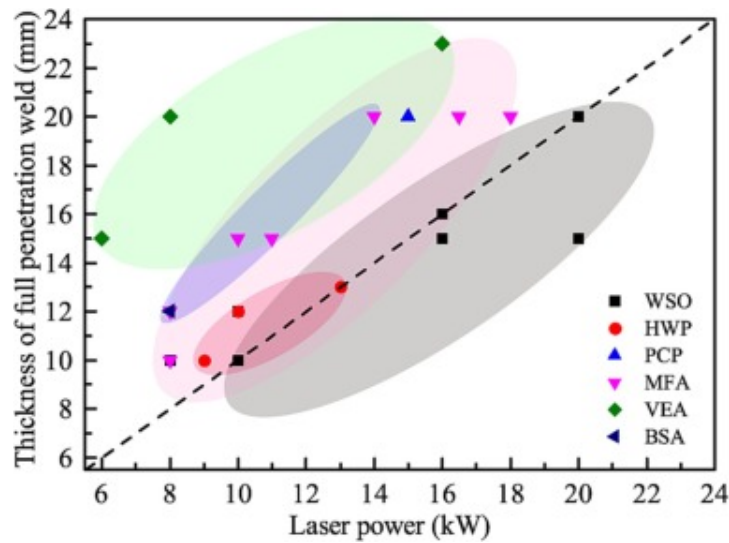
p_γ is the surface tension which can be estimated as γ/R , where γ is the surface tension coefficient and R is the curvature radius of of the weld pool root surface.

p_{up} is added upward pressure.

From the equation. decreasing p_h or p_{down} facilitates the equilibrium, and the same effect can be obtained by increasing p_γ or p_{up} . The VEA method can reduce the reaction force (p_{down}) of the plume eruption on the weld pool mainly by decreasing the number and density of the plume. It is thought that the reduction in keyhole diameter, melt pool width and volume, and the increase in surface tension (p_γ) due to the lower melt temperature also exert some positive effect. The BSA method is equivalent to adding a physical support force, while the MFA method adds a Lorentz force-induced

electromagnetic pressure. The HWP method achieves equilibrium by eliminating the contribution of gravitational acceleration to the hydrostatic pressure p_h of the weld pool. Now, we can broaden our ideas under that summary, for instance, introducing Bernoulli effect to increase the additional pressure above and below the weld pool due to the formation of pressure difference. A possible specific implementation is to add a high velocity airflow above the weld pool and load a low temperature airflow simultaneously at the back of the weld pool.

A schematic overview presented in Fig. 11 demonstrates that full penetration welds are achieved using different methods at varying laser power levels. According to the results, Without supporting optimization (WSO) method are primarily located in the diagonal and lower-right regions, while the results of the other five method are located in the upper-left region. This indicates that the latter methods can achieve deeper full penetration depth welds can be obtained with lower laser power in magnitude. The VEA method has a distinct advantage on penetration efficiency, followed by the MFA method. In terms of engineering applications, the MFA and VEA method currently requires complex supporting equipment such as movable magnetic device or mobile low vacuum chamber, while the BSA method requires tedious pre-welding preparation. As a result, the economic feasibility of these methods is limited. New ideas and approaches may expand the application of UHPLW technology for thick-section steel.



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Fig. 11. Variation of thickness of full penetration welds with respect to laser power by WSO [9], [19], [20], [40], [127], HWP [17], [20], [42], PCP [43], MFA [44], [48], [128], VEA [40], [45], [88], [89] and BSA [108] in the literatures.

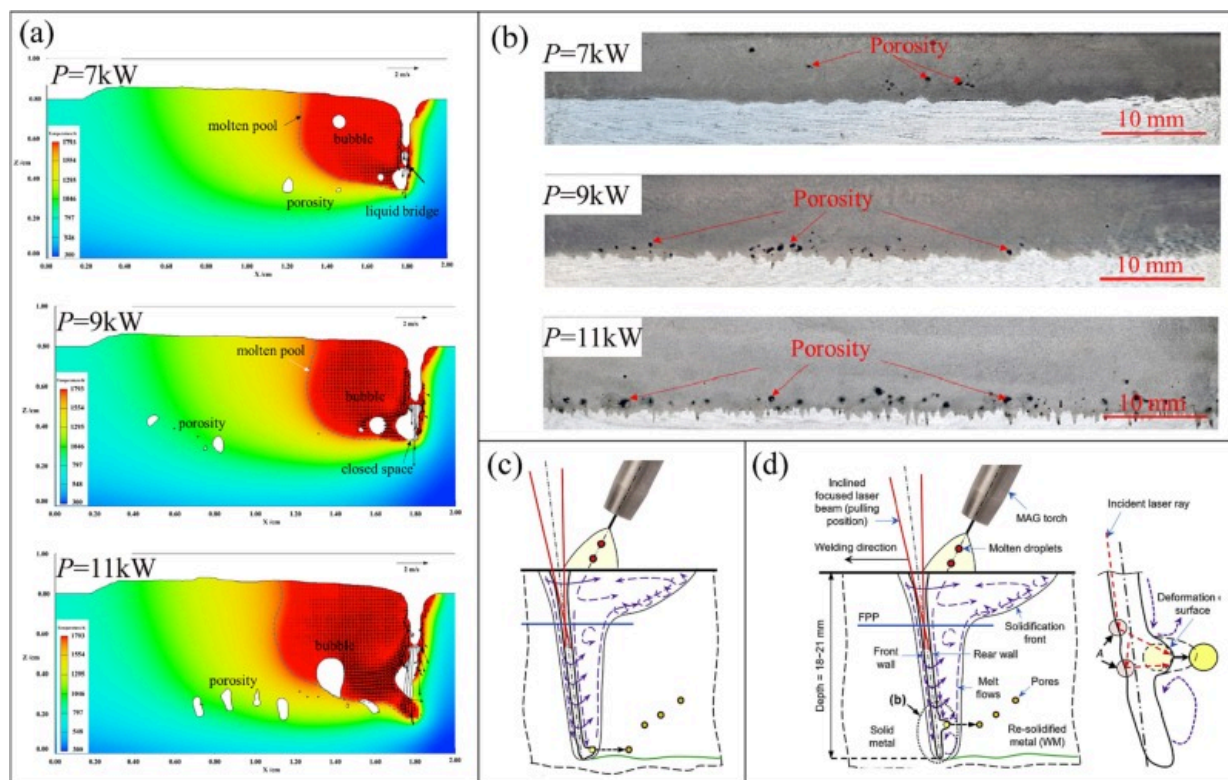
3.1.3. Porosity

Porosity defects are occasionally found in laser welded beads of thick-section steel. Porosity can be categorized as the metallurgical factor-induced type and keyhole-induced type. The metallurgical factor-induced porosity is generated due to low boiling point elements, shielding gas or surface contaminations [129]. Compared to metallurgical factor-induced porosity, Keyhole-induced porosity is a more significant issue [130], particularly in partial penetration welds. The presence of porosity could deteriorate the final mechanical properties of the weldments.

To get a deep insight on the porosity formation mechanism, in-process observations and post-process inspections are conducted to investigate keyhole and molten pool behavior

during laser welding. The in-process observations are performed by HSC [131], X-ray transmission imaging system [81], [132], [133], [134] with tracer particles [11], spectrometer and ultrahigh-speed synchrotron x-ray imaging system [135]. The post-process inspections are performed to define the relationship between porosity defect and welding parameters by means of metallographic sectional analysis [136], X-ray radiography and 3D X-ray tomography [130], [137].

The process of porosity formation during UHPLW is generally described as follows: with the laser power rapidly increases and exceeds a threshold, a cavity is drilled deeper and narrower, termed as a keyhole. The keyhole formation process is attributed to the strong recoil pressure from violent evaporation pushing the surrounding liquid metal downward [143], [144]. However, the keyhole wall is not stable and constantly fluctuate and collapse, in a non-equilibrium process that depends on the complex competition among thermal capillary force, recoil pressure, Marangoni convection, and other factors [132]. When the keyhole collapses, the bubbles generate, and they can easily become trapped by the longer solidification front due to the large escape distance, resulting in serious weld porosity defects. Kawahito et al. [11] found that porosity were not only at the tip of the keyhole but also at the middle in UHPLW. Wu et al. [145] and Bunaziv et al. [146] further subdivided the keyhole instability induced porosity into two formation mechanisms as shown in Fig. 12, one is the whole keyhole collapse between RKW and FKW in the bottom, the other is the collapse of the RKW. The explanation for the latter mechanism is that the increase absorption of laser energy reflected by the bulge at front keyhole wall causes the large depression.



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Fig. 12. The mechanism of porosity was identified by (a) numerical simulation and (b) post-weld observation with different laser power on CCS-B steels [141]; (c) and (d) schematic illustration about two formation mechanisms of keyhole induced porosity [146].

Computational fluid dynamics (CFD) models are widely acknowledged for their ability to predict the detailed thermo-fluid behavior of the melt pool and track the solidification process, making them useful for predicting porosity evolution. Table 2 lists some of the most important CFD works on keyhole-induced porosity formation. These models mainly focus on simulating the gas-liquid free interface and the laser heat source. The most commonly used methods for handling gas-liquid interfaces in numerical simulations of

the keyhole and molten pool are the Volume of Fluid (VOF) and Level Set methods. Laser source models can be divided into two categories: those that adopt a certain energy distribution combination of heat sources and those that use an advanced laser-material interaction model employing ray-tracing methods to consider multiple reflections of the laser within the keyhole and the Fresnel absorption effect. However, current multi-physics models have several limitations. For example, the majority of these models use the adiabatic bubble model [141], [142], [145], which neglects the effects of metal vapor and shielding gas on the bubble generation [147] and gas component, the heat transfer between the bubble and molten pool which influences the shape of bubble. The second major issue is the high computational expense associated with these models. As previously mentioned, the computational domains for these simulations typically involve meshes with a 0.1 mm-level. Thus, to simulate μm -level bubbles evolution, the mesh size must be even finer. Furthermore, as the melt pool depth of the UHPLW processes of thick-section steel increases from 10mm to 20mm or deeper, it will lead to an exponential increase in the computational workload, making simulations more challenging.

Table 2. An overview of some of the major CFD simulations on porosity formation.

Topic and highlights	Authors	Method	Software	Reference
2D mathematical modeling of LBW to study the mechanism of porosity suppression under MFA method	Zhou and Tsai 2007	VOF method +Ray-tracing method	In-house code	[138]
2D and 3D thermal-fluid modeling of LBW of 14mm 304 stainless steel to study the mechanism of porosity suppression	Zhao et al. 2011 Zhao et al. 2016	VOF method + Rotary Gauss heat source model	ANSYS Fluent	[139], [140]
3D thermal-fluid modeling of LBW of 12mm carbon steel to study the pore generation mechanism	Cho et al. 2012	VOF method +Ray-tracing method	Flow-3D	[142]

Topic and highlights	Authors	Method	Software	Reference
3D thermal-fluid modeling of LBW to study the sensitivity of porosity to process parameters	Lu et al. 2015 Sun et al. 2022	VOF method + Rotary Gauss heat source model, Ray- tracing method	Flow-3D	[141] , [148]
3D thermal-fluid modeling of LBW to show keyhole oscillations and the formation of porosity	Courtois et al. 2014	Level Set method +Ray-tracing method	COMSOL Multiphysics	[149]
3D thermal-fluid modeling of LBW of 304 stainless steel+study the molten pool behavior under VEA method+study the dynamics of vapor plume on the formation of the porosity	Pang et al. 2015 Pang et al. 2015 Pang et al. 2016	Level Set method +Ray-tracing method	In-house code	[117] , [147] , [150]
3D thermal-fluid modeling of LBW of 10mm Q345 steel+study two mechanisms of keyhole-induced bubble formation	Wu et al. 2018	VOF method +Ray-tracing method	Flow-3D	[145]
3D thermal-fluid modeling of LBW of 15mm Q690 steel+study the mechanisms of porosity suppression under SLIM method	Lin et al. 2021	VOF method + Rotary Gauss heat source model	ANSYS Fluent	[151]
3D thermal-fluid modeling of LBW of 10mm 304 stainless steel+study the mechanisms of porosity suppression under MFA method	Meng et al. 2022	VOF method +Ray-tracing method	ANSYS Fluent	[152]

Topic and highlights	Authors	Method	Software	Reference
3D thermal-fluid modeling of LBW of 15mm Q690 steel+study the mechanisms of porosity suppression under VEA method	Jiang et al. 2020	Ray-tracing method	In-house code	[153]

The characteristics and distribution of trapped porosity can be studied to help validate the generation mechanism. Porosity were mostly of spherical or ellipsoid geometry in steel joints [\[131\]](#), [\[136\]](#), [\[146\]](#). It was reported that three types of porosity (uniform, transitional, and root porosity) can be formed in the welds prepared by laser welding. Throughout the history of porosity in the thick-section steel welding process, it is inseparable from the influence of the surrounding environment, the characteristics of the laser beam and the introduced energy field.

In order to suppress the keyhole-induced porosity formation and improve the weld quality, various effective measures have been proposed. These include changing the surrounding environment by using a side gas [\[154\]](#), N₂ shielding gas [\[137\]](#), [\[155\]](#), adding a small amount oxygen (O₂, CO or Cr₂O₃) [\[129\]](#), [\[156\]](#), [\[157\]](#), or welding under VEA [\[89\]](#). Among these, VEA laser welding is particularly attractive as it results in a more moderate melt pool flow, a more stable keyhole, and reduced generation of bubbles. Additionally, behind the keyhole bottom of the melt pool, there is no obvious vortex movement, which facilitates bottom bubble up and reduces porosity defects finally. Other measures to reduce porosity formation with respect to SLIM include laser beam inclination [\[11\]](#), [\[158\]](#), laser beam pulse [\[132\]](#), [\[139\]](#), laser beam oscillation [\[63\]](#), [\[159\]](#), LBS and dual laser beam [\[160\]](#), [\[161\]](#). Compared to single LBW, dual LBW process is more stable due to the increased surface area of the molten pool and the reduced surface tension effect. This leads to a larger keyhole and consequently causes degassing effect. In terms of other energy field assistance, MFA can help stabilize the keyhole, facilitate bubble escape, and reduce solidification rates, thereby reducing porosity [\[138\]](#), [\[152\]](#), [\[162\]](#). However, there is

still controversy regarding the mechanism of accelerated bubble floating to reduce porosity. Whether it is due to the decreased viscosity caused by the increased temperature from the Lorentz force impeding melt flow, or the electromagnetic expulsive force caused by the Lorentz force promoting floating still requires further investigation.

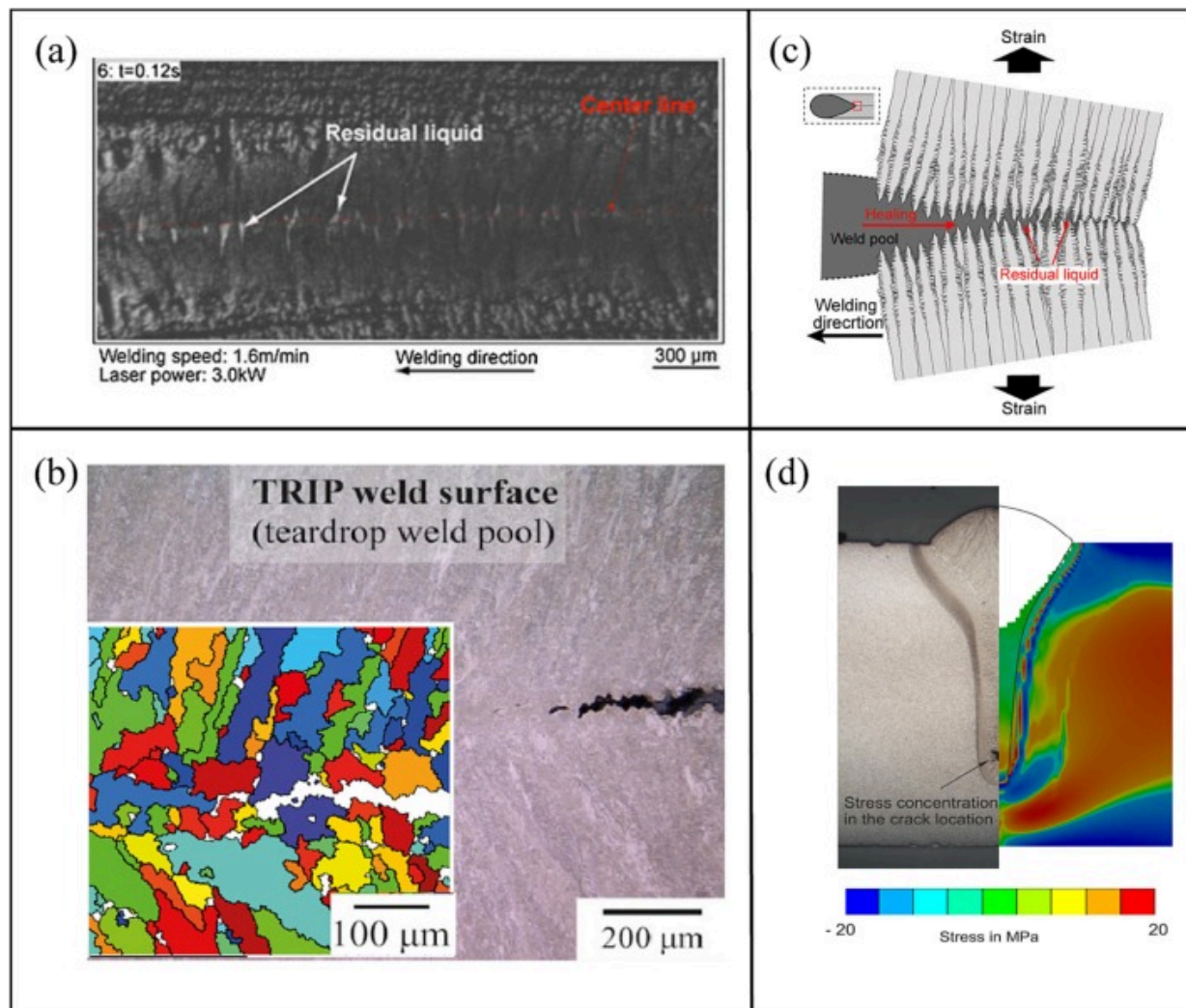
During HLAW process, adding an arc in a suitable manner can enlarge the upper part of the molten pool, which is conducive to the escape of bubbles and alleviates porosity [163]. At the same time, increased energy input can reduce the solidification rate and decrease the tendency for porosity formation. A trailing arc arrangement is recommended over a leading arc [27], [58], [130]. If the laser-arc distance is too short, molten droplets in the arc-leading mode are more likely to impinge on the keyhole opening, increasing the risk of keyhole collapse and porosity formation. Furthermore, WFLW assisted with a hot wire is an effective approach for avoiding the pore formation due to a stable molten pool [129].

3.1.4. Solidification crack

In deep penetration laser welding of thick-section steel, solidification cracking, as a type of hot cracking, is a common phenomenon in the weld zone. The crack typically occurred in the centerline of laterally symmetric welds due to high depth-to-width ratio. Thus, the quality and performance of the weld joint is unacceptable [12], [35], [164].

Under the UHPLW process of thick-section steel, that impurity elements such as phosphorus (P) and sulfur (S) can concentrate at the grain boundaries to produce a low-melting eutectics liquid film due to the extremely rapid solidification rate. As shown in Fig. 13, the liquid film will be broken under the action of tensile stresses caused by solidification shrinkage and thermal contraction, thus forming solidification cracks. In general, the two basic factors, namely critical stress/strain and solidification microstructure characteristics are direct responsible for solidification cracking [165]. Geometrical characteristic of the weld pool can play a crucial role in the solidification cracking formation. The relationship between the weld pool shape and the basic factors

for solidification cracks was further confirmed. On the weld pool surface, a teardrop shape have higher susceptibility to central solidification cracks than elliptical weld pools. Because the solid–liquid interface in the former has a slower growth rate and results in a longer duration of the liquid films [166]. On the weld pool interface, the bulge effect increases the solidification cracking susceptibility by enhancing the basic factors in the form of maximum tensile stress and the accumulation of sulfur in the bulge region [71], [167]. Currently, observation and simulation of the melt pool shape provide insights into the mechanism of solidification cracks formation induced by UHPLW. However, while the melt pool shape is a valuable intermediate parameter, it cannot fully establish the relationship between heat input and solidification cracking. Thus, a holistic modeling approach that incorporates multiple parameters is necessary to gain a more nuanced understanding of solidification cracking in UHPLW.



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Fig. 13. The mechanism of solidification cracking was identified by (a) HSC observation [172] (b) scanning electron microscope (SEM) and electron backscatter diffraction (EBSD) observation [166], (c) schematic illustration [172] and (d) finite element (FE) method simulation result [173].

The modeling of solidification cracking is very challenging because of the dynamic processes covering a wide range of physical phenomena at different time scales and length scales [168]. On a macro-scale level, solidification cracking criteria based on critical strain or strain rate are proposed. FE models are thus widely used for indirect simulations [165]. On a micro-scale level, the criteria based on the competition between the separation rate of the adjacent growing grains and the supply rates of the weld fluid [169]. A phase field (PF) model, coupling a thermodynamic and kinematic data, can be used for direct simulation. However, on a multi-scale level, which are able to bridge the length scales and can comprehensively explain the occurrence of hot cracking, are rarely reported [170], [171].

Controlling the chemical composition of the weld, adjusting the solidification structure, or using appropriate welding processes can inhibit the generation of solidification cracks. The chemical composition of the weld can be modified to reduce crack susceptibility by selecting the appropriate filler wire. A filler wire with rich primary alloying elements is normally used to provide more eutectics for crack healing during the final stage of solidification of the weld pool. Alternatively, certain elements can be added into the filler wire for fine as modifiers to promote the formation of fine equiaxed grains by creating more nucleation sites. For thick-section steel, the Manganese (Mn) content in the weld metal is a critical factor in preventing solidification cracks. High levels of Mn and sulfur (S) lead to the formation of MnS, which helps to avoid the formation of low-melting eutectics such as FeS. The welding parameter adjustment is usually considered to prevent solidification cracking in laser welding of steels. Before the welding process begins, avoidance of restraint, rigid or compressive clamping [174], [175] or preheat treatment help reduce strain to reduce cracks in welding. In the welding process, adding an additional heat source such as induction heating [176], in laser-leading HLAW [177] with lower welding speed [164], [166], [172], [173] can be used to suppress cracks through the same mechanism. Solidification structures with fine equiaxed grains are helpful to enhance interface bonding at grain boundaries, resulting in better cracking resistance, and consequently improve the strength, fracture toughness. Lateral oscillation

of the laser beam at a certain frequency seems to refine the grains and suppress the formation of solidification cracks to a significant extent [178]. If other energy field is introduced, weaving the arc or adding the magnet to change the crystallographic direction of columnar crystals can also be well reduced solidification cracks.

3.2. Microstructures and mechanical properties

In UHPLW of thick-section steel, thermal history and filler wire addition (in WFLW or HLAW) have a major effect on the microstructures of the fusion zone (FZ) and the heat affected zone (HAZ). In this subsection, the problem of insufficient filler wire distribution in the FZ and the resulting inhomogeneous microstructure and mechanical properties are described. Meanwhile, distinct differences are found between the base metal (BM), the FZ and the HAZ, such as in chemical composition, microstructure and mechanical properties.

3.2.1. Filler wire distribution in FZ

The filler wire distribution in HLAW and WFLW of thick-section steel is recognized as a potential problem which has not yet received adequate attention. The steep shape of the molten pool and the high solidification rate can result in insufficient transfer of the filler wire to the root of the weld, leading to an inhomogeneous microstructure across the thickness of the joint [179]. In low-alloyed steels, the insufficient filler wire transportation results in a lack of acicular ferrite (AF) which is essential for improving strength and toughness [180]. Additionally, it can lead to the formation of harder microstructures such as bainite and lath martensite, which may lower the mechanical properties [123], [146], [181].

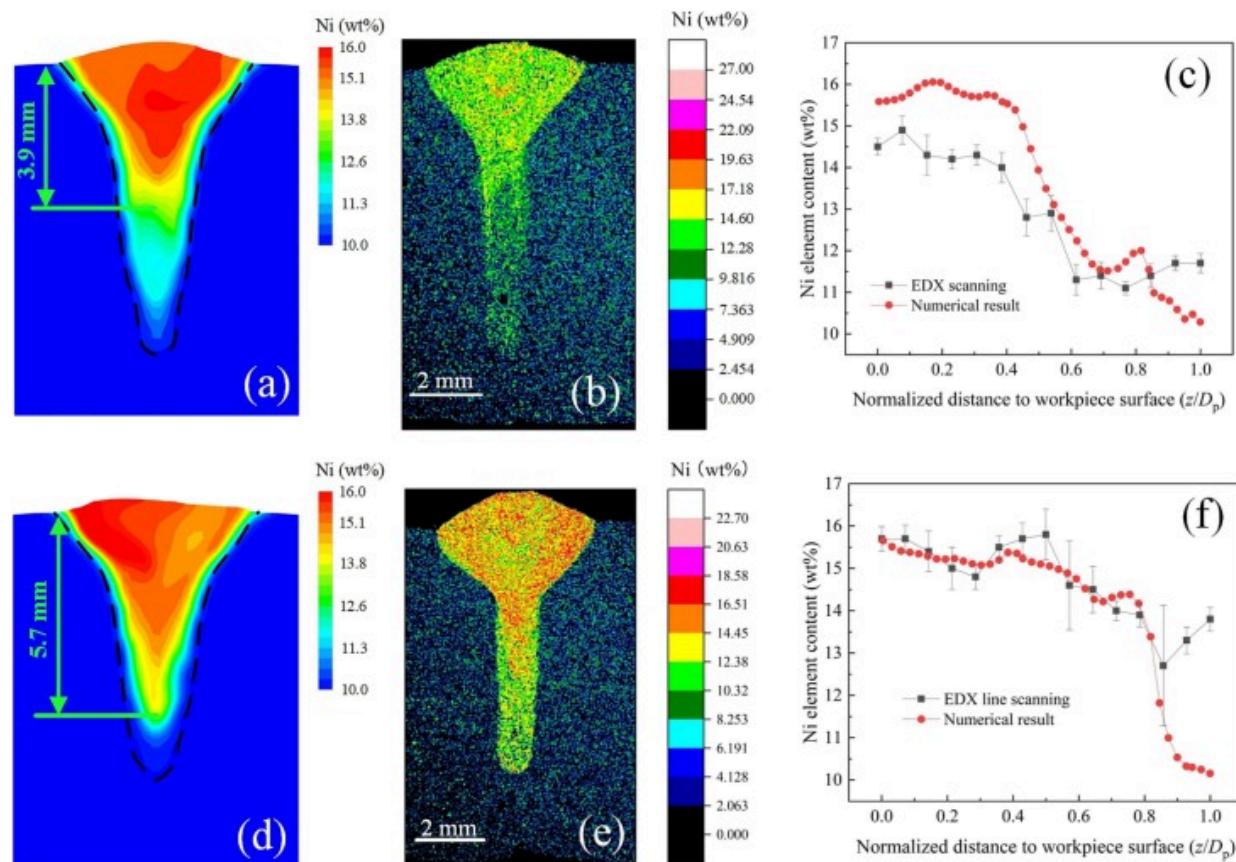
Accurate identification of filler wire distribution is a prepared work. Quantitative Analysis of filler metal distribution for steels can be done by measuring the Cr [182], [183], Ni contents [184], [185] by contrast discoloration after etched [94]; nanoparticles, non-metallic inclusions (NIMs) [146] or the nucleated AF [65], [186] by SEM with EBSD. With the help of the above method, progress was made on quantitative experimental characterization of filler wire distribution of post-weld joints. Zhao et al. [184] showed

that the wire mixing was dependent on the melt flow direction, the oxygen content in the melt and the laser-arc configuration. Karhu et al. [182] and Gook et al. [123] confirmed that the insufficient filler wire deposition rate in the roots and the difficulty in achieving homogeneous welded joint in LAHW and WFLW.

Compared to post-weld experimental characterization, capturing spatiotemporal information on in-process melt pool flow and heat transfer to study filler wire distribution is more beneficial for fine homogenous regulation. However, experimental observation is still an extremely difficult task due to the intensive light of the plasma plume, the small size of the weld pool and the non-transparent nature of the melt. Multi-physics simulations have been conducted to study the physical behavior and the filler wire transfer mechanism. Cho et al. [187] developed a 3D multi-physics model for the analysis of alloying element distributions in HLAW. The reasonable predictions can be made based on the comparison of fusion zone shape and alloying element distribution between the numerical calculation and the experimental observation. The distribution of filler wire elements in the vertical direction is directly related to the melt downward flow. The melt flow direction depends on the competition of the vapor pressure, the melt pool gravity, the frictional force of the vapor gas [188], Marangoni convection and external additional forces.

Achieving homogeneous distribution in UHPLW requires adequate supply as the first priority. Suder et al. [183] proposed that sufficient wire feeding speed is necessary to achieve fully homogenous welds. An appropriate increase of air gap can enhance the supply of the filler wire providing the condition for homogeneous weldments and enough laser power density [182]. Once the premise of sufficient supply is met, the goal of homogeneous distribution can be further achieved through enhancing the melt downward flow through processes such as Marangoni convection, electromagnetic force, etc. Zhao et al. [184] appropriately added active elements such as oxygen to change the Marangoni flow direction to achieve uniform distribution. Electromagnetic force induced by the arc can lead to a significant downward flow, which can bring the filler wire element to the root of the weld pool [59], [189], [190]. Advanced arc welding technologies

such as modified pulsed arc or CMT could promote deeper penetration of the filler material into the narrow gap [94], [123]. Bachmann et al. [48], [191], [192] confirmed that MFA helps the additional element transfer to the root of the molten pool under the downward Lorentz force, enhancing the mixing level in LBW and HLAW. As shown in Fig. 14, the added nickel from the filler wire is distributed more homogeneously in the molten pool due to the downward and forward flow in the WFLW process with the MFA method [185]. Additionally, the MFA method can stir the weld pool to further guarantee a homogeneous microstructure. The homogeneity of the weld joint through the depth can also significantly improved by pre-weld groove preparation using PCP [43], [186].



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Fig. 14. Comparison of Ni distribution in the cross-section from WFLW (a-c) and MFA-WFLW (d-f) through 3D CFD simulation and X-ray fluorescence spectrometer measurement [185].

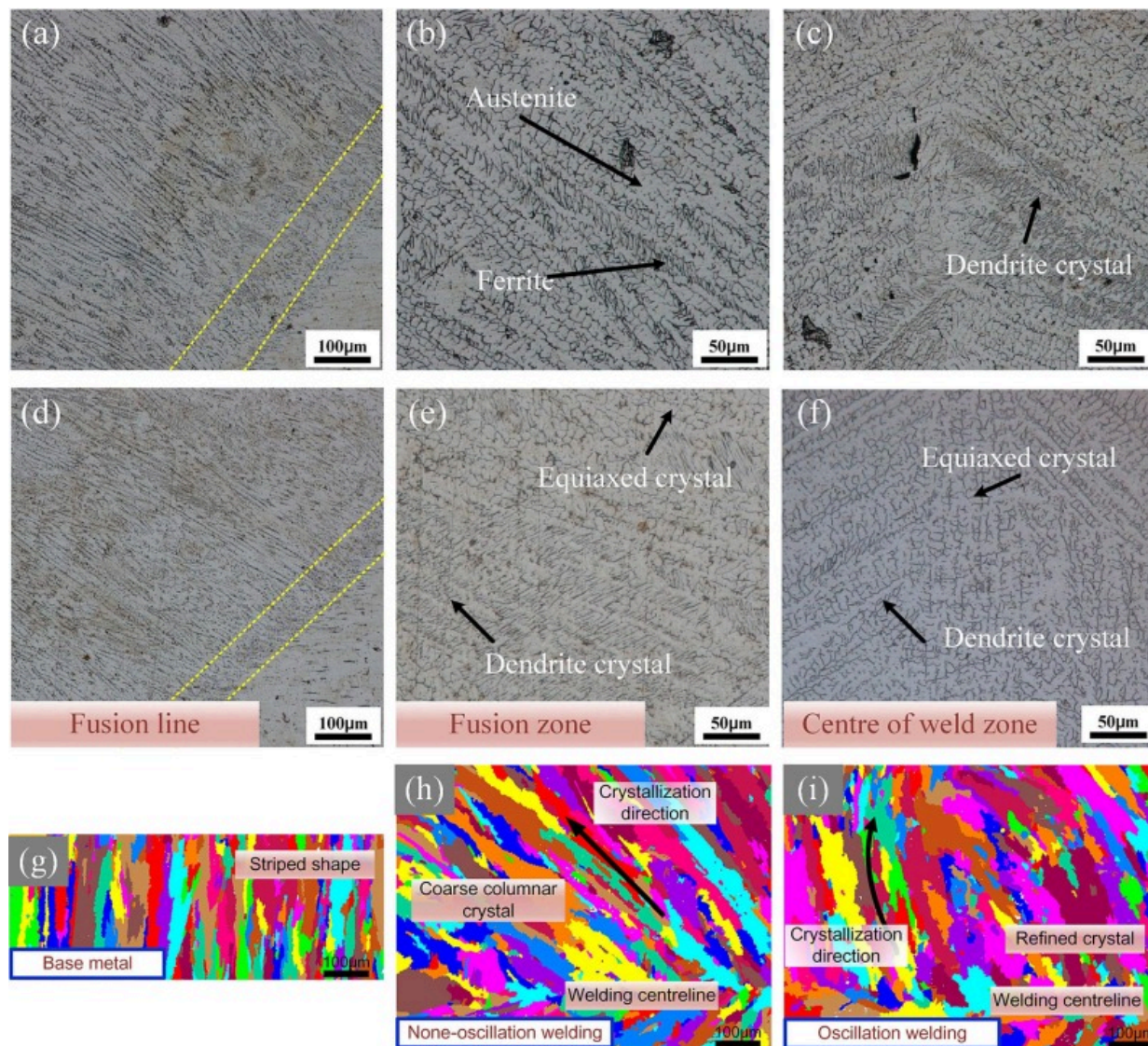
3.2.2. Microstructures of FZ and HAZ

The weld microstructure can significantly affect the mechanical properties. If the grains of the columnar grains all point towards the center of the weld, the ductility of the weld metal will be significantly reduced. In most cases, more and finer equiaxed crystals and less columnar crystals in welded joint are desired.

In LBW, the HAZ surrounding the FZ is narrow and negligible to mechanical properties [193]. Two kinds of grains can be observed in the FZ, namely equiaxed grains and columnar grains. Equiaxed grains including fine and dendritic ones are located close to the fusion boundary and in the center, respectively [193], [194]. In HLAW, the microstructures in FZ and HAZ vary with the thickness direction of thick-section steel due to the peak temperature, the cooling rate and composition at different depth. In general, the microstructure of the FZ generated by the lower part, or laser zone, is composed of fine equiaxed grains in weld center and columnar grains around the equiaxed grains. While the upper part, or arc zone, mainly consisted of coarse columnar grains and equiaxed grains between the columnar grains. Compared to arc zone, laser zone has finer grains content in the FZ and narrower HAZ. Furthermore, the microstructure of the HAZ demonstrated restrain to the grain growth, the microstructure of the arc welding process may develop coarse grain in FZ [195]. The grains growing direction is in consistent with direction of the maximum temperature gradient, which is favorable for grain growth and getting coarse columnar grains. When the molten pool metal begins to crystallize, grains always grow from the base material adjacent to the fusion line to the weld, and a part of grains can grow to the weld centerline, which is also called the parting line. Due to deep penetration in thick-section steel, the parting appears from two lines in the upper of the FZ to one line in the middle or lower of the FZ [196]. However, there is limited knowledge

of how microstructure may vary through the thickness (up to 20mm) following UHPLW of thick-section steel, which seems to have been studied only in EBW [197], [198].

Controlling the grain structure of the weld metal to produce fine equiaxed grains helps to reduce the susceptibility of the weld metal to solidification cracking during welding and can also improve the mechanical properties of the weld. Grain refinement has been achieved by SLIM such as laser beam oscillation, MFA such as magnetic stirring, nucleation agent addition or ultrasonic vibration assistance. Chen et al. [199] employed HLAW with laser beam oscillation in D406A ultra-high strength steel, found the laser beam oscillation on the one hand causes the decrease of temperature gradient which is favorable to the formation of equiaxed grains in the center of FZ, on the other hand destroys the formation of columnar dendrites and promotes microstructure homogenization. Further, increasing beam oscillation frequency may disturb weld pool solidification more severely and cause more significant grain refinement as shown in Fig. 15 [200]. The above-mentioned oscillation technique based on the galvanometer scanners can not be carried out on UHPLW of thick-section steels to realize grain refinement due to laser power limitations, while the other mechanical wobbling head technique has not been thoroughly investigated yet [201]. Chen et al. [202] used internal induced MFA and external MFA to refine grains and uniform element distribution of HLAW for 304 stainless steel. Qi et al. [203] added a suitable titanium (Ti) foil in HLAW to induce AF nucleation through the formation of a Mn depleted zone with the help of Ti_2O_3 , resulting in grain refinement. Kolubaev et al. [204] investigated the microstructural features in laser welded low carbon steel, found that ultrasonic vibration treatment facilitate the transformation of plate-like Widmanstätten ferrite into submicrosized equiaxed grains and provide a homogeneous microstructure.



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Fig. 15. Weld microstructure of the fusion line, the FZ and the center of weld zone (a)-(c) without laser beam oscillation show (d)-(f) with laser beam oscillation; (g) base metal,

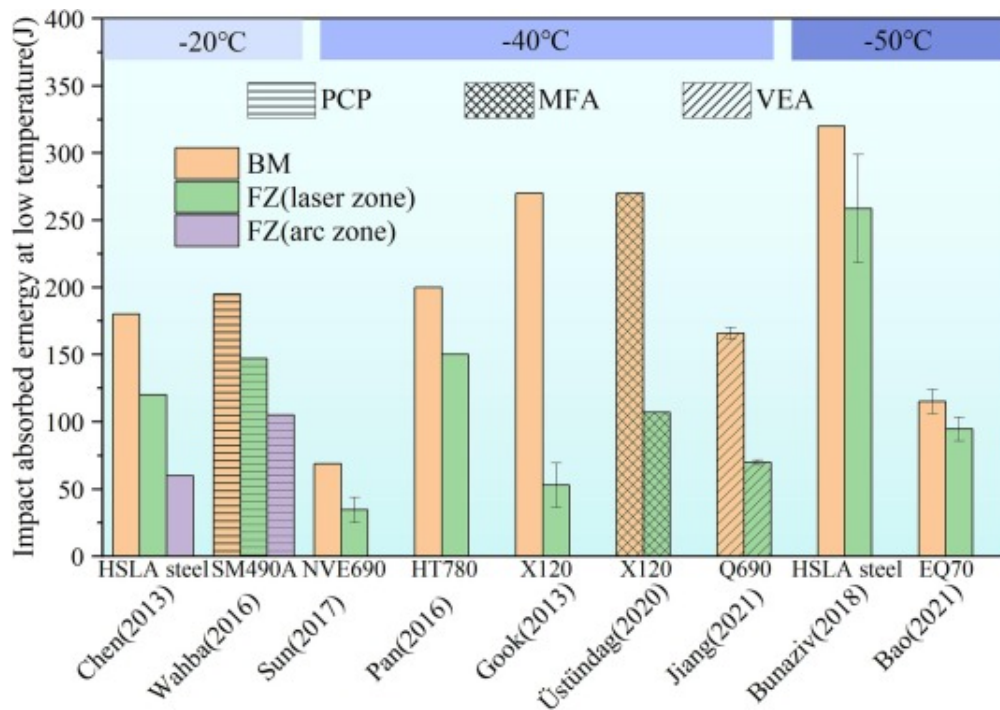
(h) coarse column grains without laser beam oscillation, (i) refined grains with laser beam oscillation at the amplitude of 0.2 mm and frequency of 100 Hz [200].

3.2.3. Mechanical properties of FZ and HAZ

Mechanical properties of welded joints is determined by critical factors such as strength, toughness, and fatigue properties. However, when exposed to harsh environment, impact toughness emerges as a crucial performance index that dictates reliability and safety. Despite notable advancements in laser welding technology aimed at improving joint toughness [202], [203], enhancing the low-temperature impact toughness of thick-section steel welded joints remains an ongoing challenge. Consequently, investigating the mechanisms responsible for the decline in impact toughness and exploring approaches to improve it have become significant research topics [108], [123], [205].

A schematic overview of low-temperature impact toughness of typical thick-section steels joints used in UHPLW processes is shown in Fig. 16. This figure intends to provide a broad overview of the results reported in the literature but does not include all classes/cases of steels. From the existing literature, the impact toughness of BM is generally higher than that in FZ. One of the most significant factors influencing joint toughness is the microstructure, which includes intrinsic properties and grain size. Understanding the relationship between microstructure and mechanical properties necessitates the use of appropriate characterization equipment to observe changes in microstructure constituents. Thus, investigating these results and establishing relationships between microstructure and mechanical properties are crucial aspects of comprehending the impact toughness of thick-section steel joints [66]. For instance, the microstructures of FZ are predominantly composed of lath martensite, which has a lower toughness than that of ferrite phase in BM [206]. According to the Hall-Petch relation, the square root of grain size is inversely proportional to the critical stress for crack propagation and proportional to the ductile–brittle transition temperature for the materials with the same microstructures [207]. This perspective of grain size can account for the higher impact toughness in the laser zone than in the arc zone, as depicted in Fig.

16. However, there has been insufficient systematic research on the difference in impact toughness between the laser zone and the arc zone. Although limited data from Fig. 16 suggest that the impact toughness in the laser zone is higher than that in the arc zone, the actual difference in heat input and added elements resulting from different welding processes leads to variations in microstructure type, grain size, and grain orientation, thereby causing different impact toughness results. Indeed, it is essential to take into account the deficiency in toughness in the laser zone, particularly in thick-section steel (with a thickness up to 20mm), as the critical factor of the parting line in the center of the weld in the laser zone can significantly reduce the impact toughness.



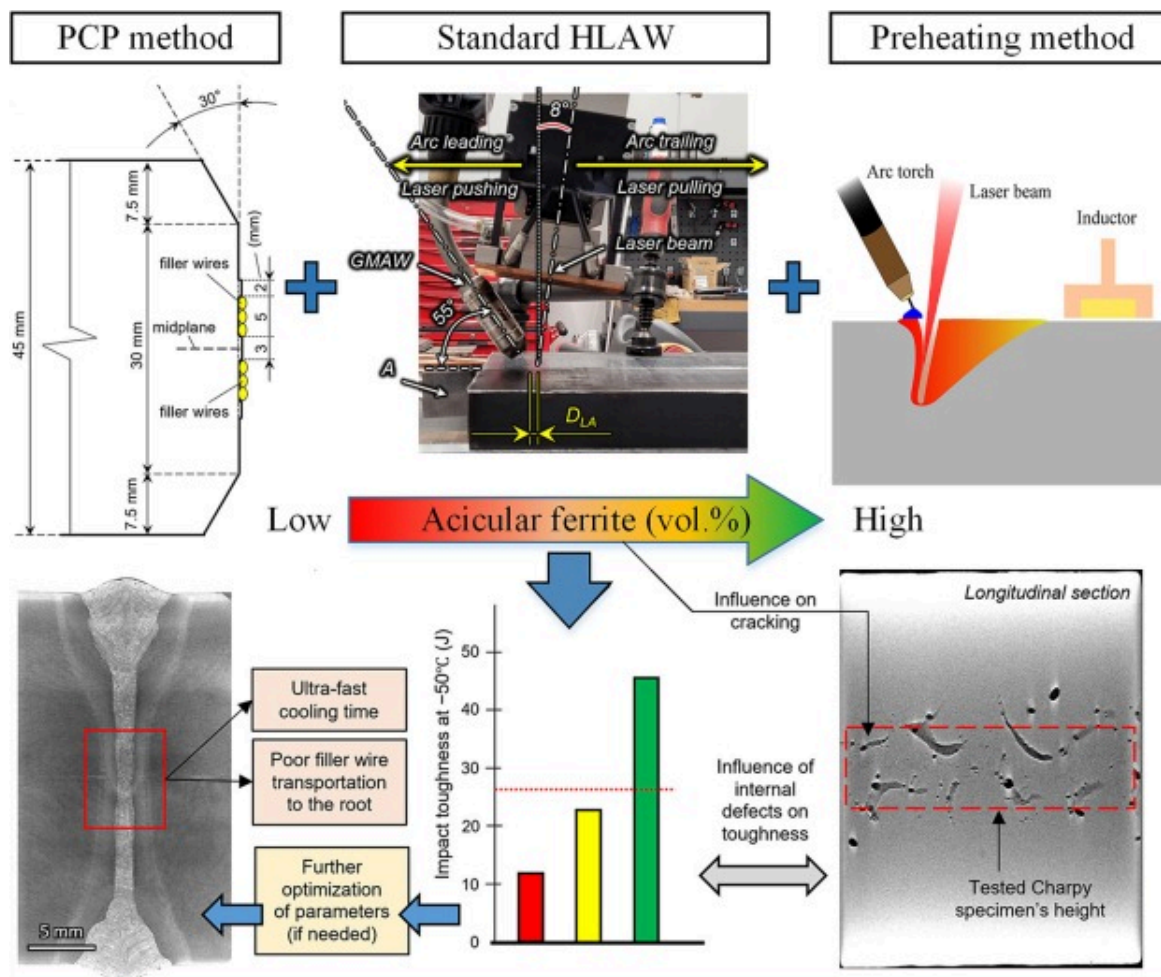
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Fig. 16. Results of Charpy V-notch impact absorption energy at low temperature [43], [65], [66], [67], [89], [95], [128], [186], [206].

In most cases, the amount of AF (the most desirable composition) is the main contributor to the toughness [208]. However, the welded joint of thick-section steel often exhibit inhomogeneous distribution of AF along the thickness direction [26], [65], [67], [206]. This is evidenced by the top zone which contains substantial amount of acicular ferrite (AF) with polygonal ferrite (PF) and some grain boundary ferrite (GBF); the middle part which includes bainite due to insufficient transfer of filler wire; and the root which consists of a mixture of bainite and lath martensite [186]. It is generally thought that the embrittlement of FZ in laser or laser-arc welded joints is attributed to the variation of AF fraction, while micro-phases or inclusions may also contribute to embrittlement. Additionally, the presence of massive Widmannstätten structures with the oriented morphology of the platelets in FZ [65] or HAZ is believed to play a significant role in joint embrittlement [209].

The improvement methods in thick-section steel laser welded joints are microstructure modulation. Increasing the nucleation of AF and promoting a uniform distribution of AF is one way to improve toughness. Controlling the titanium (Ti) content to increase the volume fraction of AF in laser-MAG hybrid welds by adding Ti foil was proven to improve impact toughness [203]. An increased air gap was found to enhance the supply of filler wire towards the root, providing high amounts of acicular ferrite formation to create softer and ductile microstructures [65]. Further, using the PCP method [43] in the air gap achieves a significant improvement in the average toughness of HLAW welded joints. However, this method does not always work as the weld plate thickness increases, thus the root cooling rate reaches a critical value and the bainite-martensite matrix nucleation is faster than AF nucleation the cut-wire provided for [186]. Applying local preheating or dual-beam technology can extend the root cooling rate and produce acceptable toughness [181]. The results in Fig. 17 show that significant improvement of impact toughness in the weld metal was achieved by combining both the PCP method and the preheating method. Alternatively, post-weld heat treatment (PWHT) can be applied, but it is not desired from a production and cost-effectiveness point.



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Fig. 17. Impact toughness values increase from 11.5 to 46J by combining both preplaced filler wire and preheating methods due to higher AF vol% [210].

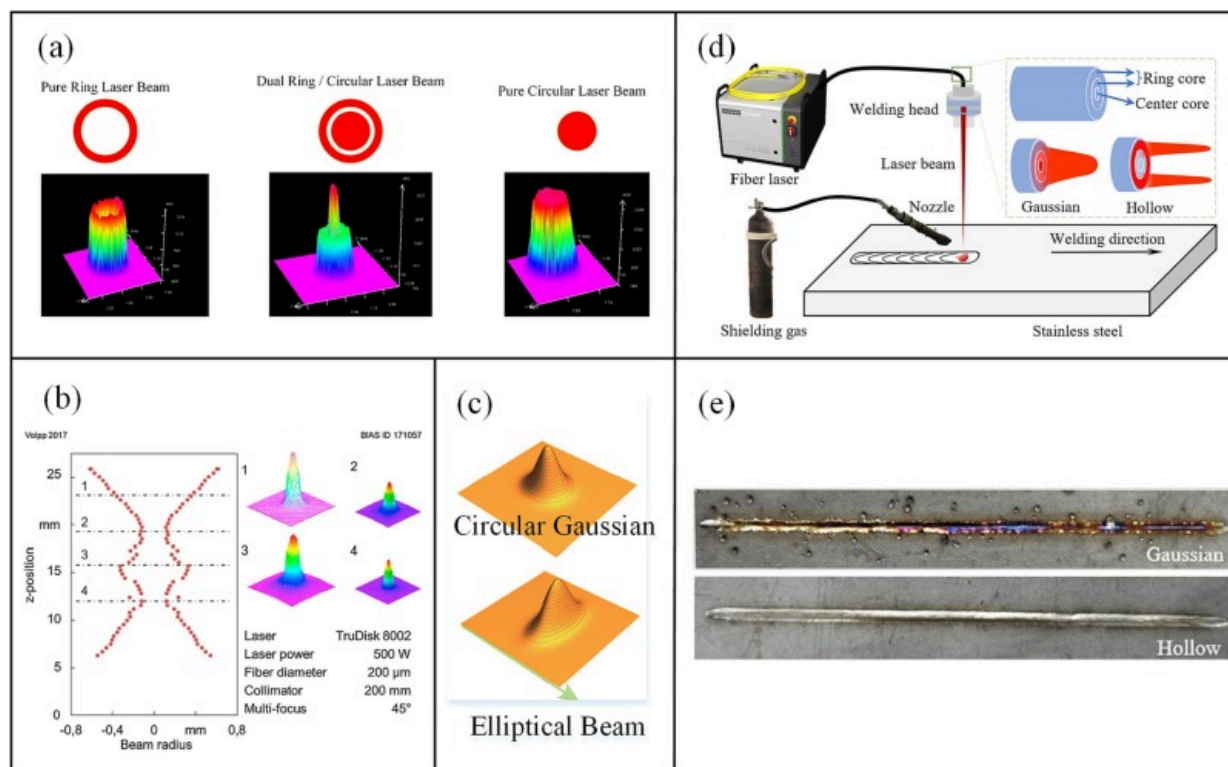
4. Future research perspectives

By far, significant progress has been made in UHPLW of thick-section steel. Nevertheless, certain bottleneck problems still need to be resolved, such as severe spattering, or

difficult-to-eliminate solidification cracks, or root hump suppression under various tedious measures, and unfavorable mechanical properties due to microstructure inhomogeneity. In order to achieve high-quality joints, future breakthroughs must focus on a deep understanding of the relationship between process, microstructure and mechanical performance. The roadmap for future research includes the exploration of advanced energy modulation processes, the explanation of mechanisms by integrated models, the improvement of defects by monitoring methods, and the evaluation and optimization of mechanical properties based on the microstructure. The future potential research points for UHPLW of thick-section steel should focus on four aspects: spatiotemporal laser intensity modulation, multi-scale integrated simulations, in-situ monitoring under interfering factors, and mechanical properties optimization through microstructure control.

4.1. Spatiotemporal laser intensity modulation

Laser beam intensity modulation, as introduced in [section 3.1.1](#), are well known to have a positive effect on process stability and can tackle the drawbacks of Gaussian mode, particularly for spatter and porosity defects. LBS such as ring mode [\[87\]](#), multi-focus mode [\[86\]](#), ellipse mode [\[211\]](#), and Bessel mode [\[212\]](#) are also very promising in terms of stability control. Wang et al. [\[87\]](#) investigated the effects of the ring mode laser beam on stabilizing the process and observed ring mode or adjustable-ring mode significantly reduce the spatter with respect to pure Gaussian mode. Volpp et al. [\[86\]](#) developed a new multi-focus beam shaping that can produce different energy distributions in the keyhole for spatters and porosity suppression, including a homogenous energy distribution. [Fig. 18](#) illustrates the application of these energy modulation strategies in UHPLW of steel with a thickness of 10mm or less.



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Fig. 18. Different laser profiles of (a) ring mode [87], (b) multi-focus mode [86] (c) elliptical mode [211] and (d) hollow mode [41]; (e) shows less spatters (laser power 3 kW, welding speed 50 mm/s) at hollow mode compared to Gaussian mode [41].

Despite its potential, this SLIM technology is rarely used to explore solutions for suppressing welding defects in thick-section steel. By developing a model under LBS and to understanding the law of the plume, molten pool, and keyhole behavior is the key to ensuring high-quality welding. The energy coupling mechanism and dynamic behavior of the plume, molten pool, and keyhole with LBS strategies in HLAW can be further exploited. The study of macroscopic morphology mechanisms under LBS strategies, together with the establishment of defect-free process windows when combined with

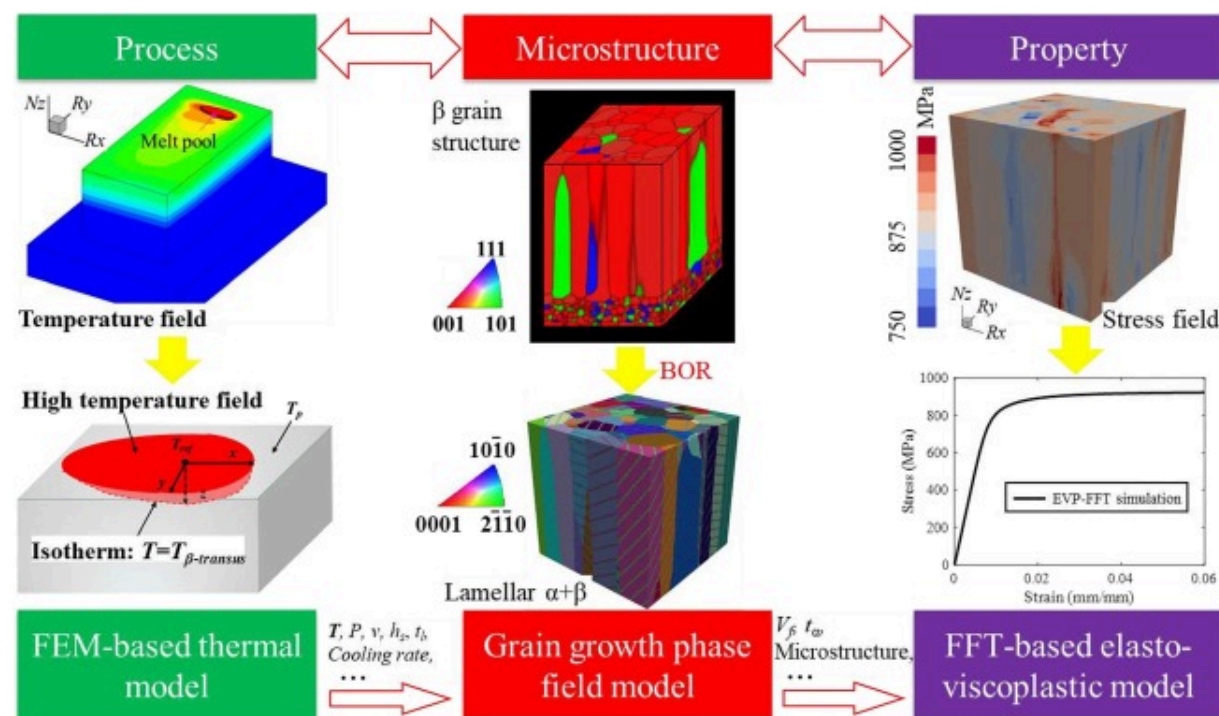
other typical defect suppression methods such as VEA and MFA, will be a fruitful area. The aforementioned measures to exploit the potential capabilities of LBS in UHPLW of thick-section steel require full and detailed explorations in both process trials and physical mechanisms.

4.2. Multi-scale integrated simulations

Despite the fact that numerical simulation techniques [213] have paved the way to efficiently predict the occurrence of welding defects and investigate the optimal process window for high-quality joints. However, these simulation techniques are often have the characteristic of using a single computational method for a single scale model. For instance, a macroscopic heat and mass transfer model [179], [214] or a microscopic solidification microstructure model, may suffer inaccuracies after complicated assumptions due to their inability to consider the interaction of microscopic factors with macroscopic components. The time and resources expense are enormous with the increasing model size of thick-section steel welding simulations, especially the size in the plate thickness direction. Hence, an efficient integrated simulation strategy of UHPLW process of thick-section steel is needed.

Currently, integrated coupling simulation is in its preliminary stage with most of the work concentrated in the field of additive manufacturing and limited in scope. The challenge lies in the integrated coupling of multiple numerical models and algorithms [215], [216], [217], [218]. Yan et al. [216] established an integrated process-structure-property multi-scale multi-physics field numerical simulation method, including CFD model for powder melting, cellular automaton (CA) model for grain growth and crystal plasticity (CP) model self-consistent clustering analysis for fatigue properties, to understand the correlation between process parameters and mechanical properties. Liu et al. [218] developed the integrated numerical simulation method by coupling a field FE model, a reduced-order PF model, and a CP fast Fourier transform (CPFFT) method for crystal plasticity, as illustrated in Fig. 19. The Existing coupled simulation algorithms can be used for thick-section steel laser welding simulation to qualitatively reveal the

monography and properties control mechanism. Nevertheless, the challenges of computational accuracy and efficiency still remains in the specific guidance of the welding process parameters optimization.



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Fig. 19. Schematic of the proposed computational framework to investigate the process-microstructure-property relationship [218].

In terms of computational accuracy, the current integrated numerical simulation method is limited to the sequential coupling method, and there is still much work to be done to develop bidirectional or full strong coupling methods. For instance, the study of the mechanism of solidification cracking formation in UHPLW of thick-section steel.

According to available experience, the research roadmap is as follows, a macro-scale finite element model was developed to calculate the strain localization in the mushy zone; then

the strain data were applied to the micro-scale model and solidification cracking sensitivity was analyzed. Thus, the occurrence of solidification cracking can be further comprehensively explained by multi-scale integrated models which are able to bridge thermal and strain field with microstructural evolution. How to realize the strong coupling of the microstructure prediction and multi-physics model to study the phase change, related residual stress and solidification cracks is an important issue to be addressed.

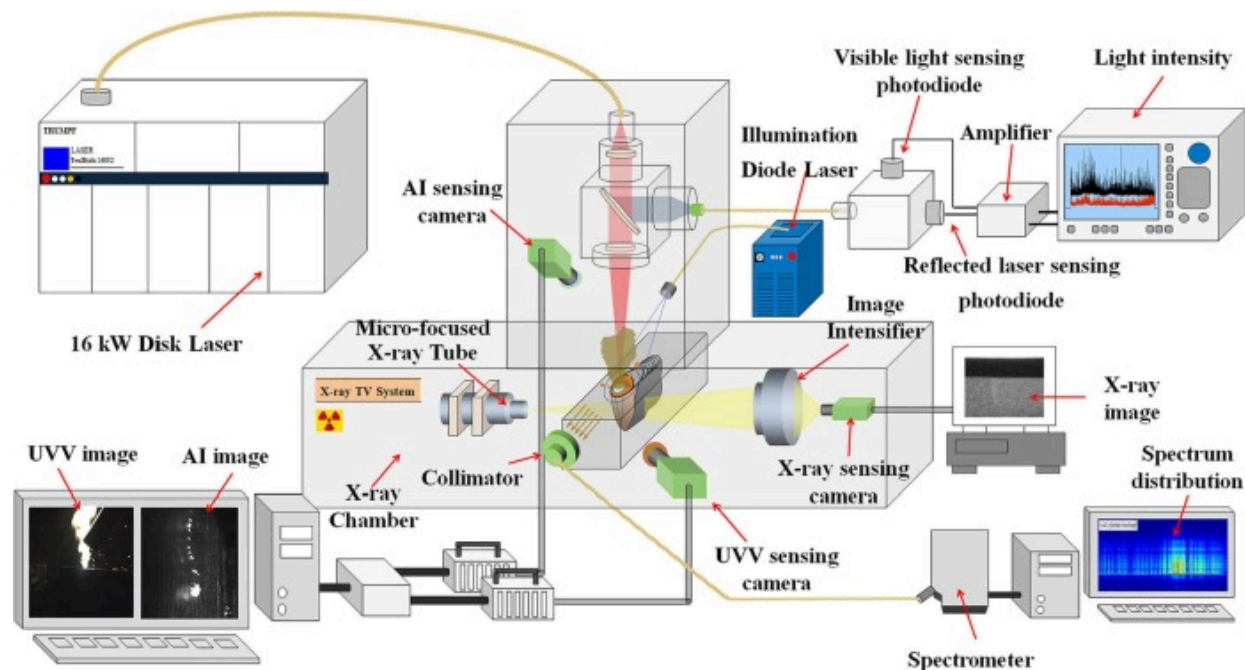
Furthermore, The computational efficiency can be improved by coupling efficient and attractive numerical models of heat and mass transfer, microstructure prediction, and mechanical property prediction models. The lattice Boltzmann method (LBM) method [219], [220], ALE3D-FE method [221], [222], and Smoothed Particles Hydrodynamics (SPH) [223] are examples of numerical models that can be used for this purpose. In particular, the LBM method has been widely used due to the advantages of simple description of fluid interactions, easy set-up of complex boundaries, and easy parallel computation for solving melting and solidification problems. It can also be coupled with macroscopic fluid models or microscopic PF models.

On the other hand, UHPLW problems have multidimensional, strongly nonlinear, massive data characteristics, making them naturally suitable for data-driven, artificial intelligence and other new calculation paradigms. Therefore, the development of a new paradigm of data-driven numerical simulation is an important research direction, in combination with the highly efficient and highly accurate real-time prediction models established by the high-precision coupled simulation algorithm described above.

4.3. In-situ monitoring under strong interfering factors

Today, laser welding in-situ monitoring has been extensively studied and has yielded significant advances in controlling the welding process and improving weld quality through real-time data. In the field of signal in-situ collecting, You et al. [224] developed a multiple monitoring system utilizing two photodiodes, a spectrometer, two visual

cameras, and an X-ray sensing system. This system as illustrated in [Fig. 20](#) could collect accurate and valid welding signals to analyze and model more comprehensively. In the field of signal processing and analysis, Zhang et al. [\[225\]](#) used principal component analysis to reduce data redundancy by analyzing the characteristics of the shadows of the molten pool which were defined and obtained in Ref. [\[226\]](#). The Genetic Algorithm improved Back Propagation Neural Network method was then established to model the relationship between the appearance of the weld seam and the obtained characteristics. In the field of feedback control of welding parameters, Mrna et al. [\[227\]](#) proposed and implemented a feedback control method to adjust the laser welding process which can quickly optimize and control the focus of the laser beam. However, there is a significant knowledge gap in research on the monitoring of UHPLW of thick-section steel. Violent eruption of metal vapor and spatter, high concentrated plume, bright electric arc, and other strong interfering factors are the reason for the difficulty of in-situ monitoring. And most critically, the exploration of UHPLW process of thick-section steel for a well-formed process window is not mature enough.



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Fig. 20. The multiple sensing system which included two photodiodes, a spectrometer, two visual cameras, and an X-ray sensing system [224].

In the future, with the breakthrough and improvement of the process, a more responsive and adaptable process control system will cooperate with the process to effectively avoid the influence of interfering factors. To further improve the welding efficiency and welding quality of thick-section steel, in situ monitoring undoubtedly plays an important role. The future research points focus on the following aspects. In situ sensing of UHPLW process under strong interfering factors needs to be developed. Multi-source welding data deep analysis and fusion is another area in which additional effort is required. Adaptive control strategy for UHPLW process parameters will be a fruitful area for further work. Furthermore, it is promising to combine the simulation data with actual sensor information by multi-fidelity surrogate model and deep data mining.

4.4. Mechanical properties evaluation based on microstructure control

UHPLW of thick-section steel inevitably suffers from microstructure inhomogeneity, which has an important impact on the mechanical properties and service reliability of thick-section steel components. This part of the research is envisaged as the ultimate goal of the work in this field: optimization of mechanical properties through microstructural control. This goal is achieved on the assumption that a process window with good joint morphology has been obtained. In the future, the weld joint can be tailored for strength and fatigue properties.

In terms of strength evaluation methods, considerable exploration has been carried out on the experimental observation and numerical simulation of dual-phase steels. Whether it is the in-situ tensile test by Tang et al. [228], or the in-situ martensitic micro-pillar axial compression test by Ghassemi-Armaki et al. [229], a lot of cost to prepare specimens and test is required. In contrast, the micromechanical model analysis by Al-Abbasi et al. [230], and the virtual RVEs numerical method by Abid et al. [231] offer more cost-effective solutions. However, to balance high computational efficiency and high fidelity, the exploration of numerical analysis methods for joint strength is still an important direction.

It is noteworthy that little attention has been paid to the systematic study of fatigue performance evaluation of thick-section steel laser welded joints. The main research points focus on the following three aspects: the mechanism of fatigue failure induced by inhomogeneous microstructure, fatigue performance characterization, and life prediction methods. The mechanism of fatigue failure induced by the peculiarity of microstructure distribution of the weld due to the increase of plate thickness is worth studying. The influence of process parameters on microstructure can be correlated by a multi-scale multi-physical field model to characterize the fatigue performance of thick-section steel joints (further details about multi-scale methods are presented in relevant subsection 4.2.). Machine learning algorithms can be used to construct the relationship between

microstructure characteristics and fatigue life, which is the frontier development direction for fatigue life prediction of the welded joints.

5. Conclusion

Thick-section steel is widely used in industries such as oil and gas pipeline, power plants, construction engineering, shipbuilding, and ocean engineering. UHPLW of thick-section steel has been the subject of extensive research and development over the past few decades. The review presented an overview of the current state-of-the-art techniques, challenges, progresses, and opportunities in this field.

To address the challenges in welding defects, microstructures, and mechanical properties, researchers have made various progresses, including process optimization, advanced innovative techniques, and high-fidelity numerical models. The main findings can be drawn:

- Spatter formation around a keyhole is related to the dynamic behavior of the elongated liquid column during the UHPLW process. Compared to welding parameters optimization, the SLIM and VEA method are considered to be more effective in reducing severe spatters due to smooth molten pool flow and an enlarged and stabilized keyhole. The evolution law of the liquid column in different areas around the keyhole can be further studied.
- UHPLW with an electromagnetic weld pool support system can achieve a single-pass full penetration for 20mm thick-section steel without root humping. The MFA method can also have beneficial effects on the suppression of porosity and solidification cracks. Despite increasing the complexity of the device, this MFA technology displays significant promise for future fabrication endeavors.
- UHPLW under a (low) vacuum environment can achieve full penetration welds without defects for thick-section steel of a thickness up to 50mm. The

VEA method not only increases penetration depth but also reduces the porosity defect. However, this VEA technology faces major challenges for industry production, such as the small size of the vacuum chamber, and limited welding flexibility for different structures.

- Porosity defects occur when bubbles induced by collapse of unstable elongated keyhole are difficult to escape from the molten pool due to the large floating distance. The SLIM, VEA, and MFA method can help reduce porosity by reducing bubble generation and promoting bubbles floating.
- The mechanisms of solidification crack generation in the root joint after UHPLW of thick-section steel may be related to the bulge effect of the melt pool. However, the real causes still need to be further explored and cross-scale modeling is a potential future option.
- Insufficient filler wire transportation results in inhomogeneous microstructure and poor mechanical properties in WFLW and HLAW of thick-section steel. The SLIM, MFA and PCP method have been shown to promote uniform distribution and increase low-temperature impact toughness. It is essential to consider the deficiency in toughness in the laser zone, particularly in thick-section steel joint with a thickness up to 20mm, as the critical factor of the parting line in the center of the weld or solidification cracks can significantly reduce the impact toughness. Currently, the inhomogeneous microstructure and mechanical properties along the thickness direction has still not been well resolved.

Despite these advances, further research is needed to fully understand the complex phenomena and develop new solutions for high-quality and high-reliability welding of thick-section steel. Future perspectives remain in new process under spatiotemporal laser intensity modulation (SLIM), multi-scale integrated simulations, in-situ monitoring under strong interfering factors, and mechanical properties evaluation based on microstructure

control. The measures to exploit the potential capabilities of SLIM in UHPLW of thick-section steel require full and detailed explorations in both process trials and physical mechanisms. The development of a new paradigm of data-driven numerical simulation is an important research direction, in combination with the highly efficient and accurate real-time prediction model established by the high-precision coupled simulation algorithm described above. In-situ sensing of the UHPLW process under strong interfering factors needs to be developed. The mechanism of fatigue failure induced by the peculiarity of microstructure distribution of the weld due to the increase of steel plate thickness is worth studying.

Overall, laser welding of thick-section steel has shown great potential to address the increasing demand for high-performance and cost-effective welding in various industrial applications. Our common vision is to achieve the ultimate goal of high-efficiency and high-quality weld joint in UHPLW of thick-section steel and realize practical applications. Attainment of the goal of high quality weld joints will require sustained research and development over several decades due to the highly complexity of UHPLW of thick-section steel.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] <https://www.thebalance.com/steel-grades-2340174> ↗.
[Google Scholar](#) ↗
- [2] S.K. Sharma, S. Maheshwari
A review on welding of high strength oil and gas pipeline steels
J. Nat. Gas Sci. Eng., 38 (2017), pp. 203-217, [10.1016/j.jngse.2016.12.039](https://doi.org/10.1016/j.jngse.2016.12.039) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [3] J. Li, Q. Sun, K. Kang, Z. Zhen, Y. Liu, J. Feng
Process stability and parameters optimization of narrow-gap laser vertical welding with hot wire for thick stainless steel in nuclear power plant
Opt. Laser Technol., 123 (2020), [10.1016/j.optlastec.2019.105921](https://doi.org/10.1016/j.optlastec.2019.105921) ↗
[Google Scholar](#) ↗
- [4] R. Wu, Y. Huang, Y. Rong, J. Xu, Q. Chen
Solid phase transformation effects on stress and strain in the thick plate EH40 welded butt joint
Opt. Laser Technol., 157 (2023), [10.1016/j.optlastec.2022.108635](https://doi.org/10.1016/j.optlastec.2022.108635) ↗
[Google Scholar](#) ↗
- [5] J. Balakrishnan, A.N. Vasileiou, J.A. Francis, M.C. Smith, M.J. Roy, M.D. Callaghan, N.M. Irvine
Residual stress distributions in arc, laser and electron-beam welds in 30 mm thick SA508 steel: a cross-process comparison
Int. J. Press. Vessels Pip., 162 (2018), pp. 59-70, [10.1016/j.ijpvp.2018.03.004](https://doi.org/10.1016/j.ijpvp.2018.03.004) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [6] H.J. Kong, C. Xu, C.C. Bu, C. Da, J.H. Luan, Z.B. Jiao, G. Chen, C.T. Liu
Hardening mechanisms and impact toughening of a high-strength steel containing low Ni and Cu additions
Acta Mater., 172 (2019), pp. 150-160, [10.1016/j.actamat.2019.04.041](https://doi.org/10.1016/j.actamat.2019.04.041) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [7] M. Sokolov, A. Salminen, M. Kuznetsov, I. Tsibulskiy
Laser welding and weld hardness analysis of thick section S355 structural steel
Mater. Des., 32 (10) (2011), pp. 5127-5131, [10.1016/j.matdes.2011.05.053](https://doi.org/10.1016/j.matdes.2011.05.053) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [8] H. Wang, M. Nakanishi, Y. Kawahito
Effects of welding speed on absorption rate in partial and full penetration welding of stainless steel with high brightness and high power laser
J. Mater. Process. Technol., 249 (2017), pp. 193-201, [10.1016/j.jmatprotec.2017.06.014](https://doi.org/10.1016/j.jmatprotec.2017.06.014) ↗
 [View PDF](#) [View article](#) [Google Scholar](#) ↗
- [9] M. Zhang, G. Chen, Y. Zhou, S. Liao
Optimization of deep penetration laser welding of thick stainless steel with a 10kW fiber laser
Mater. Des., 53 (2014), pp. 568-576, [10.1016/j.matdes.2013.06.066](https://doi.org/10.1016/j.matdes.2013.06.066) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [10] J.M. Sun, X.Z. Liu, Y.G. Tong, D. Deng
A comparative study on welding temperature fields, residual stress distributions and deformations induced by laser beam welding and CO2 gas arc welding
Mater. Des., 63 (2014), pp. 519-530, [10.1016/j.matdes.2014.06.057](https://doi.org/10.1016/j.matdes.2014.06.057) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [11] Y. Kawahito, M. Mizutani, S. Katayama

High quality welding of stainless steel with 10 kW high power fibre laser

Sci. Technol. Weld. Joining, 14 (4) (2013), pp. 288-294, [10.1179/136217108x372531](#) ↗

[Google Scholar](#) ↗

- [12] Y. Kawahito, H. Wang, S. Katayama, D. Sumimori
Ultra high power (100 kW) fiber laser welding of steel

Opt. Lett., 43 (19) (2018), pp. 4667-4670, [10.1364/OL.43.004667](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [13] S. Katayama, Y. Kawahito, M. Mizutani
Elucidation of laser welding phenomena and factors affecting weld penetration and welding defects

Phys. Procedia, 5 (2010), pp. 9-17, [10.1016/j.phpro.2010.08.024](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [14] M. Tümer, C. Schneider-Bröskamp, N. Enzinger
Fusion welding of ultra-high strength structural steels – a review

J. Manuf. Process., 82 (2022), pp. 203-229, [10.1016/j.jmapro.2022.07.049](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [15] F. Yang, G. Xia, X. Guo, C. Chen, Y. Long, B. Chen, J. Tang, G. Chen
Research progress of laser welding under subatmospheric pressure

Int. J. Adv. Manuf. Technol., 116 (3–4) (2021), pp. 803-820, [10.1007/s00170-021-07488-3](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [16] K.M. Hong, Y.C. Shin
Prospects of laser welding technology in the automotive industry: a review

J. Mater. Process. Technol., 245 (2017), pp. 46-69, [10.1016/j.jmatprotec.2017.02.008](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [17] X. Shen, L. Li, W. Guo, W. Teng, W. He

Comparison of processing window and porosity distribution in laser welding of 10 mm thick 30CrMnSiA ultrahigh strength between flat (1G) and horizontal (2G) positions

J. Laser Appl., 28 (2) (2016), [10.2351/1.4943992](#) ↗

[Google Scholar](#) ↗

- [18] T. Ohnishi, Y. Kawahito, M. Mizutani, S. Katayama
Butt welding of thick, high strength steel plate with a high power laser and hot wire to improve tolerance to gap variance and control weld metal oxygen content

Sci. Technol. Weld. Joining, 18 (4) (2013), pp. 314-322, [10.1179/1362171813y.0000000108](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [19] I. Bunaziv, C. Dørum, S.E. Nielsen, P. Suikkanen, X. Ren, B. Nyhus, M. Eriksson, O.M. Akselsen
Laser-arc hybrid welding of 12- and 15-mm thick structural steel

Int. J. Adv. Manuf. Technol., 107 (5–6) (2020), pp. 2649-2669, [10.1007/s00170-020-05192-2](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [20] C. Fang, J. Xin, W. Dai, J. Wei, J. Wu, Y. Song
Deep penetration laser welding of austenitic stainless steel thick-plates using a 20 kW fiber laser

J. Laser Appl., 32 (1) (2020), [10.2351/1.5094176](#) ↗

[Google Scholar](#) ↗

- [21] O. Seffer, S. Nothdurft, A. Hilck, M. Hustedt, J. Hermsdorf, S. Kaierle
Investigations on laser beam welding of thick steel plates using a high-power diode laser beam source

J. Laser Appl., 34 (4) (2022), [10.2351/7.0000783](#) ↗

[Google Scholar](#) ↗

- [22] V.S.M. Ramakrishna R, P.H.S.L.R. Amrutha, R.A. Rahman Rashid, S. Palanisamy, Narrow gap laser welding (NGLW) of structural steels—a technological review and future research recommendations, *Int. J. Adv. Manuf. Technol.* 111(7-8) (2020) 2277–2300, 10.1007/s00170-020-06230-9.
[Google Scholar ↗](#)
- [23] ISO 6929:2013, Steel products - Vocabulary.
[Google Scholar ↗](#)
- [24] ASTM A1011/A1011M, Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength.
[Google Scholar ↗](#)
- [25] M. Bachmann, A. Gumenyuk, M. Rethmeier
Welding with high-power lasers: trends and developments
Phys. Procedia, 83 (2016), pp. 15-25, [10.1016/j.phpro.2016.08.003 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [26] W. Guo, D. Crowther, J.A. Francis, A. Thompson, Z. Liu, L. Li
Microstructure and mechanical properties of laser welded S960 high strength steel
Mater. Des., 85 (2015), pp. 534-548, [10.1016/j.matdes.2015.07.037 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [27] X. Cao, P. Wanjara, J. Huang, C. Munro, A. Nolting
Hybrid fiber laser – arc welding of thick section high strength low alloy steel
Mater. Des., 32 (6) (2011), pp. 3399-3413, [10.1016/j.matdes.2011.02.002 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [28] G. Turichin, M. Kuznetsov, I. Tsibulskiy, A. Firsova

Hybrid laser-arc welding of the high-strength shipbuilding steels: equipment and technology

Phys. Procedia, 89 (2017), pp. 156-163, [10.1016/j.phpro.2017.08.005](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [29] I.A. Tsibulskiy, A.D. Akhmetov, R.S. Korsmik, A.A. Voropae, R.V. Mendagaliye, A.D. Ereemeev
The influence of the gap size on the formation of a welded joint in hybrid laser-arc welding of angular joints and T-joints

J. Phys.: Conf. Ser., 1109 (2018), [10.1088/1742-6596/1109/1/012033](#) ↗

[Google Scholar](#) ↗

- [30] I. Bunaziv, C. Dørum, X. Ren, M. Eriksson, O.M. Akselsen
Application of LBW and LAHW for fillet welds of 12 and 15 mm structural steel

Procedia Manuf., 36 (2019), pp. 121-130, [10.1016/j.promfg.2019.08.017](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [31] C. Churiaque, J.M. Sánchez-Amaya, M. Porrúa-Lara, A. Gumenyuk, M. Rethmeier
The effects of HLAW parameters for one side T-joints in 15 mm thickness naval steel

Metals, 11 (4) (2021), [10.3390/met11040600](#) ↗

[Google Scholar](#) ↗

- [32] A.F.H. Kaplan, J. Powell
Spatter in laser welding

J. Laser Appl., 23 (3) (2011), [10.2351/1.3597830](#) ↗

[Google Scholar](#) ↗

- [33] J. Frostevarg
Factors affecting weld root morphology in laser keyhole welding

Opt. Lasers Eng., 101 (2018), pp. 89-98, [10.1016/j.optlaseng.2017.10.005](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [34] J.J. Blecher, T.A. Palmer, T. Debroy
Porosity in thick section alloy 690 welds - experiments, modeling, mechanism, and remedy
Welding J., 95 (1) (2016), pp. 17-26
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [35] M.O. Gebhardt, A. Gumenyuk, M. Rethmeier
Solidification cracking in laser GMA hybrid welding of thick-walled parts
Sci. Technol. Weld. Joining, 19 (3) (2013), pp. 209-213, [10.1179/1362171813y.0000000171 ↗](#)
[Google Scholar ↗](#)
- [36] P.L. Moore, D.S. Howse, E.R. Wallach
Microstructures and properties of laser/arc hybrid welds and autogenous laser welds in pipeline steels
Sci. Technol. Weld. Joining, 9 (4) (2013), pp. 314-322, [10.1179/136217104225021652 ↗](#)
[Google Scholar ↗](#)
- [37] S. Webster, J.K. Kristensen, D. Petring
Joining of thick section steels using hybrid laser welding
Ironmaking Steelmaking, 35 (7) (2008), pp. 496-504, [10.1179/174328108x358505 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [38] P. Haug, V. Rominger, N. Speker, R. Weber, T. Graf, M. Weigl, M. Schmidt
Influence of laser wavelength on melt bath dynamics and resulting seam quality at welding of thick plates
Phys. Procedia, 41 (2013), pp. 49-58, [10.1016/j.phpro.2013.03.051 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [39] I. Bunaziv, C. Dørum, S.E. Nielsen, P. Suikkanen, X. Ren, B. Nyhus, M. Eriksson, O.M. Akselsen
Root formation and metallurgical challenges in laser beam and laser-arc hybrid welding of thick structural steel
Int. J. Adv. Manuf. Technol., 116 (1-2) (2021), pp. 561-578, [10.1007/s00170-021-07453-0 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [40] M. Jiang, W. Tao, Y. Chen, F. Li
Comparison of processing window in full penetration laser welding of thick high-strength steel under atmosphere and sub-atmosphere
Opt. Laser Technol., 109 (2019), pp. 449-455, [10.1016/j.optlastec.2018.08.023 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [41] Q. Li, M. Luo, Z. Mu, A. Huang, S. Pang
Improving laser welding via decreasing central beam density with a hollow beam
J. Manuf. Proces., 73 (2022), pp. 939-947, [10.1016/j.jmapro.2021.12.001 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [42] W. Guo, Q. Liu, J.A. Francis, D. Crowther, A. Thompson, Z. Liu, L. Li
Comparison of laser welds in thick section S700 high-strength steel manufactured in flat (1G) and horizontal (2G) positions
CIRP Ann., 64 (1) (2015), pp. 197-200, [10.1016/j.cirp.2015.04.070 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [43] M. Wahba, M. Mizutani, S. Katayama
Single pass hybrid laser-arc welding of 25 mm thick square groove butt joints
Mater. Des., 97 (2016), pp. 1-6, [10.1016/j.matdes.2016.02.041 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [44] V. Avilov, A. Fritzsche, M. Bachmann, A. Gumenyuk, M. Rethmeier
Full penetration laser beam welding of thick duplex steel plates with electromagnetic weld pool support
J. Laser Appl., 28 (2) (2016), [10.2351/1.4944103 ↗](#)
[Google Scholar ↗](#)

- [45] S. Katayama, R. Ido, K. Nishimoto, M. Mizutani, Y. Mizutani
Full penetration welding of thick high tensile strength steel plate with high-power disk laser in low vacuum
Weld. Int., 32 (5) (2018), pp. 289-302, [10.1080/09507116.2017.1346776](https://doi.org/10.1080/09507116.2017.1346776) ↗
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [46] O. Seffer, R. Lahdo, A. Springer, S. Kaierle
Laser-GMA hybrid welding of API 5L X70 with 23 mm plate thickness using 16 kW disk laser and two GMA welding power sources
J. Laser Appl., 26 (4) (2014), [10.2351/1.4895638](https://doi.org/10.2351/1.4895638) ↗
[Google Scholar ↗](#)
- [47] M.M. Atabaki, N. Yazdian, J. Ma, R. Kovacevic
High power laser welding of thick steel plates in a horizontal butt joint configuration
Opt. Laser Technol., 83 (2016), pp. 1-12, [10.1016/j.optlastec.2016.03.016](https://doi.org/10.1016/j.optlastec.2016.03.016) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [48] M. Bachmann, V. Avilov, A. Gumenyuk, M. Rethmeier
Experimental and numerical investigation of an electromagnetic weld pool support system for high power laser beam welding of austenitic stainless steel
J. Mater. Process. Technol., 214 (3) (2014), pp. 578-591, [10.1016/j.jmatprotec.2013.11.013](https://doi.org/10.1016/j.jmatprotec.2013.11.013) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [49] G. Mi, L. Xiong, C. Wang, X. Hu, Y. Wei
A thermal-metallurgical-mechanical model for laser welding Q235 steel
J. Mater. Process. Technol., 238 (2016), pp. 39-48, [10.1016/j.jmatprotec.2016.07.002](https://doi.org/10.1016/j.jmatprotec.2016.07.002) ↗
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [50] M.J. Torkamany, A.F.H. Kaplan, F.M. Ghaini, M. Vänskä, A. Salminen, K. Fahlström, J. Hedegård

Wire deposition by a laser-induced boiling front

Opt. Laser Technol., 69 (2015), pp. 104-112, [10.1016/j.optlastec.2014.12.013](https://doi.org/10.1016/j.optlastec.2014.12.013) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [51] S. Zhang, J. Sun, M. Zhu, L. Zhang, P. Nie, Z. Li
Fiber laser welding of HSLA steel by autogenous laser welding and autogenous laser welding with cold wire methods

J. Mater. Process. Technol., 275 (2020), [10.1016/j.jmatprotec.2019.116353](https://doi.org/10.1016/j.jmatprotec.2019.116353) ↗

[Google Scholar](#) ↗

- [52] W. Huang, S. Chen, J. Xiao, X. Jiang, Y. Jia
Investigation of filler wire melting and transfer behaviors in laser welding with filler wire

Opt. Laser Technol., 134 (2021), [10.1016/j.optlastec.2020.106589](https://doi.org/10.1016/j.optlastec.2020.106589) ↗

[Google Scholar](#) ↗

- [53] B. Acherjee
Hybrid laser arc welding: State-of-art review

Opt. Laser Technol., 99 (2018), pp. 60-71, [10.1016/j.optlastec.2017.09.038](https://doi.org/10.1016/j.optlastec.2017.09.038) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [54] B. Ribic, T.A. Palmer, T. DebRoy
Problems and issues in laser-arc hybrid welding

Int. Mater. Rev., 54 (4) (2013), pp. 223-244, [10.1179/174328009x411163](https://doi.org/10.1179/174328009x411163) ↗

[Google Scholar](#) ↗

- [55] R.G.N. Silva, C.M.M. de Paço, M.B. Rodrigues, J.M.S. de Sousa, M. Pereira, B.B. Ramos, M.B. Schwedersky, R.H. Gonçalves e Silva
A comparison between LBW and hybrid laser-GMAW processes based on microstructure and weld geometry for hardenable steels

Int. J. Adv. Manuf. Technol., 110 (9–10) (2020), pp. 2801-2814, [10.1007/s00170-020-06076-1](https://doi.org/10.1007/s00170-020-06076-1) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [56] C. Churiaque, M. Chludzinski, M. Porrua-Lara, A. Dominguez-Abecia, F. Abad-Fraga, J. Sánchez-Amaya
Laser hybrid butt welding of large thickness naval steel
Metals, 9 (1) (2019), [10.3390/met9010100](https://doi.org/10.3390/met9010100) ↗
[Google Scholar](#) ↗
- [57] S. Liu, Y. Li, F. Liu, H. Zhang, H. Ding
Effects of relative positioning of energy sources on weld integrity for hybrid laser arc welding
Opt. Lasers Eng., 81 (2016), pp. 87-96, [10.1016/j.optlaseng.2016.01.010](https://doi.org/10.1016/j.optlaseng.2016.01.010) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [58] I. Bunaziv, J. Frostevarg, O.M. Akselsen, A.F.H. Kaplan
Process stability during fiber laser-arc hybrid welding of thick steel plates
Opt. Lasers Eng., 102 (2018), pp. 34-44, [10.1016/j.optlaseng.2017.10.020](https://doi.org/10.1016/j.optlaseng.2017.10.020) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [59] J. Zhou, H.L. Tsai
Modeling of transport phenomena in hybrid laser-MIG keyhole welding
Int. J. Heat Mass Transfer, 51 (17–18) (2008), pp. 4353-4366,
[10.1016/j.ijheatmasstransfer.2008.02.011](https://doi.org/10.1016/j.ijheatmasstransfer.2008.02.011) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [60] D.W. Cho, W.I. Cho, S.J. Na
Modeling and simulation of arc: Laser and hybrid welding process
J. Manuf. Processes, 16 (1) (2014), pp. 26-55, [10.1016/j.jmapro.2013.06.012](https://doi.org/10.1016/j.jmapro.2013.06.012) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [61] C.S. Wu, H.T. Zhang, J. Chen
Numerical simulation of keyhole behaviors and fluid dynamics in laser–gas metal arc hybrid welding of ferrite stainless steel plates
J. Manuf. Processes, 25 (2017), pp. 235-245, [10.1016/j.jmapro.2016.11.009](https://doi.org/10.1016/j.jmapro.2016.11.009) ↗

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [62] X. Chen, Z. Mu, R. Hu, L. Liang, A.B. Murphy, S. Pang
A unified model for coupling mesoscopic dynamics of keyhole, metal vapor, arc plasma, and weld pool in laser-arc hybrid welding

J. Manuf. Processes, 41 (2019), pp. 119-134, [10.1016/j.jmapro.2019.03.034 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [63] L. Shi, L. Jiang, M. Gao
Numerical research on melt pool dynamics of oscillating laser-arc hybrid welding

Int. J. Heat Mass Transfer, 185 (2022), [10.1016/j.ijheatmasstransfer.2021.122421 ↗](#)

[Google Scholar ↗](#)

- [64] S. Zhang, Y. Wang, M. Zhu, Y. Feng, P. Nie, Z. Li
Effects of heat source arrangements on Laser-MAG hybrid welding characteristics and defect formation mechanism of 10CrNi3MoV steel

J. Manuf. Process., 58 (2020), pp. 563-573, [10.1016/j.jmapro.2020.08.027 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [65] I. Bunaziv, O.M. Akselsen, J. Frostevarg, A.F.H. Kaplan
Deep penetration fiber laser-arc hybrid welding of thick HSLA steel

J. Mater. Process. Technol., 256 (2018), pp. 216-228, [10.1016/j.jmatprotec.2018.02.026 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [66] L. Bao, Y. Wang, T. Han
Study on microstructure-toughness relationship in heat affected zone of EQ70 steel by laser-arc hybrid welding

Mater. Charact., 171 (2021), [10.1016/j.matchar.2020.110788 ↗](#)

[Google Scholar ↗](#)

- [67] G.F. Sun, Z.D. Wang, Y. Lu, R. Zhou, Z.H. Ni

Investigation on microstructure and mechanical properties of NV E690 steel joint by laser-MIG hybrid welding

Mater. Des., 127 (2017), pp. 297-310, [10.1016/j.matdes.2017.04.054](https://doi.org/10.1016/j.matdes.2017.04.054) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [68] Ö. Üstündağ, N. Bakir, A. Gumenyuk, M. Rethmeier
Influence of oscillating magnetic field on the keyhole stability in deep penetration laser beam welding
Opt. Laser Technol., 135 (2021), [10.1016/j.optlastec.2020.106715](https://doi.org/10.1016/j.optlastec.2020.106715) ↗
[Google Scholar](#) ↗
- [69] S. Pang, L. Chen, J. Zhou, Y. Yin, T. Chen
A three-dimensional sharp interface model for self-consistent keyhole and weld pool dynamics in deep penetration laser welding
J. Phys. D: Appl. Phys., 44 (2) (2011), [10.1088/0022-3727/44/2/025301](https://doi.org/10.1088/0022-3727/44/2/025301) ↗
[Google Scholar](#) ↗
- [70] Y. Wang, P. Jiang, S. Geng, B. Xu, J. Zhao
Influence of the wrinkle surface structures on the vapor flow and keyhole stability in 20 kW high power laser welding
Int. J. Heat Mass Transfer, 193 (2022), [10.1016/j.ijheatmasstransfer.2022.122958](https://doi.org/10.1016/j.ijheatmasstransfer.2022.122958) ↗
[Google Scholar](#) ↗
- [71] A. Artinov, X. Meng, M. Bachmann, M. Rethmeier
Study on the transition behavior of the bulging effect during deep penetration laser beam welding
Int. J. Heat Mass Transfer, 184 (2022), [10.1016/j.ijheatmasstransfer.2021.122171](https://doi.org/10.1016/j.ijheatmasstransfer.2021.122171) ↗
[Google Scholar](#) ↗
- [72] S. Zhang, J. Sun, M. Zhu, L. Zhang, P. Nie, Z. Li

Effects of shielding gases on process stability of 10CrNi3MoV steel in hybrid laser-arc welding

J. Mater. Process. Technol., 270 (2019), pp. 37-46, [10.1016/j.jmatprotec.2019.01.027](https://doi.org/10.1016/j.jmatprotec.2019.01.027) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

- [73] H. Nakamura, Y. Kawahito, K. Nishimoto, S. Katayama
Elucidation of melt flows and spatter formation mechanisms during high power laser welding of pure titanium

J. Laser Appl., 27 (3) (2015), [10.2351/1.4922383](https://doi.org/10.2351/1.4922383) ↗

[Google Scholar](#) ↗

- [74] J. Zou, B. Zhu, G. Zhang, S. Guo, R. Xiao
Power density effect on the laser beam-induced eruption of spatters in fiber laser keyhole welding

Opt. Laser Technol., 147 (2022), [10.1016/j.optlastec.2021.107651](https://doi.org/10.1016/j.optlastec.2021.107651) ↗

[Google Scholar](#) ↗

- [75] M.J. Zhang, G.Y. Chen, Y. Zhou, S.C. Li, H. Deng
Observation of spatter formation mechanisms in high-power fiber laser welding of thick plate

Appl. Surf. Sci., 280 (2013), pp. 868-875, [10.1016/j.apsusc.2013.05.081](https://doi.org/10.1016/j.apsusc.2013.05.081) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [76] Y. Li, S. Geng, S. Gu, D. Huang, Y. Wang, G. Mi, P. Jiang
Effect of liquid column on process stability and weld formation under ultra-high power fiber laser-arc hybrid welding of thick plates

Int. J. Adv. Manuf. Technol., 121 (11–12) (2022), pp. 8243-8255, [10.1007/s00170-022-09712-0](https://doi.org/10.1007/s00170-022-09712-0) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [77] G. Zhang, B. Zhu, J. Zou, Q. Wu, R. Xiao

Correlation between the spatters and evaporation vapor on the front keyhole wall during fiber laser keyhole welding

J. Mater. Res. Technol., 9 (6) (2020), pp. 15143-15152, [10.1016/j.jmrt.2020.10.103](https://doi.org/10.1016/j.jmrt.2020.10.103) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [78] Y. Huang, X. Hua, F. Li, C. Shen, G. Mou, B. Tang
Spatter feature analysis in laser welding based on motion tracking method

J. Manuf. Process., 55 (2020), pp. 220-229, [10.1016/j.jmapro.2020.04.016](https://doi.org/10.1016/j.jmapro.2020.04.016) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [79] T. Liu, R. Hu, X. Chen, S. Gong, S. Pang
Localized boiling-induced spatters in the high-power laser welding of stainless steel: three-dimensional visualization and physical understanding

Appl. Phys. A, 124 (9) (2018), [10.1007/s00339-018-2058-7](https://doi.org/10.1007/s00339-018-2058-7) ↗

[Google Scholar](#) ↗

- [80] M. Zhang, S. Chen, Y. Zhang, G. Chen, Z. Bi
Mechanisms for improvement of weld appearance in autogenous fiber laser welding of thick stainless steels

Metals, 8 (8) (2018), [10.3390/met8080625](https://doi.org/10.3390/met8080625) ↗

[Google Scholar](#) ↗

- [81] S. Li, G. Chen, S. Katayama, Y. Zhang
Relationship between spatter formation and dynamic molten pool during high-power deep-penetration laser welding

Appl. Surf. Sci., 303 (2014), pp. 481-488, [10.1016/j.apsusc.2014.03.030](https://doi.org/10.1016/j.apsusc.2014.03.030) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [82] A.F.H. Kaplan, E.M. Westin, G. Wiklund, P. Norman, Imaging in cooperation with modeling of selected defect mechanisms during fiber laser welding of stainless steel, in: International Congress on Applications of Lasers & Electro-Optics, 2008.
[Google Scholar ↗](#)
- [83] D. Wu, X. Hua, F. Li, L. Huang
Understanding of spatter formation in fiber laser welding of 5083 aluminum alloy
Int. J. Heat Mass Transfer, 113 (2017), pp. 730-740, [10.1016/j.ijheatmasstransfer.2017.05.125 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [84] Z. Wan, H.-P. Wang, J. Li, J. Solomon, N. Chen, B. Carlson
Effect of beam oscillation frequency on spattering in remote laser stitch welding of thin-gage zinc-coated steel with keyhole penetration
J. Mater. Process. Technol., 302 (2022), [10.1016/j.jmatprotec.2021.117482 ↗](#)
[Google Scholar ↗](#)
- [85] S. Pang, W. Chen, J. Zhou, D. Liao
Self-consistent modeling of keyhole and weld pool dynamics in tandem dual beam laser welding of aluminum alloy
J. Mater. Process. Technol., 217 (2015), pp. 131-143, [10.1016/j.jmatprotec.2014.11.013 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [86] J. Volpp, F. Vollertsen
Impact of multi-focus beam shaping on the process stability
Opt. Laser Technol., 112 (2019), pp. 278-283, [10.1016/j.optlastec.2018.11.032 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [87] L. Wang, X. Gao, F. Kong
Keyhole dynamic status and spatter behavior during welding of stainless steel with adjustable-ring mode laser beam
J. Manuf. Processes, 74 (2022), pp. 201-219, [10.1016/j.jmapro.2021.12.011 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [88] C. Börner, K. Dilger, V. Rominger, T. Harrer, T. Krüssel, T. Löwer
Influence of ambient pressure on spattering and weld seam quality in
laser beam welding with the solid-state laser
Int. Congr. Appl. Lasers Electro-Optics (2011), pp. 621-629
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [89] M. Jiang, N. Jiang, X. Chen, S. Ma, Y. Chen, Y. Chen, Z. Lei
Experimental and numerical investigation of single-pass laser welding of
20 mm-thick high-strength steel under reduced ambient pressure

J. Mater. Res. Technol., 15 (2021), pp. 2317-2331, [10.1016/j.jmrt.2021.09.030 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [90] M. Wahba, M. Mizutani, S. Katayama
Hybrid welding with fiber laser and CO2 gas shielded arc
J. Mater. Process. Technol., 221 (2015), pp. 146-153, [10.1016/j.jmatprotec.2015.02.004 ↗](#)
[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [91] C. Zhang, M. Gao, X. Zeng
Influences of synergy effect between laser and arc on laser-arc hybrid
welding of aluminum alloys
Opt. Laser Technol., 120 (2019), [10.1016/j.optlastec.2019.105766 ↗](#)
[Google Scholar ↗](#)

- [92] S. Liu, F. Zhang, S. Dong, H. Zhang, F. Liu
Characteristics analysis of droplet transfer in laser-MAG hybrid welding
process
Int. J. Heat Mass Transfer, 121 (2018), pp. 805-811, [10.1016/j.ijheatmasstransfer.2018.01.047 ↗](#)
[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [93] F. Liu, B. Yang, H. Sun, H. Wang, C. Tan, G. Wang, B. Chen

Mechanism investigation for the influence of laser power on droplet transfer behaviors in laser-MIG hybrid welding

Opt. Laser Technol., 157 (2023), [10.1016/j.optlastec.2022.108750](https://doi.org/10.1016/j.optlastec.2022.108750) ↗

[Google Scholar](#) ↗

- [94] J. Frostevarg
Comparison of three different arc modes for laser-arc hybrid welding steel

J. Laser Appl., 28 (2) (2016), [10.2351/1.4944098](https://doi.org/10.2351/1.4944098) ↗

[Google Scholar](#) ↗

- [95] Q. Pan, M. Mizutani, Y. Kawahito, S. Katayama
Effect of shielding gas on laser-MAG arc hybrid welding results of thick high-tensile-strength steel plates

Weld. World, 60 (4) (2016), pp. 653-664, [10.1007/s40194-016-0333-9](https://doi.org/10.1007/s40194-016-0333-9) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [96] Y. Zhu, Y. Cai, M. Wang
Effects of He content in shielding gases on high-efficient hybrid laser arc welding with C-276 filler metal

J. Mater. Process. Technol., 299 (2022), [10.1016/j.jmatprotec.2021.117367](https://doi.org/10.1016/j.jmatprotec.2021.117367) ↗

[Google Scholar](#) ↗

- [97] Y. Zhu, Y. Cai, H. Dong, M. Wang
Tailoring droplet transfer and molten pool flow during hybrid laser arc welding of nickel base alloy

Opt. Laser Technol., 147 (2022), [10.1016/j.optlastec.2021.107620](https://doi.org/10.1016/j.optlastec.2021.107620) ↗

[Google Scholar](#) ↗

- [98] Y. Li, S. Geng, Z. Zhu, Y. Wang, G. Mi, P. Jiang
Effects of heat source configuration on the welding process and joint formation in ultra-high power laser-MAG hybrid welding

J. Manuf. Process., 77 (2022), pp. 40-53, [10.1016/j.jmapro.2022.02.045](https://doi.org/10.1016/j.jmapro.2022.02.045) ↗

[View PDF](#)[View article](#)[Google Scholar ↗](#)

- [99] M. Rethmeier, S. Gook, M. Lammers, A. Gumenyuk, Laser-hybrid welding of thick plates up to 32 mm using a 20 kW fibre laser, J. Quarte. J. Jpn. Weld. Soc. 27(2) (2009) 74s–79s, 10.2207/qjjws.27.74s.

[Google Scholar ↗](#)

- [100] F. Vollertsen, S. Grunenwald, M. Rethmeier, A. Gumenyuk, U. Reisgen, S. Olschok
Welding thick steel plates with fibre lasers and GMAW
Weld. World, 54 (3–4) (2010), pp. R62-R70, 10.1007/bf03263489 ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [101] F. Farrokhi, R.M. Larsen, M. Kristiansen
Single-pass hybrid laser welding of 25 mm thick steel
Phys. Procedia, 89 (2017), pp. 49-57, 10.1016/j.phpro.2017.08.019 ↗

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [102] E. Iso
Welding - Laser-arc Hybrid Welding of Steels, Nickel and Nickel Alloys -
Quality Levels for Imperfections
ISO (2013)

[Google Scholar ↗](#)

- [103] J.J. Blecher, T.A. Palmer, T. DebRoy
Mitigation of root defect in laser and hybrid laser-arc welding
Weld. J., 94 (3) (2015), pp. 73-82

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [104] M. Zhang, Z. Zhang, K. Tang, C. Mao, Y. Hu, G. Chen
Analysis of mechanisms of underfill in full penetration laser welding of
thick stainless steel with a 10 kW fiber laser
Opt. Laser Technol., 98 (2018), pp. 97-105, 10.1016/j.optlastec.2017.07.037 ↗

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[105] J. Frostevarg, A.F.H. Kaplan

Undercuts in laser arc hybrid welding

Phys. Procedia, 56 (2014), pp. 663-672, [10.1016/j.phpro.2014.08.071 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[106] J. Karlsson, P. Norman, A.F.H. Kaplan, P. Rubin, J. Lamas, A. Yañez

Observation of the mechanisms causing two kinds of undercut during laser hybrid arc welding

Appl. Surf. Sci., 257 (17) (2011), pp. 7501-7506, [10.1016/j.apsusc.2011.03.068 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[107] T. Ilar, I. Eriksson, J. Powell, A. Kaplan

Root humping in laser welding - an investigation based on high speed imaging

Phys. Procedia, 39 (2012), pp. 27-32, [10.1016/j.phpro.2012.10.010 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

[108] Q. Pan, M. Mizutani, Y. Kawahito, S. Katayama

High power disk laser-metal active gas arc hybrid welding of thick high tensile strength steel plates

J. Laser Appl., 28 (1) (2016), [10.2351/1.4934939 ↗](#)

[Google Scholar ↗](#)

[109] S. Li, G. Chen, C. Zhou

Effects of welding parameters on weld geometry during high-power laser welding of thick plate

Int. J. Adv. Manuf. Technol., 79 (1–4) (2015), pp. 177-182, [10.1007/s00170-015-6813-z ↗](#)

[Google Scholar ↗](#)

[110] J. Powell, T. Ilar, J. Frostevarg, M.J. Torkamany, S.-J. Na, D. Petring, L. Zhang, A.F.H. Kaplan

Weld root instabilities in fiber laser welding

J. Laser Appl., 27 (S2) (2015), [10.2351/1.4906390](#) ↗

[Google Scholar](#) ↗

[111] Y. Ai, P. Jiang, C. Wang, G. Mi, S. Geng

Experimental and numerical analysis of molten pool and keyhole profile during high-power deep-penetration laser welding

Int. J. Heat Mass Transfer, 126 (2018), pp. 779-789, [10.1016/j.ijheatmasstransfer.2018.05.031](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[112] M. Zhang, T. Liu, R. Hu, Z. Mu, S. Chen, G. Chen

Understanding root humping in high-power laser welding of stainless steels: a combination approach

Int. J. Adv. Manuf. Technol., 106 (11–12) (2020), pp. 5353-5364, [10.1007/s00170-020-05021-6](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[113] R. Zhang, X. Tang, L. Xu, F. Lu, H. Cui

Mechanism study of thermal fluid flow and weld root hump suppression in full penetration laser welding of Al alloy with alternating magnetic field support

Int. J. Heat Mass Transfer, 166 (2021), [10.1016/j.ijheatmasstransfer.2020.120759](#) ↗

[Google Scholar](#) ↗

[114] Y. Qi, G. Chen, S. Deng, D. Zhou

Periodic root humps in thick-plate laser welding using steady electromagnetic force

J. Mater. Process. Technol., 273 (2019), [10.1016/j.jmatprotec.2019.05.028](#) ↗

[Google Scholar](#) ↗

[115] M. Bachmann, R. Kunze, V. Avilov, M. Rethmeier

Finite element modeling of an alternating current electromagnetic weld pool support in full penetration laser beam welding of thick duplex stainless steel plates

J. Laser Appl., 28 (2) (2016), [10.2351/1.4943906](#) ↗

[Google Scholar](#) ↗

- [116] U. Reisgen, S. Olschok, S. Jakobs, C. Turner
Laser beam welding under vacuum of high grade materials

Weld. World, 60 (3) (2016), pp. 403-413, [10.1007/s40194-016-0302-3](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [117] S. Pang, K. Hirano, R. Fabbro, T. Jiang
Explanation of penetration depth variation during laser welding under variable ambient pressure

J. Laser Appl., 27 (2) (2015), [10.2351/1.4913455](#) ↗

[Google Scholar](#) ↗

- [118] D. Petring, C. Fuhrmann, N. Wolf, R. Poprawe
Progress in laser-mag hybrid welding of high-strength steels up to 30 mm thickness

Int. Congr. Appl. Lasers Electro-Opt. (2007)

[Google Scholar](#) ↗

- [119] R. Michael, G. Sergej, L. Marco, G. Andrey
Laser-hybrid welding of thick plates up to 32 mm using a 20 kw fibre laser

Quart. J. Jpn. Weld. Soc., 27 (2) (2009), pp. 74-79

[Google Scholar](#) ↗

- [120] G. Tang, X. Zhao, R. Li, Y. Liang, Y. Jiang, H. Chen
The effect of arc position on laser-arc hybrid welding of 12-mm-thick high strength bainitic steel

Opt. Laser Technol., 121 (2020), [10.1016/j.optlastec.2019.105780](#) ↗

[Google Scholar ↗](#)

- [121] S. Webster, J.K. Kristensen, D. Petring
Joining of thick section steels using hybrid laser welding
Ironmaking Steelmaking, 35 (7) (2013), pp. 496-504, [10.1179/174328108x358505 ↗](#)
[Google Scholar ↗](#)
- [122] P. Tao, G. Mi, C. Wang, X. Zhang
Formation, microstructure and mechanical properties of double-sided fiber laser welded ultra-high strength steel
Opt. Laser Technol., 153 (2022), [10.1016/j.optlastec.2022.108208 ↗](#)
[Google Scholar ↗](#)
- [123] S. Gook, A. Gumenyuk, M. Rethmeier
Hybrid laser arc welding of X80 and X120 steel grade
Sci. Technol. Weld. Joining, 19 (1) (2013), pp. 15-24, [10.1179/1362171813y.00000000154 ↗](#)
[Google Scholar ↗](#)
- [124] M. Wu, Z. Luo, S. Ao
A novel welding method for extra-thick high-strength steel: double-sided narrow gap oscillating laser and oscillating laser-TIG hybrid welding
Opt. Laser Technol., 164 (2023), [10.1016/j.optlastec.2023.109432 ↗](#)
[Google Scholar ↗](#)
- [125] F. Chaussé, P. Paillard, E. Bertrand, G. Rückert
Hybrid laser-metal active gas welding of S460ML steel plates: weldability and consequences of dropping on weld quality
J. Laser Appl., 28 (2) (2016), [10.2351/1.4943991 ↗](#)
[Google Scholar ↗](#)
- [126] J. Frostevarg, M.J. Torkamany, J. Powell, A.F.H. Kaplan
Improving weld quality by laser re-melting

J. Laser Appl., 26 (4) (2014), [10.2351/1.4895562 ↗](#)

[Google Scholar ↗](#)

- [127] L.J. Zhang, J.X. Zhang, A. Gumenyuk, M. Rethmeier, S.J. Na
Numerical simulation of full penetration laser welding of thick steel plate
with high power high brightness laser

J. Mater. Process. Technol., 214 (8) (2014), pp. 1710-1720, [10.1016/j.jmatprotec.2014.03.016 ↗](#)



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [128] Ö. Üstündağ, S. Gook, A. Gumenyuk, M. Rethmeier
Hybrid laser arc welding of thick high-strength pipeline steels of grade
X120 with adapted heat input

J. Mater. Process. Technol., 275 (2020), [10.1016/j.jmatprotec.2019.116358 ↗](#)

[Google Scholar ↗](#)

- [129] M. Mazar Atabaki, J. Ma, W. Liu, R. Kovacevic
Pore formation and its mitigation during hybrid laser/arc welding of
advanced high strength steel

Mater. Des., 67 (2015), pp. 509-521, [10.1016/j.matdes.2014.10.072 ↗](#)



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [130] Y. Zhang, G. Chen, C. Zhou, Y. Jiang, P. Zhong, S. Li
Pores formation in laser–MAG welding of 42CrMo steel

J. Mater. Process. Technol., 245 (2017), pp. 309-317, [10.1016/j.jmatprotec.2017.02.029 ↗](#)



[View PDF](#) [View article](#) [Google Scholar ↗](#)

- [131] J. Xu, Y. Rong, Y. Huang, P. Wang, C. Wang
Keyhole-induced porosity formation during laser welding

J. Mater. Process. Technol., 252 (2018), pp. 720-727, [10.1016/j.jmatprotec.2017.10.038 ↗](#)



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [132] A. Matsunawa, J.D. Kim, N. Seto, M. Mizutani, S. Katayama

Dynamics of keyhole and molten pool in laser welding

J. Laser Appl., 10 (6) (1998), pp. 247-254, [10.2351/1.521858](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[133] S. Katayama, A. Matsunawa

Microfocused X-ray transmission real-time observation of laser welding phenomena

Weld. Int., 16 (6) (2002), pp. 425-431, [10.1080/09507110209549555](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[134] M. Boley, P. Webster, A. Heider, R. Weber, T. Graf

Investigating the keyhole behavior by using x-ray and optical depth measurement techniques

Int. Congr. Appl. Lasers Electro-Opt. (2014), pp. 426-430

[Google Scholar](#) ↗

[135] R. Cunningham, C. Zhao, N. Parab, C. Kantzos, J. Pauza, K. Fezzaa, T. Sun, A.D. Rollett

Keyhole threshold and morphology in laser melting revealed by ultrahigh-speed x-ray imaging

Science, 363 (6429) (2019), pp. 849-852, [10.1126/science.aav4687](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[136] J.D. Madison, L.K. Aagesen

Quantitative characterization of porosity in laser welds of stainless steel

Scr. Mater., 67 (9) (2012), pp. 783-786, [10.1016/j.scriptamat.2012.06.015](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[137] J.W. Elmer, J. Vaja, H.D. Carlton, R. Pong

The effect of Ar and N2 shielding gas on laser weld porosity in steel, stainless steel, and nickel

Weld. J., 94 (10) (2015), pp. 313-325

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [138] J. Zhou, H.L. Tsai
Effects of electromagnetic force on melt flow and porosity prevention in pulsed laser keyhole welding
Int. J. Heat Mass Transfer, 50 (11–12) (2007), pp. 2217–2235,
[10.1016/j.ijheatmasstransfer.2006.10.040](https://doi.org/10.1016/j.ijheatmasstransfer.2006.10.040) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [139] H. Zhao, L. He, W.C. Niu, B. Zhang
Investigation on porosity suppression in deep-penetration laser welding by using computational fluid dynamics
J. Laser Appl., 28 (3) (2016), [10.2351/1.4953065](https://doi.org/10.2351/1.4953065) ↗
[Google Scholar](#) ↗
- [140] H. Zhao, W. Niu, B. Zhang, Y. Lei, M. Kodama, T. Ishide
Modelling of keyhole dynamics and porosity formation considering the adaptive keyhole shape and three-phase coupling during deep-penetration laser welding
J. Phys. D: Appl. Phys., 44 (48) (2011), [10.1088/0022-3727/44/48/485302](https://doi.org/10.1088/0022-3727/44/48/485302) ↗
[Google Scholar](#) ↗
- [141] F. Lu, X. Li, Z. Li, X. Tang, H. Cui
Formation and influence mechanism of keyhole-induced porosity in deep-penetration laser welding based on 3D transient modeling
Int. J. Heat Mass Transfer, 90 (2015), pp. 1143–1152, [10.1016/j.ijheatmasstransfer.2015.07.041](https://doi.org/10.1016/j.ijheatmasstransfer.2015.07.041) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [142] W.I. Cho, S.J. Na, C. Thomy, F. Vollertsen
Numerical simulation of molten pool dynamics in high power disk laser welding
J. Mater. Process. Technol., 212 (1) (2012), pp. 262–275, [10.1016/j.jmatprotec.2011.09.011](https://doi.org/10.1016/j.jmatprotec.2011.09.011) ↗

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [143] W.E. King, H.D. Barth, V.M. Castillo, G.F. Gallegos, J.W. Gibbs, D.E. Hahn, C. Kamath, A.M. Rubenchik

Observation of keyhole-mode laser melting in laser powder-bed fusion additive manufacturing

J. Mater. Process. Technol., 214 (12) (2014), pp. 2915-2925, [10.1016/j.jmatprotec.2014.06.005 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [144] A. Kaplan
A model of deep penetration laser welding based on calculation of the keyhole profile

J. Phys. D: Appl. Phys., 27 (9) (1994), p. 1805

[Crossref ↗](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [145] D. Wu, X. Hua, L. Huang, F. Li, Y. Cai
Elucidation of keyhole induced bubble formation mechanism in fiber laser welding of low carbon steel

Int. J. Heat Mass Transfer, 127 (2018), pp. 1077-1086, [10.1016/j.ijheatmasstransfer.2018.07.107 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [146] I. Bunaziv, S. Wenner, X. Ren, J. Frostevarg, A.F.H. Kaplan, O.M. Akselsen
Filler metal distribution and processing stability in laser-arc hybrid welding of thick HSLA steel

J. Manuf. Process., 54 (2020), pp. 228-239, [10.1016/j.jmapro.2020.02.048 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [147] S. Pang, X. Chen, J. Zhou, X. Shao, C. Wang
3D transient multiphase model for keyhole, vapor plume, and weld pool dynamics in laser welding including the ambient pressure effect

Opt. Lasers Eng., 74 (2015), pp. 47-58, [10.1016/j.optlaseng.2015.05.003 ↗](#)

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [148] Y. Sun, L. Li, Y. Hao, S. Lin, X. Tang, F. Lu
Numerical modeling on formation of periodic chain-like pores in high power laser welding of thick steel plate
J. Mater. Process. Technol., 306 (2022), [10.1016/j.jmatprotec.2022.117638 ↗](#)
[Google Scholar ↗](#)
- [149] M. Courtois, M. Carin, P. Le Masson, S. Gaied, M. Balabane
A complete model of keyhole and melt pool dynamics to analyze instabilities and collapse during laser welding
J. Laser Appl., 26 (4) (2014), [10.2351/1.4886835 ↗](#)
[Google Scholar ↗](#)
- [150] S. Pang, X. Chen, X. Shao, S. Gong, J. Xiao
Dynamics of vapor plume in transient keyhole during laser welding of stainless steel: Local evaporation, plume swing and gas entrapment into porosity
Opt. Lasers Eng., 82 (2016), pp. 28-40, [10.1016/j.optlaseng.2016.01.019 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [151] L. Shi, X. Li, L. Jiang, M. Gao
Numerical study of keyhole-induced porosity suppression mechanism in laser welding with beam oscillation
Sci. Technol. Weld. Joining, 26 (5) (2021), pp. 349-355, [10.1080/13621718.2021.1913562 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [152] X. Meng, M. Bachmann, A. Artinov, M. Rethmeier
A study of the magnetohydrodynamic effect on keyhole dynamics and defect mitigation in laser beam welding
J. Mater. Process. Technol., 307 (2022), [10.1016/j.jmatprotec.2022.117636 ↗](#)
[Google Scholar ↗](#)

- [153] M. Jiang, T. Debroy, M. Jiang, Y.B. Chen, X. Chen, W. Tao, Enhanced penetration depth during reduced pressure keyhole-mode laser welding, *Weld. J.* 99(4) (2020) 110s–123s, 10.29391/2020.99.011.

[Google Scholar ↗](#)

- [154] R. Fabbro, S. Slimani, I. Doudet, F. Coste, F. Briand
Experimental study of the dynamical coupling between the induced vapour plume and the melt pool for Nd–Yag CW laser welding
J. Phys. D: Appl. Phys., 39 (2) (2006), pp. 394–400, 10.1088/0022-3727/39/2/023 ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [155] J. Sun, P. Nie, K. Feng, Z. Li, B. Guo, E. Jiang
The elimination of pores in laser welds of AISI 304 plate using different shielding gases
J. Mater. Process. Technol., 248 (2017), pp. 56–63, 10.1016/j.jmatprotec.2017.05.011 ↗



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [156] L. Zhao, S. Tsukamoto, G. Arakane, T. Sugino
Prevention of porosity by oxygen addition in fibre laser and fibre laser–GMA hybrid welding
Sci. Technol. Weld. Joining, 19 (2) (2013), pp. 91–97, 10.1179/1362171813y.00000000160 ↗

[Google Scholar ↗](#)

- [157] L. Mei, D. Yan, S. Xie, Z. Lei, X. Ge
Effects of Cr₂O₃ active agent on the weld process dynamic behavior and joint comprehensive properties of fiber laser welded stainless steel thick plate
Opt. Lasers Eng., 128 (2020), 10.1016/j.optlaseng.2020.106027 ↗

[Google Scholar ↗](#)

- [158] Y. Wang, P. Jiang, J. Zhao, S. Geng, B. Xu

Effects of energy density attenuation on the stability of keyhole and molten pool during deep penetration laser welding process: a combined numerical and experimental study

Int. J. Heat Mass Transfer, 176 (2021), [10.1016/j.ijheatmasstransfer.2021.121410](https://doi.org/10.1016/j.ijheatmasstransfer.2021.121410) ↗

[Google Scholar](#) ↗

[159] C. Cai, L. Li, W. Tao, X. Chen

Effects of weaving laser on scanning laser-MAG hybrid welding characteristics of high-strength steel

Sci. Technol. Weld. Joining, 22 (2) (2016), pp. 104-109, [10.1080/13621718.2016.1199126](https://doi.org/10.1080/13621718.2016.1199126) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[160] X. Chen, X. Zhang, S. Pang, R. Hu, J. Xiao

Vapor plume oscillation mechanisms in transient keyhole during tandem dual beam fiber laser welding

Opt. Lasers Eng., 100 (2018), pp. 239-247, [10.1016/j.optlaseng.2017.09.010](https://doi.org/10.1016/j.optlaseng.2017.09.010) ↗



[View PDF](#)

[View article](#)

[Google Scholar](#) ↗

[161] L. Li, H. Xia, G. Ma, G. Peng

Flow dynamics during single- and dual-spot laser welding with one common keyhole of 321 stainless steel

J. Mater. Process. Technol., 255 (2018), pp. 841-852, [10.1016/j.jmatprotec.2018.01.041](https://doi.org/10.1016/j.jmatprotec.2018.01.041) ↗



[View PDF](#)

[View article](#)

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[162] L. Huang, P. Liu, S. Zhu, X. Hua, S. Dong

Experimental research on formation mechanism of porosity in magnetic field assisted laser welding of steel

J. Manuf. Process., 50 (2020), pp. 596-602, [10.1016/j.jmapro.2020.01.007](https://doi.org/10.1016/j.jmapro.2020.01.007) ↗



[View PDF](#)

[View article](#)

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[163] Z. Shenghai, S. Yifu, Q. Huijuan

The technology and welding joint properties of hybrid laser-tig welding on thick plate

Opt. Laser Technol., 48 (2013), pp. 381-388, [10.1016/j.optlastec.2012.11.014](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [164] G. Wiklund, O. Akselsen, A.J. Sørgerd, A.F.H. Kaplan
Geometrical aspects of hot cracks in laser-arc hybrid welding
J. Laser Appl., 26 (1) (2014), [10.2351/1.4827135](#) ↗
[Google Scholar](#) ↗

- [165] N. Bakir, A. Artinov, A. Gumenyuk, M. Bachmann, M. Rethmeier
Numerical simulation on the origin of solidification cracking in laser welded thick-walled structures
Metals, 8 (6) (2018), [10.3390/met8060406](#) ↗
[Google Scholar](#) ↗

- [166] G. Agarwal, A. Kumar, I.M. Richardson, M.J.M. Hermans
Evaluation of solidification cracking susceptibility during laser welding in advanced high strength automotive steels
Mater. Des., 183 (2019), [10.1016/j.matdes.2019.108104](#) ↗
[Google Scholar](#) ↗

- [167] A. Artinov, M. Bachmann, X. Meng, V. Karkhin, M. Rethmeier
On the relationship between the bulge effect and the hot cracking formation during deep penetration laser beam welding
Procedia CIRP, 94 (2020), pp. 5-10, [10.1016/j.procir.2020.09.002](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [168] H. Gao, G. Agarwal, M. Amirthalingam, M.J.M. Hermans
Hot cracking investigation during laser welding of high-strength steels with multi-scale modelling approach
Sci. Technol. Weld. Joining, 23 (4) (2017), pp. 287-294, [10.1080/13621718.2017.1384884](#) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[169] S. Kou

A criterion for cracking during solidification

Acta Mater., 88 (2015), pp. 366-374, [10.1016/j.actamat.2015.01.034 ↗](#)



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[170] X. Wang, F. Lu, H.P. Wang, Z. Qu, L. Xia

Micro-scale model based study of solidification cracking formation mechanism in Al fiber laser welds

J. Mater. Process. Technol., 231 (2016), pp. 18-26, [10.1016/j.jmatprotec.2015.12.006 ↗](#)



[View PDF](#) [View article](#) [Google Scholar ↗](#)

[171] H.R. Zareie Rajani, A.B. Phillion

3D multi-scale multi-physics modelling of hot cracking in welding

Mater. Des., 144 (2018), pp. 45-54, [10.1016/j.matdes.2018.02.007 ↗](#)



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[172] K. Kadoi, A. Fujinaga, M. Yamamoto, K. Shinozaki

The effect of welding conditions on solidification cracking susceptibility of type 310S stainless steel during laser welding using an in-situ observation technique, 57 (2013), pp. 383-390, [10.1007/s40194-013-0023-9 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

[173] N. Bakir, Ö. Üstündağ, A. Gumenyuk, M. Rethmeier

Experimental and numerical study on the influence of the laser hybrid parameters in partial penetration welding on the solidification cracking in the weld root

Weld. World, 64 (3) (2020), pp. 501-511, [10.1007/s40194-020-00847-w ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [174] E. Cicală, G. Duffet, H. Andrzejewski, D. Grevey, S. Ignat, Hot cracking in Al–Mg–Si alloy laser welding – operating parameters and their effects, *Mater. Sci. Eng., A* 395(1-2) (2005) 1–9, [10.1016/j.msea.2004.11.026](https://doi.org/10.1016/j.msea.2004.11.026).
[Google Scholar ↗](#)
- [175] M. Karhu, V. Kujanpää
Solidification cracking studies in multi pass laser hybrid welding of thick section austenitic stainless steel
Hot Crack. Phenom. Welds, III (2011), pp. 161-182
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [176] G. Göbel, B. Brenner
Avoiding hot cracking by induction based change of thermal strains during laser welding
Int. Cong. Appl. Lasers Electro-Opt. (2006)
[Google Scholar ↗](#)
- [177] Y. Zhang, G. Chen, B. Chen, J. Wang, C. Zhou
Experimental study of hot cracking at circular welding joints of 42CrMo steel
Opt. Laser Technol., 97 (2017), pp. 327-334, [10.1016/j.optlastec.2017.07.018](https://doi.org/10.1016/j.optlastec.2017.07.018) ↗
 [View PDF](#) [View article](#) [Google Scholar ↗](#)
- [178] C. Kim, M. Kang, N. Kang
Solidification crack and morphology for laser weave welding of Al 5J32 alloy
Sci. Technol. Weld. Joining, 18 (1) (2013), pp. 57-61, [10.1179/1362171812y.0000000073](https://doi.org/10.1179/1362171812y.0000000073) ↗
[Google Scholar ↗](#)
- [179] F. Farrokhi, B. Endelt, M. Kristiansen

A numerical model for full and partial penetration hybrid laser welding of thick-section steels

Opt. Laser Technol., 111 (2019), pp. 671-686, [10.1016/j.optlastec.2018.08.059](https://doi.org/10.1016/j.optlastec.2018.08.059) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [180] J.S. Byun, J.H. Shim, Y.W. Cho, D.N. Lee
Non-metallic inclusion and intragranular nucleation of ferrite in Ti-killed C-Mn steel

Acta Mater., 51 (6) (2003), pp. 1593-1606, [10.1016/s1359-6454\(02\)00560-8](https://doi.org/10.1016/s1359-6454(02)00560-8) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [181] I. Bunaziv, O.M. Akselsen, J. Frostevarg, A.F.H. Kaplan
Application of laser-arc hybrid welding of steel for low-temperature service

Int. J. Adv. Manuf. Technol., 102 (5–8) (2019), pp. 2601-2613, [10.1007/s00170-019-03304-1](https://doi.org/10.1007/s00170-019-03304-1) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [182] M. Karhu, V. Kujanpää, H. Eskelinen, A. Salminen
Filler metal mixing behaviour of 10 mm thick stainless steel butt-joint welds produced with laser-arc hybrid and laser cold-wire processes

Appl. Sci., 9 (8) (2019), [10.3390/app9081685](https://doi.org/10.3390/app9081685) ↗

[Google Scholar](#) ↗

- [183] W. Suder, S. Ganguly, S. Williams, B.Y.B. Yudodibroto
Penetration and mixing of filler wire in hybrid laser welding

J. Mater. Process. Technol., 291 (2021), [10.1016/j.jmatprotec.2020.117040](https://doi.org/10.1016/j.jmatprotec.2020.117040) ↗

[Google Scholar](#) ↗

- [184] L. Zhao, T. Sugino, G. Arakane, S. Tsukamoto
Influence of welding parameters on distribution of wire feeding elements in CO₂laser GMA hybrid welding

Sci. Technol. Weld. Joining, 14 (5) (2013), pp. 457-467, [10.1179/136217109x434252](https://doi.org/10.1179/136217109x434252) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [185] X. Meng, A. Artinov, M. Bachmann, M. Rethmeier
Numerical and experimental investigation of thermo-fluid flow and element transport in electromagnetic stirring enhanced wire feed laser beam welding
Int. J. Heat Mass Transfer, 144 (2019), [10.1016/j.ijheatmasstransfer.2019.118663 ↗](#)
[Google Scholar ↗](#)
- [186] I. Bunaziv, O.M. Akselsen, J. Frostevarg, A.F.H. Kaplan
Laser-arc hybrid welding of thick HSLA steel
J. Mater. Process. Technol., 259 (2018), pp. 75-87, [10.1016/j.jmatprotec.2018.04.019 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [187] W.I. Cho, S.J. Na, M.H. Cho, J.S. Lee
Numerical study of alloying element distribution in CO2 laser–GMA hybrid welding
Comput. Mater. Sci, 49 (4) (2010), pp. 792-800, [10.1016/j.commatsci.2010.06.025 ↗](#)
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [188] R. Rai, S.M. Kelly, R.P. Martukanitz, T. DebRoy
A convective heat-transfer model for partial and full penetration keyhole mode laser welding of a structural steel
Metall. Mater. Trans. A, 39 (1) (2007), pp. 98-112, [10.1007/s11661-007-9400-6 ↗](#)
[Google Scholar ↗](#)
- [189] J. Zhou, H.L. Tsai
Investigation of mixing and diffusion processes in hybrid spot laser–MIG keyhole welding
J. Phys. D: Appl. Phys., 42 (9) (2009), [10.1088/0022-3727/42/9/095502 ↗](#)
[Google Scholar ↗](#)

- [190] X. Chen, G. Yu, X. He, S. Li, Z. Li
Numerical study of heat transfer and solute distribution in hybrid laser-MIG welding
Int. J. Therm. Sci., 149 (2020), [10.1016/j.ijthermalsci.2019.106182](https://doi.org/10.1016/j.ijthermalsci.2019.106182) ↗
[Google Scholar](#) ↗
- [191] X. Meng, M. Bachmann, A. Artinov, M. Rethmeier
Experimental and numerical assessment of weld pool behavior and final microstructure in wire feed laser beam welding with electromagnetic stirring
J. Manuf. Processes, 45 (2019), pp. 408-418, [10.1016/j.jmapro.2019.07.021](https://doi.org/10.1016/j.jmapro.2019.07.021) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [192] Ö. Üstündağ, A. Fritzsche, V. Avilov, A. Gumenyuk, M. Rethmeier
Hybrid laser-arc welding of thick-walled ferromagnetic steels with electromagnetic weld pool support
Weld. World, 62 (4) (2018), pp. 767-774, [10.1007/s40194-018-0597-3](https://doi.org/10.1007/s40194-018-0597-3) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [193] J. Yan, M. Gao, X. Zeng
Study on microstructure and mechanical properties of 304 stainless steel joints by TIG, laser and laser-TIG hybrid welding
Opt. Lasers Eng., 48 (4) (2010), pp. 512-517, [10.1016/j.optlaseng.2009.08.009](https://doi.org/10.1016/j.optlaseng.2009.08.009) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [194] C.Y. Cui, X.G. Cui, X.D. Ren, T.T. Liu, J.D. Hu, Y.M. Wang
Microstructure and microhardness of fiber laser butt welded joint of stainless steel plates
Mater. Des., 49 (2013), pp. 761-765, [10.1016/j.matdes.2013.02.059](https://doi.org/10.1016/j.matdes.2013.02.059) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [195] M. Gao, X. Zeng, J. Yan, Q. Hu

Microstructure characteristics of laser–MIG hybrid welded mild steel

Appl. Surf. Sci., 254 (18) (2008), pp. 5715-5721, [10.1016/j.apsusc.2008.03.070](https://doi.org/10.1016/j.apsusc.2008.03.070) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [196] Y. Li, P. Jiang, Y. Li, G. Mi, S. Geng
Microstructure evolution and mechanical properties in the depth
direction of ultra-high power laser-arc hybrid weld joint of 316L stainless
steel

Opt. Laser Technol., 160 (2023), [10.1016/j.optlastec.2022.109093](https://doi.org/10.1016/j.optlastec.2022.109093) ↗

[Google Scholar](#) ↗

- [197] M. Alali, I. Todd, B.P. Wynne
Through-thickness microstructure and mechanical properties of electron
beam welded 20 mm thick AISI 316L austenitic stainless steel

Mater. Des., 130 (2017), pp. 488-500, [10.1016/j.matdes.2017.05.080](https://doi.org/10.1016/j.matdes.2017.05.080) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [198] X. Xia, J. Wu, Z. Liu, H. Ji, X. Shen, J. Ma, P. Zhuang
Correlation between microstructure evolution and mechanical properties
of 50 mm 316L electron beam welds

Fusion Eng. Des., 147 (2019), [10.1016/j.fusengdes.2019.111245](https://doi.org/10.1016/j.fusengdes.2019.111245) ↗

[Google Scholar](#) ↗

- [199] C. Chen, W. Wang, D. Li, Y. Cai, Y. Zhang, K. Zhang
Effect of beam oscillation on microstructure and properties of laser-TIG
hybrid welding of D406A ultra-high strength steel

J. Manuf. Processes, 57 (2020), pp. 798-805, [10.1016/j.jmapro.2020.07.048](https://doi.org/10.1016/j.jmapro.2020.07.048) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [200] J. Li, Q. Sun, Y. Liu, Z. Zhen, Q. Sun, J. Feng

Melt flow and microstructural characteristics in beam oscillation superimposed laser welding of 304 stainless steel

J. Manuf. Process., 50 (2020), pp. 629-637, [10.1016/j.jmapro.2019.12.053](https://doi.org/10.1016/j.jmapro.2019.12.053) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

- [201] R. Lahdo, S. Nothdurft, J. Hermsdorf, P. Urbanek, M. Puschmann, F. Riedel, L. Overmeyer, S. Kaierle

Investigations on high-power laser beam-submerged arc hybrid welding of thick duplex stainless steel using laser beam wobbling

J. Laser Appl., 33 (4) (2021), [10.2351/7.0000524](https://doi.org/10.2351/7.0000524) ↗

[Google Scholar](#) ↗

- [202] R. Chen, P. Jiang, X.Y. Shao, G.Y. Mi, C.M. Wang, S.N. Geng, S. Gao, L.C. Cao
Improvement of low-temperature impact toughness for 304 weld joint produced by laser-MIG hybrid welding under magnetic field

J. Mater. Process. Technol., 247 (2017), pp. 306-314, [10.1016/j.jmatprotec.2017.04.004](https://doi.org/10.1016/j.jmatprotec.2017.04.004) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [203] X. Qi, X. Wang, H. Di, X. Shen, Z. Liu, P. Huan, L. Chen
Effect of Ti content on the inclusions, microstructure and fracture mechanism of X100 pipeline steel laser-MAG hybrid welds

Mater. Sci. Eng., A 831 (2022), [10.1016/j.msea.2021.142207](https://doi.org/10.1016/j.msea.2021.142207) ↗

[Google Scholar](#) ↗

- [204] A.V. Kolubaev, O.V. Sizova, S.V. Fortuna, A.V. Vorontsov, A.N. Ivanov, E.A. Kolubaev
Weld structure of low-carbon structural steel formed by ultrasonic-assisted laser welding

J. Construct. Steel Res., 172 (2020), [10.1016/j.jcsr.2020.106190](https://doi.org/10.1016/j.jcsr.2020.106190) ↗

[Google Scholar](#) ↗

- [205] A.R. Pavan, B. Arivazhagan, M. Vasudevan, T.N. Prasanthi, C. Sudha

Study on the microstructure and mechanical properties of hybrid laser + MIG welded joints of 316LN stainless steel

Opt. Laser Technol., 163 (2023), [10.1016/j.optlastec.2023.109410](https://doi.org/10.1016/j.optlastec.2023.109410) ↗

[Google Scholar](#) ↗

[206] Y. Chen, J. Feng, L. Li, S. Chang, G. Ma

Microstructure and mechanical properties of a thick-section high-strength steel welded joint by novel double-sided hybrid fibre laser-arc welding

Mater. Sci. Eng., A 582 (2013), pp. 284-293, [10.1016/j.msea.2013.05.077](https://doi.org/10.1016/j.msea.2013.05.077) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[207] N.J. Petch

The influence of grain boundary carbide and grain size on the cleavage strength and impact transition temperature of steel

Acta Metall., 34 (7) (1986), pp. 1387-1393, [10.1016/0001-6160\(86\)90026-X](https://doi.org/10.1016/0001-6160(86)90026-X) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[208] J.C.F. Jorge, L.F.G.d Souza, M.C. Mendes, I.S. Bott, L.S. Araújo, V.R.d. Santos, J.M.A.

Rebello, G.M. Evans, Microstructure characterization and its relationship with impact toughness of C–Mn and high strength low alloy steel weld metals – a review, J. Mater. Res. Technol. 10 (2021) 471–501, [10.1016/j.jmrt.2020.12.006](https://doi.org/10.1016/j.jmrt.2020.12.006).

[Google Scholar](#) ↗

[209] S. Zhen, Z. Duan, D. Sun, Y. Li, D. Gao, H. Li

Study on microstructures and mechanical properties of laser–arc hybrid welded S355J2W+N steel

Opt. Laser Technol., 59 (2014), pp. 11-18, [10.1016/j.optlastec.2013.11.021](https://doi.org/10.1016/j.optlastec.2013.11.021) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[210] I. Bunaziv, G. Langelandsvik, X. Ren, I. Westermann, G. Rørvik, C. Dørum, M.H. Danielsen, M. Eriksson

Effect of preheating and preplaced filler wire on microstructure and toughness in laser-arc hybrid welding of thick steel

J. Manuf. Process., 82 (2022), pp. 829-847, [10.1016/j.jmapro.2022.08.026](https://doi.org/10.1016/j.jmapro.2022.08.026) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [211] R. Shi, S.A. Khairallah, T.T. Roehling, T.W. Heo, J.T. McKeown, M.J. Matthews
Microstructural control in metal laser powder bed fusion additive manufacturing using laser beam shaping strategy

Acta Mater., 184 (2020), pp. 284-305, [10.1016/j.actamat.2019.11.053](https://doi.org/10.1016/j.actamat.2019.11.053) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [212] T.U. Tumkur, T. Voisin, R.P. Shi, P.J. Depond, T.T. Roehling, S. Wu, M.F. Crumb, J.D. Roehling, G. Guss, S.A. Khairallah, M.J. Matthews
Nondiffractive beam shaping for enhanced optothermal control in metal additive manufacturing

Sci. Adv., 7 (38) (2021), [10.1126/sciadv.abg9358](https://doi.org/10.1126/sciadv.abg9358) ↗

[Google Scholar](#) ↗

- [213] M. Dal, R. Fabbro
An overview of the state of art in laser welding simulation

Opt. Laser Technol., 78 (2016), pp. 2-14, [10.1016/j.optlastec.2015.09.015](https://doi.org/10.1016/j.optlastec.2015.09.015) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [214] A. Artinov, M. Bachmann, M. Rethmeier
Equivalent heat source approach in a 3D transient heat transfer simulation of full-penetration high power laser beam welding of thick metal plates

Int. J. Heat Mass Transfer, 122 (2018), pp. 1003-1013, [10.1016/j.ijheatmasstransfer.2018.02.058](https://doi.org/10.1016/j.ijheatmasstransfer.2018.02.058) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [215] A. Ahmadi, R. Mirzaeifar, N.S. Moghaddam, A.S. Turabi, H.E. Karaca, M. Elahinia

Effect of manufacturing parameters on mechanical properties of 316L stainless steel parts fabricated by selective laser melting: a computational framework

Mater. Des., 112 (2016), pp. 328-338, [10.1016/j.matdes.2016.09.043](https://doi.org/10.1016/j.matdes.2016.09.043) ↗

[View PDF](#)[View article](#)[View in Scopus](#) ↗[Google Scholar](#) ↗

- [216] W. Yan, Y. Lian, C. Yu, O.L. Kafka, Z. Liu, W.K. Liu, G.J. Wagner
An integrated process–structure–property modeling framework for additive manufacturing

Comput. Methods Appl. Mech. Eng., 339 (2018), pp. 184-204, [10.1016/j.cma.2018.05.004](https://doi.org/10.1016/j.cma.2018.05.004) ↗

[View PDF](#)[View article](#)[View in Scopus](#) ↗[Google Scholar](#) ↗

- [217] C. Herriott, X. Li, N. Kouraytem, V. Tari, W. Tan, B. Anglin, A.D. Rollett, A.D. Spear
A multi-scale, multi-physics modeling framework to predict spatial variation of properties in additive-manufactured metals

Modell. Simul. Mater. Sci. Eng., 27 (2) (2019), [10.1088/1361-651X/aaf753](https://doi.org/10.1088/1361-651X/aaf753) ↗

[Google Scholar](#) ↗

- [218] P.W. Liu, Z. Wang, Y.H. Xiao, R.A. Lebensohn, Y.C. Liu, M.F. Horstemeyer, X.Y. Cui, L. Chen
Integration of phase-field model and crystal plasticity for the prediction of process-structure-property relation of additively manufactured metallic materials

Int. J. Plast., 128 (2020), [10.1016/j.ijplas.2020.102670](https://doi.org/10.1016/j.ijplas.2020.102670) ↗

[Google Scholar](#) ↗

- [219] C. Körner, E. Attar, P. Heinel
Mesoscopic simulation of selective beam melting processes

J. Mater. Process. Technol., 211 (6) (2011), pp. 978-987, [10.1016/j.jmatprotec.2010.12.016](https://doi.org/10.1016/j.jmatprotec.2010.12.016) ↗

[View PDF](#)[View article](#)[View in Scopus](#) ↗[Google Scholar](#) ↗

- [220] M. Zheng, L. Wei, J. Chen, Q. Zhang, C. Zhong, X. Lin, W. Huang

A novel method for the molten pool and porosity formation modelling in selective laser melting

Int. J. Heat Mass Transfer, 140 (2019), pp. 1091-1105, [10.1016/j.ijheatmasstransfer.2019.06.038](https://doi.org/10.1016/j.ijheatmasstransfer.2019.06.038)

[View PDF](#)[View article](#)[View in Scopus](#)[Google Scholar](#)

- [221] S.A. Khairallah, A.T. Anderson, A. Rubenchik, W.E. King
Laser powder-bed fusion additive manufacturing: physics of complex melt flow and formation mechanisms of pores, spatter, and denudation zones

Acta Mater., 108 (2016), pp. 36-45, [10.1016/j.actamat.2016.02.014](https://doi.org/10.1016/j.actamat.2016.02.014)

[View PDF](#)[View article](#)[View in Scopus](#)[Google Scholar](#)

- [222] M. Bayat, A. Thanki, S. Mohanty, A. Witvrouw, S. Yang, J. Thorborg, N.S. Tiedje, J.H. Hattel
Keyhole-induced porosities in Laser-based Powder Bed Fusion (L-PBF) of Ti6Al4V: High-fidelity modelling and experimental validation

Addit. Manuf., 30 (2019), [10.1016/j.addma.2019.100835](https://doi.org/10.1016/j.addma.2019.100835)

[Google Scholar](#)

- [223] M.A. Russell, A. Souto-Iglesias, T.I. Zohdi
Numerical simulation of laser fusion additive manufacturing processes using the SPH method

Comput. Methods Appl. Mech. Eng., 341 (2018), pp. 163-187, [10.1016/j.cma.2018.06.033](https://doi.org/10.1016/j.cma.2018.06.033)

[View PDF](#)[View article](#)[View in Scopus](#)[Google Scholar](#)

- [224] D. You, X. Gao, S. Katayama
Data-driven based analyzing and modeling of MIMO laser welding process by integration of six advanced sensors

Int. J. Adv. Manuf. Technol., 82 (5–8) (2015), pp. 1127-1139, [10.1007/s00170-015-7455-x](https://doi.org/10.1007/s00170-015-7455-x)

[Google Scholar](#)

- [225] Y. Zhang, X. Gao, S. Katayama

Weld appearance prediction with BP neural network improved by genetic algorithm during disk laser welding

J. Manuf. Syst., 34 (2015), pp. 53-59, [10.1016/j.jmsy.2014.10.005](#) ↗



[View PDF](#) [View article](#) [Google Scholar](#) ↗

[226] Y. Zhang, X. Gao

Analysis of characteristics of molten pool using cast shadow during high-power disk laser welding

Int. J. Adv. Manuf. Technol., 70 (9–12) (2013), pp. 1979-1988, [10.1007/s00170-013-5442-7](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[227] L. Mrňa, M. Šarbort, Š. Řeřucha, P. Jedlička

Feedback control of laser welding based on frequency analysis of light emissions and adaptive beam shaping

Physics Procedia, 39 (2012), pp. 784-791, [10.1016/j.phpro.2012.10.101](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[228] A. Tang, H. Liu, R. Chen, G. Liu, Q. Lai, Y. Zhong, L. Wang, J. Wang, Q. Lu, Y. Shen

Mesoscopic origin of damage nucleation in dual-phase steels

Int. J. Plast., 137 (2021), [10.1016/j.ijplas.2020.102920](#) ↗

[Google Scholar](#) ↗

[229] H. Ghassemi-Armaki, R. Maaß, S.P. Bhat, S. Sriram, J.R. Greer, K.S. Kumar

Deformation response of ferrite and martensite in a dual-phase steel

Acta Mater., 62 (2014), pp. 197-211, [10.1016/j.actamat.2013.10.001](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[230] F.M. Al-Abbasi

Predicting the effect of ultrafine ferrite on the deformation behavior of DP-steels

Comput. Mater. Sci., 119 (2016), pp. 90-107, [10.1016/j.commatsci.2016.03.048](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[231] N.H. Abid, R.K. Abu Al-Rub, A.N. Palazotto
Micromechanical finite element analysis of the effects of martensite morphology on the overall mechanical behavior of dual phase steel

Int. J. Solids Struct., 104–105 (2017), pp. 8–24, [10.1016/j.ijsolstr.2016.11.005](https://doi.org/10.1016/j.ijsolstr.2016.11.005) ↗

[View PDF](#)[View article](#)[View in Scopus](#) ↗[Google Scholar](#) ↗[View Abstract](#)

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...Multi-pass laser filler welding primarily involves filling the groove through the melting and transition of filler wire [6]. On the contrary, single-pass laser penetration welding relies on the strong penetrability of high-power lasers, eliminating the need for grooves and filler materials [7]. Compared to laser filler welding, laser penetration welding significantly improves welding efficiency and reduces heat input [8]....

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Research article

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Hu Y., ..., Wang C.

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Stable production of dissimilar steel joints in construction machinery by narrow gap oscillating laser welding

Fan Y., ..., Zheng C.

Citation Excerpt:

...The laser moved along a specific trajectory so that the forced convection stirred the molten pool and made it fluctuate periodically [25,30]. Under this mode, the dendrites were periodically impacted by the flowing liquid metal, which inhibited the growth of columnar crystals [18,22,29,37] as shown in Fig. 7d. In NGLW, the liquid metal solidified along the fusion zone of the weld towards the central line of the weld....

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Falodun O.E., ..., Bodunrin M.O.

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